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Microwave engineering

- * for satellite and nuclear communication we should have knowledge on microwave engineering
- * micro-tiny (small) $\rightarrow$ related to wavelength not freq.
microwave means very small wavelength signals $\rightarrow$ 1 meter to 1mm
- * freq of microwave varies from 300 MHz to 300 GHz.

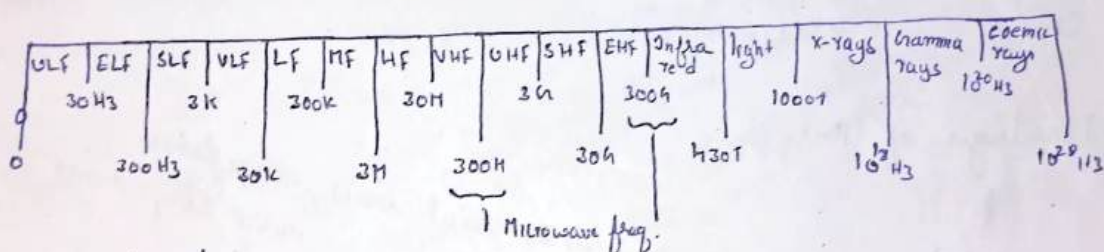
$$f = 300 \text{ MHz to } 300 \text{ GHz} \quad \lambda = 1 \text{ m to } 1 \text{ mm}$$

we know $f = c/\lambda \Rightarrow \lambda = c/f$

for 300 MHz $\lambda = \frac{3 \times 10^8 \text{ m/sec}}{300 \times 10^6 \text{ 1/sec}} = 1 \text{ m}$

for 300 GHz $\lambda = \frac{3 \times 10^8 \text{ m/sec}}{300 \times 10^9 \text{ 1/sec}} = 10^{-3} \text{ m} = 1 \text{ mm}$

Freq spectrum:-



band allocation

if we want to launch nuclear or satellite we can't allocate entire freq. bcoz cross commⁿ occur or diff problems.
so for each satellite or nuclear specific freq band is allocated
eg: * satellite $\rightarrow$ 3 G to 4 G (range).

Two satellite can't have same freq because of miscommunication.

* International telecommunication union (ITU)

ITU launched IEEE standard for better band design

Designation	freq range	wavelength range
L band	1-2 GHz	15 cm to 30 cm
S	2-4	7.5 to 15 cm
C	4-8	3.75 to 7.5 cm
X	8-12	2.5 mm to 3.75 mm

In our lab $\rightarrow$ we use X-band freq

K _u band	12-18	16.7 mm to 25 mm
K	18-26.5	11.3 to 16.7
K _a	26.5 to 40	5.0 mm to 11.3 mm
Q	33 to 50	6.0 to 9.0
U	40 to 60	5.0 to 7.5 mm
V	50 to 75	4.0 to 6.0
W	75 to 110	2.7 to 4.0
F	90 to 140	2.1 to 3.3
D	110 to 170	1.8 to 2.7

K under K_u
K above K_a

Advantages of Microwave

Two types
 multi conductor guided media $\rightarrow$ co-axial lines, strip lines, micro strip lines
 single conductor " " (only TE & TM) modes
 eg Rectangular, circular

microwaves
 $\downarrow$
 mill
 $\rightarrow$ This are
 $\rightarrow$ freq

Micro waves

small electromagnetic wave (signal)

→ This are " " with a wavelength range 1m to 1mm

→ freq range 300MHz to 300GHz. → $\lambda = 1\text{mm}$ to 30cm

→ freq can be categorized into 3 bands if freq 1Hz to 11GHz then $\lambda = 1\text{m}$ to 1mm

- 1) Ultra high freq range (0.3GHz to 3GHz)
- 2) Super high " " (3GHz to 30GHz)
- 3) Extra high " " (30GHz to 300GHz).

Wave guides

at high freq transmission of electromagnetic wave along Tx lines & cables is diff due to losses such as ohms losses, dielectric losses

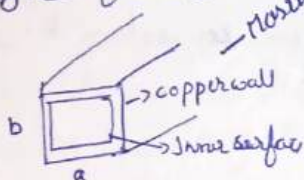
A hollow metallic tube of uniform cross-section for transmitting electromagnetic waves by successive reflections from inner walls of tube is called waveguide

power losses occur, to minimize this waveguide wall resistance is made as low as possible so inner surface is coated with either gold or silver to improve conductivity usually filled with air

NO TEM waves

Types of waveguides

→ mostly used



rectangular



circular



elliptical



ridge



Double ridge

→ used to twist waves used at rotating antennas or radars

→ used for flexibility of wave used for bending, twisting

↓ used to reduce size of wave guide so inture λ is ↑ but distortions occur but useful freq range can be increased

alikes (B)

- 71, 72, 73, 75, 77, 78, 79,
81, 82, 84, 85, 86, 88,
92, 94, 97, 98, 99, A0
A2, A3, A4, A8, B0
B1, B3, B4, B6, B7
C1, C2, C3, C4, C7, C8, C9,
D1, D2, D3, D4, D5, D8, D9, E0
E1, E2,

(A)

- 5, 6, 8, 13, 18, 19, 22, 23, 24
25, 26, 27, 28, 30, 31, 34,
37, 40, 43, 44, 49, 50, 53
56, 58, 59, 60, 61, 63, 69

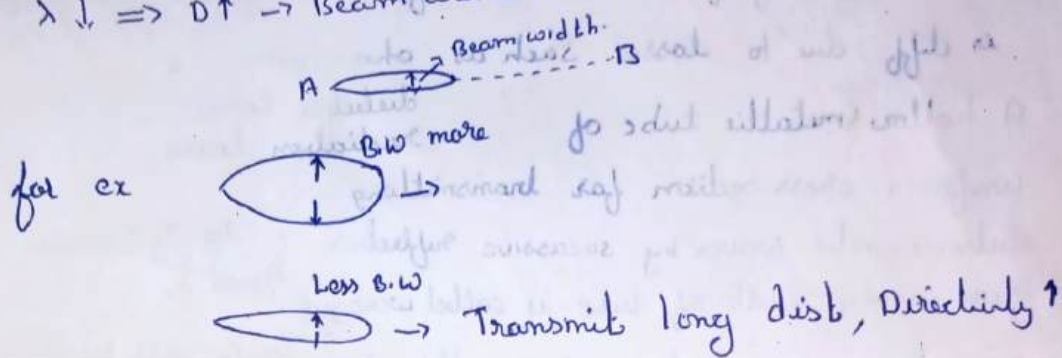
Advantages of Microwave

→ Large bandwidth

MW has large B.W when compared to short waves, medium waves and other waves. point to point communication is possible

→ Better directivity → towards the particular target

$f \uparrow \lambda \downarrow \Rightarrow D \uparrow \rightarrow \text{Beam width} \downarrow$



→ Small size antenna: M.W allows to decrease the size of antenna $f \propto 1/\text{size of antenna}$

In older days one cable operator and he give connection to everyone is TV connection

But now every one has dish antenna

→ Low power consumption

for high freq, power consumption is very low

→ Effect of fading: It is minimized by using line of sight (LOS) propagation of MW freq.

ations

used in long distance communication

- (i) Telecommunication $\rightarrow$ Telephone, TV, space communication, telemetry communication
- (ii) Radars $\rightarrow$ detect aircrafts, tracking missiles, weather forecasting, speed detection
bugler alarms, garage door openers, air traffic control
- (iii) Commercial and industrial applications
 - $\rightarrow$ microwave oven (2450 MHz, 600W); Drying machines - textiles, drying clothes, chips
 - food and paper industry
 - $\rightarrow$ Food processing industry - pre-cooking / cooling, roasting beans, grains
 - $\rightarrow$ Rubber industry $\rightarrow$ plastics chemicals
 - $\rightarrow$ mining / public works, breaking rocks, tunnel boring
 - $\rightarrow$ Drying inks, textiles, leather, tobacco.
 - $\rightarrow$ Medical application

since it has only one conductor
TEM wave mode is not supported
in wave guide.

(iv) Identifying objects by non-contact method

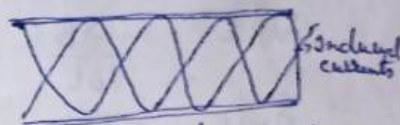
Waveguide: A hollow metallic tube of uniform cross section for transmitting electromagnetic waves by successive reflections from their inner walls of tube

No TEM TE, TM - exists

Due to induced currents - losses occurs

inner walls - coated with silver, gold,

$\rightarrow$ should be smooth and moist free so manufacture using



copper, brass, aluminium,

Comparison of waveguide with 2-wire transmission lines

Similarities

- $\rightarrow$ wave will be attenuated as in tx line
- $\rightarrow$ impedances should be matched to observe the wave at end or else it will be reflected
- $\rightarrow$ irregularities produce reflections

Dissimilarities

- $\rightarrow$ In wave guide there is cut off freq f_c , only freq above f_c will be transmitted whereas in a transmission line all freq will be transmitted ^{2-wire loss less}
- $\rightarrow$ wave propagate through multiple reflection
- $\rightarrow$ velocity of wave in inside waveguide is diff from the wave in free space
- propagation in waveguide is in accordance with field theory
- " " " " " "
- " " " " " "

Electromagnetic plane waves
 principle is based on relationship b/w electricity & magnetism
 A change in magnetic field produces electric field
 " " " electric " " " magnetic "
 Sinusoidal form of wave produces energy to be interchanged b/w
 magnetic and electric field in direction of wave propagation
 In free space electric & magnetic are perpendicular to each other
 and both are normal to direction of propagation. This wave is
 called as TEM wave
 If electric wave exists TE-mode wave $E_z = 0, H_z \neq 0$ (direction of
 " mag " " " TM mode wave $E_z \neq 0, H_z = 0$

Electric and magnetic field equations
 can be derived from Maxwell equations

In time domain equations are.

$$\nabla \times E = - \frac{\partial B}{\partial t} \quad - (1)$$

$$\nabla \cdot D = \rho_v \quad - (3)$$

$$\nabla \times H = J + \frac{\partial D}{\partial t} \quad - (2)$$

$$\nabla \cdot B = 0 \quad - (4)$$

vector operator ∇ (del) is expressed as

$$\nabla = \frac{\partial}{\partial x} u_x + \frac{\partial}{\partial y} u_y + \frac{\partial}{\partial z} u_z \quad (\text{Cartesian}) \quad - (5)$$

$$\nabla = \frac{\partial}{\partial r} u_r + \frac{1}{r} \frac{\partial}{\partial \phi} u_\phi + \frac{\partial}{\partial z} u_z \quad (\text{cylindrical}) \quad - (6)$$

$$\nabla = \frac{\partial}{\partial r} u_r + \frac{1}{r} \frac{\partial}{\partial \theta} u_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} u_\phi \quad (\text{spherical}) \quad - (7)$$

E - electric field intensity in V/m; H - magnetic field intensity amperes/meter

D - electric flux density in coulombs/sq.m

B - mag " " " webers/sq.m

J - conduction current density in amperes/sq.meter

$\partial D / \partial t \rightarrow$ displacement current density

(4)

$$D = \epsilon E \quad (9)$$

$$J_c = \sigma E \quad (11)$$

$$B = \mu H \quad (10)$$

$$E = E_r E_0 \quad (12)$$

$$\mu = \mu_r \mu_0 \quad (13)$$

$\epsilon \rightarrow$ Dielectric permittivity Farads/meter

$\epsilon_r \rightarrow$ relative " constant (dimensionless)

$\epsilon_0 \rightarrow 8.854 \times 10^{-12} \text{ F/m}$ dielectric permittivity of free space.

$\mu \rightarrow$ magnetic permeability ; $\mu_r \rightarrow$ relative permeability

$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ is permeability of free space

$\sigma =$ conductivity of medium in mho/m

assume sinusoidal time fun is in form $e^{j\omega t}$ $\partial/\partial t \rightarrow j\omega$

$$\therefore \nabla \times E = -j\omega \mu H \quad (14)$$

$$\nabla \times H = (\sigma E) + j\omega \epsilon E \Rightarrow (\sigma + j\omega \epsilon) E \quad (15)$$

$$\nabla \cdot D = \rho_v \quad (16) \quad \nabla \cdot B = 0 \quad (17)$$

Taking curl of eq (14) on both sides we get

$$\nabla \times \nabla \times E = -j\omega \mu \nabla \times H \quad (18)$$

eq 15 in 18

$$\nabla \times \nabla \times E = -j\omega \mu (\sigma + j\omega \epsilon) E \quad (19)$$

The vector identity for curl of curl of a vector quantity E is expressed as

$$\nabla \times \nabla \times E = -\nabla^2 E + \nabla(\nabla \cdot E) \quad (20)$$

In free space space density is zero, and in perfect conductor time varying or static fields do not exist so

$$\nabla \cdot D = \rho_v = 0 \quad (21)$$

$$\nabla \cdot E = 0 \quad (22)$$

eq (20) (22) in left hand side of (19) and

$$-\nabla^2 E + \nabla(\nabla \cdot E) = -j\omega \mu (\sigma + j\omega \epsilon) E$$

$$-\nabla^2 E = \gamma^2 E \quad \text{where} \quad \gamma = \sqrt{j\omega \mu (\sigma + j\omega \epsilon)} = \alpha + j\beta$$

$\hookrightarrow$ propagation constants

α = attenuation constant of medium
 β = phase constant in radians per meter
 Hly magnetic wave eq. is

$$\nabla^2 H = \gamma^2 H \quad \text{--- (24)}$$

double del ∇^2 is second order operator in 3 diff co-ordinate system

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad \text{--- (25) - In rectangular co-ordinate}$$

so eq (1) and (2) can be written as

$$\nabla^2 E = \gamma^2 E = (j\omega\mu\sigma + (j\omega)^2\mu\epsilon)E$$

$$\nabla^2 E = \mu\sigma \frac{\partial E}{\partial t} + \mu\epsilon \frac{\partial^2 E}{\partial t^2} \quad \text{--- (26)}$$

$$\nabla^2 H = \mu\sigma \frac{\partial H}{\partial t} + \mu\epsilon \frac{\partial^2 H}{\partial t^2} \quad \text{--- (30)}$$

in free space $\sigma = 0$

$$\nabla^2 E = \mu\epsilon \frac{\partial^2 E}{\partial t^2} = \mu\epsilon (\omega^2 E)$$

$$\therefore \boxed{\nabla^2 E = -\omega^2 \mu\epsilon E} \quad \text{for TM wave } H_z = 0 \\ E_z \neq 0$$

$$\text{Hly } \boxed{\nabla^2 H = -\omega^2 \mu\epsilon H} \quad \text{for TE wave } E_z = 0 \\ H_z \neq 0$$

(5)

propagation of waves in rectangular wave guide

propagation along z-direction

TE TM equations are

$$\nabla^2 E_z = -\omega^2 \mu \epsilon H_z \quad (E_z = 0)$$

$$\nabla^2 H_z = -\omega^2 \mu \epsilon E_z \quad (H_z = 0)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\nabla^2 E_z = \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} = -\omega^2 \mu \epsilon E_z \quad \text{--- (1)}$$

Since wave propagating in z direction we can write

$$\frac{\partial^2}{\partial z^2} = \gamma^2 \quad \text{--- (2)}$$

(2) in (1)

$$\therefore \nabla^2 E_z = \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \gamma^2 E_z = -\omega^2 \mu \epsilon E_z$$

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + E_z (\gamma^2 + \omega^2 \mu \epsilon)$$

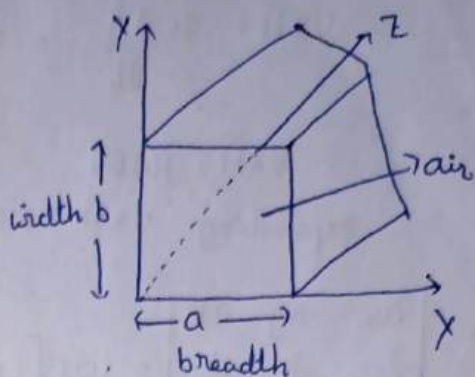
For TM wave

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + h^2 E_z = 0 \quad [h^2 = \gamma^2 + \omega^2 \mu \epsilon]$$

For TE wave

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + h^2 H_z = 0$$

By solving abv TE, TM eq, we get E_z, H_z value



Using Maxwell's 1st eq

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} = \sigma \vec{E} + j\omega \epsilon \vec{E} \quad \text{in free space } \sigma = 0$$

$$\therefore \nabla \times \vec{H} = j\omega \epsilon \vec{E}$$

expanding $\nabla \times \vec{H}$ we get

$$\begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = j\omega \epsilon [E_x \vec{a}_x + E_y \vec{a}_y + E_z \vec{a}_z]$$

replace $\partial/\partial z = -\gamma$

$$\begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ H_x & H_y & H_z \end{vmatrix} = j\omega \epsilon [E_x \vec{a}_x + E_y \vec{a}_y + E_z \vec{a}_z]$$

$$\vec{a}_x \left(\frac{\partial H_z}{\partial y} + \gamma H_y \right) - \vec{a}_y \left(\frac{\partial H_z}{\partial x} + \gamma H_x \right) + \vec{a}_z \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) = j\omega \epsilon E_x \vec{a}_x + j\omega \epsilon E_y \vec{a}_y + j\omega \epsilon E_z \vec{a}_z$$

$$\frac{\partial H_z}{\partial y} + \gamma H_y = j\omega \epsilon E_x \quad - (3)$$

$$\frac{\partial H_z}{\partial x} + \gamma H_x = j\omega \epsilon E_y \quad - (4)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega \epsilon E_z \quad - (5)$$

From Maxwell eq, $\nabla \times \vec{E} = -j\omega \mu \vec{H}$

$$\begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = -j\omega \mu (H_x \vec{a}_x + H_y \vec{a}_y + H_z \vec{a}_z) \Rightarrow \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ E_x & E_y & E_z \end{vmatrix} = -j\omega \mu (\dots)$$

by expanding and equating we get

$$\frac{\partial}{\partial y} E_z + \gamma E_y = -j\omega \mu H_x \quad - (6)$$

$$\frac{\partial}{\partial x} E_z + \gamma E_x = -j\omega \mu H_y \quad - (7)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega \mu H_z \quad - (8)$$

(6)

From (7) $H_y = \frac{1}{j\omega\mu} \left[\frac{\partial E_z}{\partial x} + \gamma E_x \right] \Rightarrow \frac{1}{j\omega\mu} \frac{\partial E_z}{\partial x} + \frac{\gamma E_x}{j\omega\mu} \quad \text{--- (9)}$

eq (9) in (5)

$$\frac{\partial H_z}{\partial y} + \gamma \left[\frac{1}{j\omega\mu} \frac{\partial E_z}{\partial x} + \frac{\gamma E_x}{j\omega\mu} \right] = j\omega\epsilon E_x$$

$$\frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial x} + \frac{\gamma^2 E_x}{j\omega\mu} = j\omega\epsilon E_x$$

$$E_x \left[j\omega\epsilon - \frac{\gamma^2}{j\omega\mu} \right] = \frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial x}$$

$$E_x \left[j\omega\epsilon - \frac{\gamma^2}{j\omega\mu} \right] = \frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial x}$$

$$E_x \left[\frac{-\omega^2\epsilon\mu - \gamma^2}{j\omega\mu} \right] = \frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial x}$$

$$E_x \left[\frac{-h^2}{j\omega\mu} \right] = \frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial x}$$

$$E_x = \frac{-j\omega\mu}{h^2} \cdot \frac{\partial H_z}{\partial y} - \frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} \rightarrow (10)$$

from eq (6) $H_x = -\frac{1}{j\omega\mu} \frac{\partial E_z}{\partial y} - \frac{\gamma}{j\omega\mu} E_y \quad \text{--- (11)}$

(11) in (4) $\frac{\partial H_z}{\partial x} + \gamma \left[-\frac{1}{j\omega\mu} \frac{\partial E_z}{\partial y} - \frac{\gamma}{j\omega\mu} E_y \right] = -j\omega\epsilon E_y$

$$\frac{\partial H_z}{\partial x} - \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial y} - \frac{\gamma^2}{j\omega\mu} E_y = -j\omega\epsilon E_y$$

$$E_y \left[\frac{\gamma^2}{j\omega\mu} - j\omega\epsilon \right] = \frac{\partial H_z}{\partial x} - \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial y}$$

$$E_y \left[\frac{\gamma^2 + \omega^2\epsilon\mu}{j\omega\mu} \right] = \frac{\partial H_z}{\partial x} - \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial y}$$

$$E_y \left[\frac{h^2}{j\omega\mu} \right] = \frac{\partial H_z}{\partial x} - \frac{\gamma}{j\omega\mu} \frac{\partial E_z}{\partial y}$$

$$E_y = \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} - \frac{\gamma}{h^2} \frac{\partial E_z}{\partial y} \rightarrow (12)$$

From

From eq (11)

$$H_x = \frac{-1}{j\omega\mu} \frac{\partial E_z}{\partial y} - \frac{\gamma}{j\omega\mu} E_y$$

substitute eq (12) in eq (11)

$$H_x = \frac{-1}{j\omega\mu} \frac{\partial E_z}{\partial y} - \frac{\gamma}{j\omega\mu} \left[\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} - \frac{\gamma}{h^2} \frac{\partial E_z}{\partial y} \right]$$

$$= \frac{-1}{j\omega\mu} \frac{\partial E_z}{\partial y} - \frac{\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{\gamma^2}{h^2 j\omega\mu} \frac{\partial E_z}{\partial y}$$

$$= \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{\partial E_z}{\partial y} \left[\frac{-1}{j\omega\mu} + \frac{\gamma^2}{h^2 j\omega\mu} \right]$$

$$= \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{\partial E_z}{\partial y} \left[\frac{-h^2 + \gamma^2}{j\omega\mu h^2} \right]$$

$$= \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{\partial E_z}{\partial y} \left[\frac{-\gamma^2 - \omega^2\mu\epsilon + \gamma^2}{h^2 j\omega\mu} \right]$$

$$= \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{\partial E_z}{\partial y} \left[\frac{-\omega\epsilon}{h^2 j} \right]$$

$$H_x = \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial x} + \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y} \quad - (13)$$

From eq (9)

$$H_y = \frac{1}{j\omega\mu} \frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega\mu} E_x$$

eq (9) in (10)

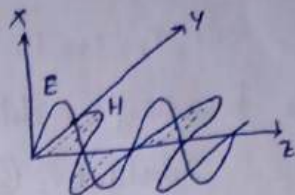
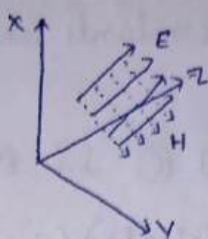
$$H_y = \frac{1}{j\omega\mu} \frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega\mu} \left[\frac{-j\omega\mu}{h^2} \frac{\partial H_z}{\partial y} - \frac{\gamma}{h^2} \frac{\partial E_z}{\partial x} \right]$$

$$H_y = \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x} \quad - (14)$$

TM modes

EM wave in waveguide has no. of patterns due to multiple reflections called as modes

TE and TM are $\perp$



conditions $E \perp$ conductor
 H not $\perp$ " (surface of waveguide)

propagation of TEH waves

For TEH If $E_z = H_z = 0$ and all components E_x, E_y, H_x, H_y vanish

so TEH does not exist in waveguide

Propagation of TM waves

For TM wave $H_z = 0$; $E_z \neq 0$

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + h^2 E_z = 0 \quad \text{--- (1) } \rightarrow \text{partial differential eq.}$$

when solved we get E_x, E_y, E_z, H_x, H_y values

Assume $E_z = xy$ (x, y independent variables)

$$\frac{\partial^2 E_z}{\partial x^2} = \frac{\partial^2 (xy)}{\partial x^2} = y \frac{\partial^2 x}{\partial x^2} \quad \text{|| } y \frac{\partial^2 E_z}{\partial y^2} = x \frac{\partial^2 y}{\partial y^2}$$

substitute in eq (1)

$$y \frac{\partial^2 x}{\partial x^2} + x \frac{\partial^2 y}{\partial y^2} + h^2 xy = 0$$

Divide by xy we get

$$\frac{1}{x} \frac{\partial^2 x}{\partial x^2} + \frac{1}{y} \frac{\partial^2 y}{\partial y^2} + h^2 = 0 \quad \text{--- (2)}$$

x, y are independent variables, each term must be equal

to constant and sum also should be constant

$$\text{Let } \frac{1}{x} \frac{\partial^2 x}{\partial x^2} = -B^2 \quad \text{--- (3)} \quad \frac{1}{y} \frac{\partial^2 y}{\partial y^2} = -A^2 \quad \text{--- (4)}$$

where $-A^2, -B^2 = \text{constant}$

$$\text{(3) (4) in (2)} \Rightarrow -A^2 - B^2 + h^2 = 0$$

$$h^2 = A^2 + B^2$$

⑤ and ⑥ are second order diff eq and their solution

$$x = C_1 \cos Bx + C_2 \sin Bx \quad \text{--- (5)}$$

$$y = C_3 \cos Ay + C_4 \sin Ay \quad \text{--- (6)}$$

C_1, C_2, C_3, C_4 are constants and can be calculated by boundary conditions

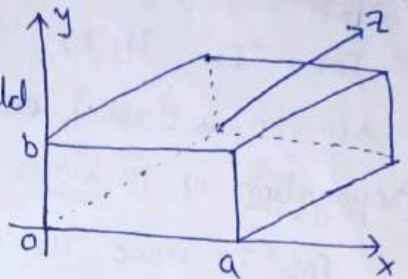
Substituting ⑤ and ⑥ in E_z

$$E_z = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay) \quad \text{--- (7)}$$

This is characteristic eq of TM wave.

Boundary conditions for TM waves

entire surface of rectangular waveguide act as a short ckt or ground of electric field $E_z = 0$ all along boundary walls becoz electric field produced inside waveguide not on surface.



Since there are four walls we have four conditions

→ 1st boundary condition (Bottom plane)

$$E_z = 0 \text{ at } y=0 \quad \forall x, x \rightarrow 0 \text{ to } a$$

→ 2nd boundary condition (left side)

$$E_z = 0 \text{ at } x=0 \quad \forall y \rightarrow 0 \text{ to } b$$

→ 3rd boundary condition (Top plane)

$$E_z = 0 \text{ at } y=b \quad \forall x \rightarrow 0 \text{ to } a$$

→ 4th boundary condition (right side)

$$E_z = 0 \text{ at } x=a \quad \forall y \rightarrow 0 \text{ to } b$$

1st condⁿ in ⑦

$$E_z = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay)$$

$$0 = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos 0 + C_4 \sin 0)$$

$$(C_1 \cos Bx + C_2 \sin Bx) C_3 = 0$$

$$\text{Since } C_1 \cos Bx + C_2 \sin Bx \neq 0 \quad C_3 = 0$$

Sub in eq 7 we get

$$E_z = [C_1 \cos Bx + C_2 \sin Bx][C_4 \sin Ay] \rightarrow \text{⑧}$$

sub 2nd condⁿ in eq (8)

$$0 = (C_1 \cos 0 + C_2 \sin 0) [C_4 \sin \pi y]$$

$$C_1 C_4 \sin \pi y = 0 \quad \forall y \rightarrow 0 \text{ to } b$$

$$\text{since } \sin \pi y \neq 0 \quad C_1 = 0$$

sub in eq (8)

$$E_z = C_2 C_4 \sin Bx \sin \pi y \rightarrow (9)$$

sub 3rd cond in eq (9)

$$0 = C_2 C_4 \sin Bx \sin \pi b$$

$$C_2 = C_4 = \sin Bx \neq 0 \quad \text{so}$$

$$\sin \pi b = 0$$

$$\pi b = n\pi$$

$$\pi b = \text{multiple of } \pi$$

$$\pi b = n\pi \Rightarrow b = n\pi/\pi$$

$$n = \text{integer constant} \\ = 0, 1, 2, 3$$

sub 4th cond in eq 9

$$0 = C_2 C_4 \sin Bx \sin \pi y$$

$$\sin Bx = 0$$

$$Bx = m\pi$$

$$Bx = m\pi$$

$$B = m\pi/a$$

$$m = \text{another integer constant} \\ = 0, 1, 2, 3$$

now complete solution is

$$E_z = C_2 C_4 \sin Bx \sin \pi y$$

$$E_z = C_2 C_4 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

since, wave is propagating along z-direction with sinusoidal

variation w.r.t t we can include term $e^{-\gamma z}$ & $e^{j\omega t}$

$$\therefore E_z = C_2 C_4 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{j\omega t - \gamma z} \quad (10)$$

since E_z known and $H_z = 0$ we can calculate E_x, E_y, H_x, H_y by using 10, 12, 13, 14 eq.

$$E_x = \frac{-\gamma^2}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$= \frac{-\gamma^2}{h^2} \frac{\partial}{\partial x} \left[c \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right]$$

$$E_x = \frac{-\gamma^2}{h^2} \left[c \left(\frac{m\pi}{a}\right) \cos\left(\frac{m\pi}{a}\right)x \cdot \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right] \quad (11)$$

From eq (12)

$$E_y = \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} - \frac{\gamma}{h^2} \frac{\partial E_z}{\partial y}$$

$$E_y = \frac{-\gamma}{h^2} \frac{\partial}{\partial y} \left[c \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right]$$

$$E_y = \frac{-\gamma}{h^2} \left[c \left(\frac{n\pi}{b}\right) \cos\left(\frac{n\pi}{b}\right)y \cdot \sin\left(\frac{m\pi}{a}\right)x \cdot e^{j\omega t - \gamma z} \right] \quad (12)$$

From eq (13)

$$H_x = \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial z} + \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y}$$

$$= \frac{j\omega\epsilon}{h^2} \frac{\partial}{\partial y} \left[c \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right]$$

$$H_x = \frac{j\omega\epsilon}{h^2} \left[c \left(\frac{n\pi}{b}\right) \sin\left(\frac{m\pi}{a}\right)x \cdot \cos\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right] \quad (13)$$

From eq (14)

$$H_y = \frac{-\gamma}{h^2} \frac{\partial H_z}{\partial x} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x}$$

$$H_y = \frac{-j\omega\epsilon}{h^2} \frac{\partial}{\partial x} \left[c \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right]$$

$$H_y = \frac{-j\omega\epsilon}{h^2} \left[c \left(\frac{m\pi}{a}\right) \cdot \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cdot e^{j\omega t - \gamma z} \right] \quad (14)$$

E_z, E_x, E_y, H_x, H_y are field components of TM mode

(9)

Various TM_{mn} modes

TM₀₀ mode:- $m=n=0$ is substituted in E_x, E_y, H_x, H_y all components becomes 0 hence TM₀₀ mode cannot exist

TM₀₁ mode:- cannot exist

TM₁₀ mode:- cannot exist

TM₁₁ mode:- E_x, E_y, H_x, H_y exists so higher values of m and n can exist like TM₂₂, TM₂₁, TM₃₃ can exist.

Cut of freq of waveguide (waveguide act a high pass filter)

we know

$$h^2 = \gamma^2 + \omega^2 \mu \epsilon = A^2 + B^2$$

$$\gamma^2 + \omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\gamma^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon$$

$$\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon}$$

$$\gamma = \alpha + j\beta$$

At low freq $\omega \mu \epsilon < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

so $\gamma = \alpha$ (attenuation const) i.e. wave is attenuated and no phase change. Hence wave cannot propagate

At high freq

$$\omega \mu \epsilon > \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

γ = imaginary, phase change occur and wave propagate
freq at which γ is just 0 is def as cut-off freq

At $f = f_c$ $\gamma = 0$ or $\omega = \omega_c = 2\pi f_c$

$$0 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega_c^2 \mu \epsilon$$

$$\omega_c^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\omega_c = \frac{1}{\sqrt{\mu \epsilon}} \left[\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \right]$$

$$f_c = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$f_c = \frac{c}{2\pi} \times \pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$c = 1/\sqrt{\mu\epsilon}$$

$$f_c = c/2 \sqrt{\left(m/a\right)^2 + \left(n/b\right)^2}$$

at $f > f_c$ wave propagate ; $f < f_c \Rightarrow$ attenuation

cut of wavelength (λ_c)

$$\lambda_c = c/f_c = c / \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\lambda_c = \frac{1}{\frac{1}{2} \sqrt{\frac{m^2 b^2 + n^2 a^2}{a^2 b^2}}} = \frac{1}{\frac{1}{2} \sqrt{\frac{m^2 b^2 + n^2 a^2}{a^2 b^2}}}$$

$$\lambda_c =$$

$$\lambda_c = \frac{2ab}{\sqrt{m^2 b^2 + n^2 a^2}}$$

all wavelength $> \lambda_c$ are attenuated and $< \lambda_c$ are propagated

Mode Characteristics

Guide wavelength (λ_g), Group velocity (v_g) phase velocity (v_p)

Guide wavelength (λ_g)

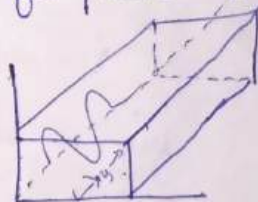
dist travelled by wave along the axis of waveguide for phase shift of 2π radians.

related to phase const by relation $\lambda_g = 2\pi/\beta$.

λ in waveguide is diff from λ in free space

relation b/w $\lambda_g, \lambda_0, \lambda_c \xrightarrow{\text{free space } \lambda} \text{cut off } \lambda$

$$\frac{1}{\lambda_g^2} = \frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}$$



$$\frac{1}{\lambda_g^2} = \frac{\lambda_c^2 - \lambda_0^2}{(\lambda_0 \lambda_c)^2}$$

$$\lambda_g^2 = \frac{\lambda_0^2 \lambda_c^2}{\lambda_c^2 - \lambda_0^2}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\frac{\lambda_c^2 - \lambda_0^2}{\lambda_c^2}}} = \boxed{\frac{\lambda_0}{\sqrt{1 - \frac{\lambda_0^2}{\lambda_c^2}}}} = \lambda_g$$

$\sqrt{1 - \frac{\lambda_0^2}{\lambda_c^2}}$

case (i) $\lambda_0 < \lambda_c$ then $\lambda_g = \frac{\lambda_0}{\sqrt{1 - \frac{\lambda_0^2}{\lambda_c^2}}} = \lambda_g > \lambda_0$

case (ii) $\lambda_0 = \lambda_c$ then $\lambda_g = \frac{\lambda_0}{\sqrt{1 - 1}} \Rightarrow \lambda_g = \infty$

case (iii) $\lambda_0 > \lambda_c$ then it is imaginary no wave propagation

Phase velocity and group velocity

• wave propagating in waveguide has 2 velocities v_p, v_g
 → one velocity is used for propagating and other for phase shifting

→ In free space both are same called as velocity of light

Phase velocity (v_p)

wave propagates when $\lambda_g > \lambda_0$

$v_p = \lambda_g f$ (velocity of propagation is product of λ and f)

$$v = \lambda / t = \lambda f$$

but $v_c = \lambda_0 f$ (v_c = speed of light)

This means $v_p > v_c \because \lambda_g > \lambda_0$

$$\lambda_g = c / f$$

bt not possible

however wavelength in guide is length of cycle

def as rate at which wave changes its phase in term of guide wavelength

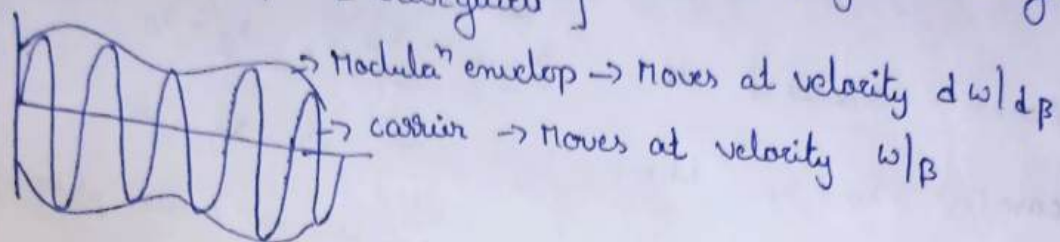
$$v_p = \lambda_g / \text{unit time} \Rightarrow \lambda_g \cdot f = \frac{2\pi f}{2\pi} \cdot \lambda_g = \frac{2\pi f}{2\pi / \lambda_g}$$

$$\therefore v_p = \omega / \beta$$

If there is modulaⁿ in envelop carrier, the modulaⁿ envelop actually travels at velocity slower than carrier and speed of light. The velocity of modulaⁿ envelop is called v_g .

This happens when modulated signal travel in w.h, Modulaⁿ goes on slipping backward w.r.t carrier

$$v_g = d\omega/d\beta \quad \left[\text{rate at which wave propagates through waveguide} \right]$$



Expression for phase velocity (v_p)

$$v_p = \omega/\beta$$

$$h^2 = \gamma^2 + \omega^2 \mu \epsilon = A^2 + B^2$$

$$\gamma^2 = (m\pi/a)^2 + (n\pi/b)^2 - \omega^2 \mu \epsilon \quad (1)$$

$$\gamma = \alpha + j\beta$$

$$\gamma = j\beta \quad (\alpha = 0, \text{ for wave propagation})$$

$$\text{At } f = f_c \quad \omega = \omega_c, \gamma = 0$$

$$0 = (m\pi/a)^2 + (n\pi/b)^2 - \omega_c^2 \mu \epsilon$$

$$\omega_c^2 \mu \epsilon = (m\pi/a)^2 + (n\pi/b)^2 \quad (2)$$

$$(2) \text{ in } (1)$$

$$\gamma^2 = \omega_c^2 \mu \epsilon - \omega^2 \mu \epsilon$$

$$\gamma = j\beta \Rightarrow -\beta^2 = \omega_c^2 \mu \epsilon - \omega^2 \mu \epsilon$$

$$\beta^2 = \omega^2 \mu \epsilon - \omega_c^2 \mu \epsilon$$

$$\beta = \sqrt{\omega^2 \mu \epsilon - \omega_c^2 \mu \epsilon}$$

speed of light
develop

(11)

$$\beta = \sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}$$

$$= \sqrt{\mu\epsilon} \sqrt{\omega^2 - \omega_c^2}$$

$$v_p = \omega / \beta = \omega / \sqrt{\mu\epsilon} \sqrt{\omega^2 - \omega_c^2}$$

$$= \frac{1}{\sqrt{\mu\epsilon}} \cdot \frac{1}{\sqrt{\frac{\omega^2 - \omega_c^2}{\omega^2}}}$$

$$= \frac{1}{\sqrt{\mu\epsilon}} \cdot \frac{1}{\sqrt{1 - \frac{\omega_c^2}{\omega^2}}}$$

$$v_p = \frac{c}{\sqrt{1 - \frac{\omega_c^2}{\omega^2}}} = \boxed{v_p = \frac{c}{\sqrt{1 - \frac{f_c^2}{f^2}}}}$$

we know $f = c/\lambda_0$ $f_c = c/\lambda_c$

$$\frac{f}{f_c} = \frac{c}{\lambda_c} \cdot \frac{\lambda_0}{c} = \lambda_0 / \lambda_c$$

$$\therefore \boxed{v_p = \frac{c}{\sqrt{1 - (\lambda_0 / \lambda_c)^2}}}$$

Expression for group velocity (v_g)

$$v_g = d\omega / d\beta$$

$$\beta = \sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}$$

$$\frac{d\beta}{d\omega} = \frac{d}{d\omega} \sqrt{\mu\epsilon(\omega^2 - \omega_c^2)} = \frac{1}{2\sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}} \cdot 2\omega\mu\epsilon$$

$$= \frac{1}{\sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}} \cdot \frac{\omega\mu\epsilon}{\omega^2} = \frac{\sqrt{\mu\epsilon}}{\sqrt{1 - \frac{\omega_c^2}{\omega^2}}} = \sqrt{\frac{\mu\epsilon}{1 - \left(\frac{f_c}{f}\right)^2}}$$

$$\frac{d\omega}{d\beta} = \sqrt{\frac{1 - (f_c/f)^2}{\mu\epsilon}} \quad \#$$

$$\frac{1}{\sqrt{\mu\epsilon}} = c$$

$$= c \cdot \sqrt{1 - (f_c/f)^2} \quad f_c/f = \lambda_0/\lambda_c$$

$$\therefore v_g = c \cdot \sqrt{1 - (\lambda_0/\lambda_c)^2}$$

$$v_p \cdot v_g = \frac{c}{\sqrt{1 - (\lambda_0/\lambda_c)^2}} \cdot c \cdot \sqrt{1 - (\lambda_0/\lambda_c)^2}$$

$$\boxed{v_p \cdot v_g = c^2}$$

Relation b/w λ_g , λ_0 and λ_c

$$v_p = \lambda_g \cdot f = \lambda_g \cdot c/\lambda_0$$

$$v_p = c / \sqrt{1 - (\lambda_0/\lambda_c)^2}$$

$$\text{So } \frac{c}{\sqrt{1 - (\lambda_0/\lambda_c)^2}} = \frac{\lambda_g}{\lambda_0} \cdot c$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - (\lambda_0/\lambda_c)^2}}$$

$$\lambda_g^2 = \frac{\lambda_0^2}{1 - (\lambda_0/\lambda_c)^2}$$

$$= \frac{\lambda_0^2}{\frac{\lambda_c^2 - \lambda_0^2}{\lambda_c^2}} \Rightarrow \frac{\lambda_0^2 \lambda_c^2}{\lambda_c^2 - \lambda_0^2} = \frac{1}{\frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}}$$

$$\boxed{\therefore \frac{1}{\lambda_g^2} = \frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}}$$

Dominant mode

mode having highest cut-off wavelength $\rightarrow$ distortion less Tx
 other modes $\rightarrow$ higher modes $\rightarrow$ loss of power, harmonic distortion
 used for all electro-magnetic transmission

Dominant mode in TM_{mn}

highest wavelength

Let us consider various modes

$$\text{we know: } \lambda_c = \frac{2ab}{\sqrt{m^2 b^2 + n^2 a^2}}$$

$$\text{TM}_{11} \quad \lambda_{c11} = \frac{2ab}{\sqrt{a^2 + b^2}} \quad \checkmark \quad 0.707 \quad 0.24$$

$$\text{TM}_{12} \quad \lambda_{c12} = \frac{2ab}{\sqrt{b^2 + 4a^2}} \quad \text{TM}_{21} \quad \lambda_{c21} = \frac{2ab}{\sqrt{4b^2 + a^2}}$$

$$\text{TM}_{22} \quad \lambda_{c22} = \frac{2ab}{\sqrt{4(a^2 + b^2)}}$$

$$\text{it is clear that } \lambda_{c11} > \lambda_{c12}$$

$$\lambda_{c11} > \lambda_{c21}$$

$$\lambda_{c11} > \lambda_{c22}$$

λ_{c11} has highest value so TM₁₁ is dominant mode.

lowest cut-off freq (dominant mode)

$$f_{c11} = \frac{c}{2} \sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2}\right)} = \frac{c\sqrt{a^2 + b^2}}{2ab}$$

Field components in TM₁₁ mode:

$m=n=1$ in eq (11), (10), (12), (13), (14)

$$E_z = c \sin\left(\frac{\pi}{a}\right)x \sin\left(\frac{\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$E_x = \frac{-j}{h^2} c \left(\frac{\pi}{a}\right) \cos\left(\frac{\pi}{a}\right)x \sin\left(\frac{\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$E_y = \frac{-j}{h^2} c \left(\frac{\pi}{b}\right) \sin\left(\frac{\pi}{a}\right)x \cos\left(\frac{\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$H_x = \frac{j\omega\epsilon}{h^2} c \left(\frac{\pi}{b}\right) \cdot \sin\left(\frac{\pi}{a}\right)x \cos\left(\frac{\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

$$H_y = \frac{j\omega\epsilon}{h^2} c \left(\frac{\pi}{a}\right) \cos\left(\frac{\pi}{a}\right)x \sin\left(\frac{\pi}{b}\right)y \cdot e^{j\omega t - \gamma z}$$

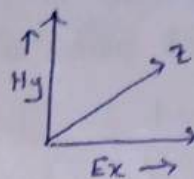
$$H_z = 0$$

Wave impedance (Z_z) in TM wave

def as ratio of strength of electric field in one transverse direction
the strength of magnetic field along other direction

$$Z_z = \frac{E_x}{H_y} = -\frac{E_y}{H_x}$$

$$Z_z = \frac{\sqrt{E_x^2 + E_y^2}}{\sqrt{H_x^2 + H_y^2}}$$



For TM wave

$$Z_z = Z_{TM} = \frac{E_x}{H_y} = \frac{-\frac{\partial}{\partial h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}}{-\frac{\partial}{\partial h^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x}}$$

for TM wave $\gamma = j\beta$ $\alpha = 0$

$$Z_{TM} = \frac{\frac{j\beta}{h^2} \frac{\partial E_z}{\partial x}}{\frac{-j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x}} = \frac{\beta}{\omega\epsilon} \quad \beta = \sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}$$

$$\therefore Z_{TM} = \frac{\sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}}{\omega\epsilon}$$

$$= \frac{\sqrt{\mu\epsilon} \sqrt{\omega^2 - \omega_c^2}}{\sqrt{\epsilon^2} \sqrt{\omega^2}}$$

$$= \sqrt{\frac{\mu}{\epsilon}} \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} = \sqrt{\frac{\mu}{\epsilon}} \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = \sqrt{\frac{\mu}{\epsilon}} \sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}$$

For air $\mu_r = 1, \epsilon_r = 1$ for air

$$\sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} = \sqrt{\frac{4\pi \times 10^{-7}}{\frac{1}{36\pi} \times 10^9}} = 120\pi \Omega = 377\Omega = \eta$$

$$\therefore Z_{TM} = \eta \sqrt{1 - \left(\lambda_0/\lambda_c\right)^2}$$

since $\lambda_0 < \lambda_c$ for wave propagation

so $Z_{TM} < \eta$

For TM wave wave impedance < free space impedance

$$\begin{aligned} &10^2 \\ &144 \times \pi^2 \times 10^2 \\ &12\pi \times 10^2 \end{aligned}$$

Propagation of TE waves

$$E_z = 0 ; H_z \neq 0$$

wave equation is $\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + h^2 H_z = 0$ $[h^2 = \gamma^2 + \omega^2 \mu \epsilon]$

(1)

Let $H_z = xy$. — (2)

$$\frac{\partial^2 H_z}{\partial x^2} = \frac{\partial^2}{\partial x^2} (xy) = y \cdot \frac{\partial^2 x}{\partial x^2} \text{ — (3)}$$

$$\frac{\partial^2 H_z}{\partial y^2} = \frac{\partial^2}{\partial y^2} (xy) = x \cdot \frac{\partial^2 y}{\partial y^2} \text{ — (4)}$$

(3) (4) in (1)

$$y \cdot \frac{\partial^2 x}{\partial x^2} + x \cdot \frac{\partial^2 y}{\partial y^2} + h^2 xy = 0$$

$$\frac{1}{x} \cdot \frac{\partial^2 x}{\partial x^2} + \frac{1}{y} \cdot \frac{\partial^2 y}{\partial y^2} + h^2 = 0 \text{ — (5)}$$

Let $\frac{1}{x} \cdot \frac{\partial^2 x}{\partial x^2} = -B^2$ — (6)

$$\frac{1}{y} \cdot \frac{\partial^2 y}{\partial y^2} = -A^2 \text{ — (7)}$$

(6) and (7) in (5)

$$h^2 = A^2 + B^2$$

Solutions of 6 and 7 eq are

$$x = C_1 \cos Bx + C_2 \sin Bx \text{ — (8)}$$

$$y = C_3 \cos Ay + C_4 \sin Ay \text{ — (9)}$$

(8) & (9) in (2)

$$H_z = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay) \text{ — (10)}$$

C_1, C_2, C_3, C_4 can be evaluated by apply boundary cond

Boundary conditions

$E_z = 0$ b/c we have other components x and y .

$E_x = 0$ all along top and bottom walls

$E_y = 0$ all along left and right walls

1st boundary condition (bottom plane)

$E_x = 0$ at $y=0$ $\forall x$ $x \rightarrow 0$ to a

2nd boundary condition (Top plane)

$E_y = 0$ at $y=b$ $\forall x$ $y \rightarrow 0$ to b $x \rightarrow 0$ to a

3rd boundary condition (left side)

$E_y = 0$ at $y=b$ $x=0$ $\forall y$ $y \rightarrow 0$ to b

4th boundary condition (right side)

$E_y = 0$ at $x=a$ $\forall y$ $y \rightarrow 0$ to b

1st condition in eq (16) of propagation of waves in Rectangular waveguide

is

$$E_x = \frac{-\gamma}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_x = \frac{-j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_x = \frac{-j\omega\mu}{h^2} \frac{\partial}{\partial y} ((C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay))$$

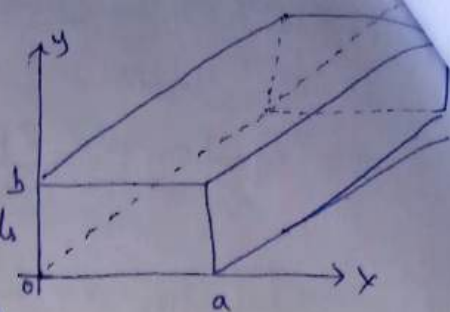
$$E_x = \frac{-j\omega\mu}{h^2} [(C_1 \cos Bx + C_2 \sin Bx)(-A(C_3 \sin Ay + C_4 \cos Ay))]$$

Substitute eq (16) in E_x (1st condⁿ)

$$0 = \frac{-j\omega\mu}{h^2} ((C_1 \cos Bx + C_2 \sin Bx)(A C_4))$$

$$(C_1 \cos Bx + C_2 \sin Bx) \neq 0 \quad A \neq 0 \quad C_4 = 0$$

$$\therefore \text{eq (16) is } H_z = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay) \quad \text{--- (17)}$$



3rd boundary condⁿ is

$E_y = 0$ at $x=0$ $\therefore$ from eq of RWG

$$E_y = \frac{-\gamma}{h^2} \cancel{\frac{\partial E_z}{\partial y}} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$E_y = \frac{j\omega\mu}{h^2} \frac{\partial}{\partial x} [(C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay)]$$

$$E_y = \frac{j\omega\mu}{h^2} [(-BC_1 \sin Bx + BC_2 \cos Bx)(C_3 \cos Ay)]$$

from 3rd condⁿ

$$0 = \frac{j\omega\mu}{h^2} [BC_2 C_3 \cos Ay]$$

$$\cos Ay \neq 0, B \neq 0, C_3 \neq 0 \therefore C_2 = 0$$

Substitute eq $C_2 = 0$ in eq 11

$$H_z = C_1 C_3 \cos Bx \cos Ay \rightarrow (12)$$

2nd boundary condition is

$E_x = 0$ at $y=b$

$$\therefore E_x = \frac{-\gamma}{h^2} \cancel{\frac{\partial E_z}{\partial x}} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$= -\frac{j\omega\mu}{h^2} \frac{\partial}{\partial y} (C_1 C_3 \cos Bx \cos Ay)$$

$$= -\frac{j\omega\mu}{h^2} [-C_1 C_3 A \cos Bx \sin Ay]$$

$$E_x = \frac{j\omega\mu}{h^2} C_1 C_3 A \cos Bx \sin Ay.$$

2nd condⁿ in abv eq.

$$0 = \frac{j\omega\mu}{h^2} C_1 C_3 A \cos Bx \sin Ab$$

$$\cos Bx \neq 0, A \neq 0, C_3 \neq 0$$

$$\sin Ab = 0$$

$$Ab = \sin^{-1} 0$$

$$Ab = n\pi \Rightarrow A = n\pi/b$$

4th boundary condition

$$E_y = 0 \text{ at } x = a$$

$$E_y = \frac{-\partial}{h^2} \frac{\partial \cancel{E_z}}{\partial y} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$E_y = \frac{j\omega\mu}{h^2} \frac{\partial}{\partial x} [C_1 C_3 \cos Bx \cos Ay]$$

$$E_y = \frac{-j\omega\mu}{h^2} [C_1 C_3 B \sin Bx \cos Ay]$$

4th condⁿ in E_y term

$$0 = \frac{-j\omega\mu}{h^2} [C_1 C_3 B \sin Ba \cos Ay]$$

$$\sin Ba = 0$$

$$Ba = m\pi \Rightarrow B = m\pi/a$$

A, B values in eq (1) in Hz

$$H_z = C_1 C_3 \cos Bx \cos Ay$$

$$H_z = C_1 C_3 \cos(m\pi/a)x \cos(n\pi/b)y$$

$C_1 C_3 = C$, due to sinusoidal variations $e^{-\gamma z}$ $e^{j\omega t}$ is added

$$H_z = C \cos(m\pi/a)x \cos(n\pi/b)y e^{j\omega t - \gamma z}$$

↳ characteristic equation of TE wave

field components of TE wave.

$$E_x = -\frac{\partial}{\partial x} \frac{\partial E_z^0}{\partial y} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}$$

$$E_x = \frac{\partial}{\partial x} \left[-\frac{j\omega\mu}{h^2} \frac{\partial}{\partial y} \left[C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right] \right]$$

$$E_x = \frac{j\omega\mu}{h^2} \left[C \cos\left(\frac{m\pi}{a}\right)x \left(\frac{n\pi}{b}\right) \sin\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right] \quad (13)$$

$$E_y = -\frac{\partial}{\partial y} \frac{\partial E_z^0}{\partial x} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$= \frac{j\omega\mu}{h^2} \frac{\partial}{\partial x} \left[C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right]$$

$$E_y = -\frac{j\omega\mu}{h^2} \left[C \left(\frac{m\pi}{a}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right] \quad (14)$$

$$H_x = -\frac{\partial}{\partial x} \frac{\partial H_z}{\partial y} + \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial y}$$

$$= -\frac{\partial}{\partial x} \frac{\partial}{\partial y} \left[C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right]$$

$$H_x = \frac{\gamma}{h^2} \left[C \left(\frac{m\pi}{a}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right] \quad (15)$$

$$H_y = -\frac{\partial}{\partial y} \frac{\partial H_z}{\partial x} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial x}$$

$$= -\frac{\partial}{\partial y} \frac{\partial}{\partial x} \left[C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right]$$

$$= \frac{\gamma}{h^2} \left[C \left(\frac{n\pi}{b}\right)y \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y e^{j\omega t - \gamma z} \right] \quad (16)$$

Various TE_{mn} Modes

(i) TE₀₀ Mode $\rightarrow$ doesn't exist, all components are zero

(ii) TE₀₁ Mode $m=0, n=1$

$$E_x = \frac{j\omega\mu}{h^2} c \left(-\frac{\pi}{b}\right) \sin\left(\frac{\pi}{b}\right) y e^{j\omega t - \gamma z}$$

$$E_y = H_x = 0$$

$$H_y = \frac{\gamma}{h^2} c \left(\frac{\pi}{b}\right) \sin\left(\frac{\pi}{b}\right) y e^{j\omega t - \gamma z}$$

$$H_z = c \cos\left(\frac{\pi}{b}\right) y e^{j\omega t - \gamma z}$$

$\therefore$ TE₀₁ mode exist and TE_{0n} can also exist

(iii) TE₁₀ Mode $m=1; n=0$

$$E_x = H_y = 0$$

$$E_y = -\frac{j\omega\mu}{h^2} c \left(\frac{\pi}{a}\right) \sin\left(\frac{\pi}{a}\right) x e^{j\omega t - \gamma z}$$

$$H_x = \frac{\gamma}{h^2} c \left(\frac{\pi}{a}\right) \sin\left(\frac{\pi}{a}\right) x e^{j\omega t - \gamma z}$$

$$H_z = c \cos\left(\frac{\pi}{a}\right) x e^{j\omega t - \gamma z}$$

TE₁₀ also exist

(iv) TE₁₁ mode also exist, all higher order modes exist.

cut-off freq of TE-mode

Same as TM-mode becoz h^2 value is same

$$\lambda_c = 2ab / \sqrt{m^2 b^2 + n^2 a^2} \quad f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Mode characteristics are also same.

Wave impedance (Z_z) in TE wave.

$$Z_z = Z_{TE} = E_x / H_y.$$

$$= \frac{-\frac{\partial}{\partial x} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}}{-\frac{\partial}{\partial y} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_x}{\partial x}} = \frac{\frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y}}{\frac{j\omega\epsilon}{h^2} \frac{\partial E_x}{\partial x}}$$

$$= \frac{j\omega\mu}{\sqrt{\mu\epsilon(\omega^2 - \omega_c^2)}} = \frac{\omega\mu\sqrt{\mu}}{\sqrt{\mu\epsilon}\sqrt{\omega^2 - \omega_c^2}}$$

$$= \sqrt{\frac{\mu}{\epsilon}} \cdot \frac{1}{\sqrt{1 - \frac{\omega_c^2}{\omega^2}}} = \sqrt{\frac{\mu}{\epsilon}} \cdot \frac{1}{\sqrt{1 - \frac{\omega_c^2}{\omega^2}}}$$

$$Z_{TE} = \sqrt{\frac{\mu}{\epsilon}} \cdot \frac{1}{\sqrt{1 - (f_c/f)^2}} = \frac{\eta}{\sqrt{1 - (\lambda_0/\lambda_c)^2}}$$

for wave propagation $\lambda_0 < \lambda_c$

For TE $Z_{TE} > \eta$.

For TM $Z_{TM} < \eta$.

For TEM $Z_{TEM} = \eta$. (but doesn't exist in w.c)

Dominant mode (highest λ_c value)

$$\lambda_{cmn} = 2ab / \sqrt{m^2 b^2 + n^2 a^2}$$

$$TE_{00} = 2ab / \sqrt{0} = \infty \text{ doesn't exist}$$

$$TE_{01} \quad \lambda_{01} = 2ab / \sqrt{a^2} = 2ab/a = 2b$$

$$TE_{10} \quad \lambda_{10} = 2ab / \sqrt{b^2} = 2ab/b = 2a$$

$$TE_{11} \quad \lambda_{11} = 2ab / \sqrt{a^2 + b^2}$$

$$TE_{22} \quad \lambda_{22} = 2ab / \sqrt{4a^2 + 4b^2}$$

$$\lambda_{10} > \lambda_{01}$$

$$\lambda_{10} > \lambda_{11}$$

$$\lambda_{10} > \lambda_{22}$$

becoz a has large dimension so TE_{10} is dominant mode

$$f_{c10} = \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(0/b\right)^2} = \frac{c}{2a}$$

$$\therefore f_{10} = c/2a$$

Cavity Resonators

→ At low freq. oscillaⁿ and amplification carried by conventional wires, ^{2nd} ^{2nd}

Transistor ^{2nd}

→ oscillators can be designed with coil and capacitor

→ amplifier by transistor and FET

At high freq. ^{abv} 3MHz. all methods not applicable due to skin effect and stray L/C. So cavity resonator is used for ampⁿ & oscⁿ at 3MHz.

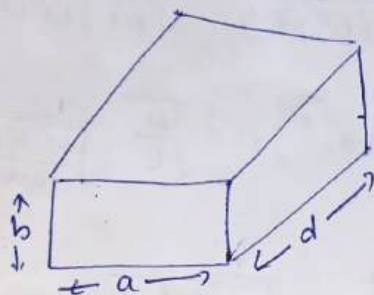
* Cavity resonator saves electromagnetic waves by reflecting them back and forth b/w cavity boundaries.

Two types

(1) Rectangular cavity resonator

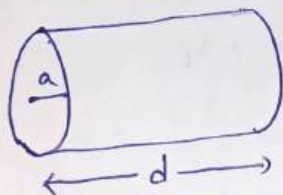
(2) circular " "

(1) RCR



6 surfaces all are closed

(2)



dimensions should be finite length and width must be multiples of $\lambda/2$.

for resonator m, n, a, b values const, λ_c is fixed becoz it depends on m, n, a, b values

λ_0 is fixed $\lambda_0 = c/f_0 \rightarrow$ Resonant freq (f_0 depends on m, n, a, b)

Appⁿ → used to select one particular freq

" " produce oscillation at microwave freq.

we know

$$h^2 = \gamma^2 + \omega^2 \mu \epsilon = A^2 + B^2$$

$$\gamma^2 + \omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \gamma^2$$

$$\gamma = j\beta = -\beta^2$$

$$\omega^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \beta^2$$

for cavity resonator for wave to exist there should be phase change $\therefore \beta = 2\pi/\lambda_g$

condition for resonator for resonance

$$\beta = p\pi/d$$

$p = \text{constant} = 1, 2, 3, \dots$

p indicates half wave variation of either E field or H field along z -dir

$d = \text{Length of resonator}$

when $\beta = p\pi/d$; $f = f_0$

$$\omega = 2\pi f_0 = \omega_0$$

$$\therefore \omega_0^2 \mu \epsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2$$

$$\omega_0^2 = \frac{1}{\mu \epsilon} \left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2 \right]$$

$$\omega_0 = \frac{1}{2\pi \sqrt{\mu \epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$$

$$f_0 = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

mode of propagation for CR is TM_{mnp} or TE_{mnp}

for both TE and TM $\rightarrow$ Resonant freq is same in RWG

Expression for f_0 in circular cavity Resonator

circular end plates are used to short both ends

a = radius of w.g

d = length / h.t. of cylindrical w.g

condⁿ for resonance $\beta = P\pi/d$.

$$h^2 = \gamma^2 + \omega^2 \mu \epsilon =$$

Dominant mode for cavity Resonator

least resonating freq $\rightarrow$ dominant mode

min value of P is 0 since $P \neq 0$

$$TE_{mnp} = f_0 = c/2 \sqrt{(m/a)^2 + (n/b)^2 + (P/d)^2}$$

$$P = 1, 2, 3, \dots$$

$$TE_{01} = TE_{101} = f_0 = c/2 \sqrt{(m/a)^2 + (n/b)^2 + (1/d)^2} \rightarrow \text{minimum}$$

TM TE_{101} is dominant mode

TM TE_{111} is " "

Q-factor of cavity resonator

* Standing waves occur in RCR

* LC tank is used to store E & M field

* Because of standing wave energy won't propagate bt stored

* C.R is reactive element and produce reactance

$$Q = \text{Reactance} / \text{Resistance} = \omega L / R$$

$Q = \infty$ becoz $R = 0$ for C.R (for L and C no resistance)

bt not possible becoz of losses. This losses measured by $Q \rightarrow$ Quality

$Q \rightarrow$ determine how much losses are there

Q -factor measures freq selectivity

$$Q = 2\pi \text{ max energy stored in cycle} / \text{energy dissipated / cycle}$$

$$= \frac{\omega_0 \cdot W}{P} \rightarrow \text{energy stored}$$

$$\downarrow \rightarrow \text{resonating freq.}$$

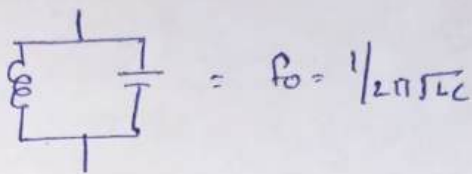
Q-factor of rectangular cavity resonator

$$W = U_{\text{stored}} = \int_V \epsilon E^2 dv \quad (\text{or}) \quad \frac{1}{2} \int_V \mu H^2 dv$$

Cavity resonator $\rightarrow$ hollow metallic enclosed tube exhibit resonance behaviour when excited by EM field

It is similar to parallel resonating ckt at low freq

The freq at which ckt produce resonance that freq is called resonant freq.



consider a waveguide, at one end source is placed, after some time source is removed and that end is shorted. and at dist $\lambda/2$ other end is shorted, the EM waves produced travel back and forth and produce standing waves



$$d = d\lambda_g/2$$

a, b, m, n are fixed so

$\lambda_c = \text{const}, \lambda_0 = \text{const}$

$\lambda_0 = c/f \quad f = f_0 \rightarrow \text{resonant freq}$

Introduction

At micro freq, size of electronic devices req for generating energy is very low, due to this it has low power handling capacity and more noise.

E.D. devices such as tubes and transistors req at H.W. freq.

Tube - high o/p power (control electric current flow) in vacuum tubes, electron tube, valve

S.C. devices - less noise, better reliability with reduced o/p power level.

Conventional tubes - used at low microwave freq.

Special tubes req at high freq, even UHF tubes can't be operated at high freq.

Microwave tubes → make use of transit time effect
 ↳ time taken by e^- to move from one electrode to other
 for microwave freq.
 accelerating e^- produce E.H.W.

Principle of Microwave tube:-

Transfer of power from source of d.c. voltage to the source of d.c. voltage by means of high current density modulated electron beam. Achieved by accelerating e^- in static electric field and retarding them in ac field. The density modulation e^- beam allows more e^- to be retarded in ac field than accelerating.

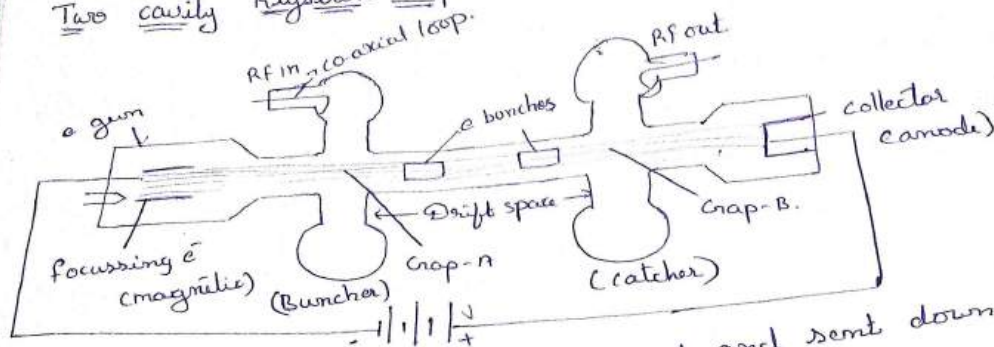
∴ producing more energy to ac field.

Various tubes differ from.

- (i) Mechanism of producing e^- beam
- (ii) acceleration & retardation of e^- in ac field
- (iii) retardation of e^- in short gap or over an extended region

Klystron $\rightarrow$ vacuum tube that can be used either as a generator or as an amplifier of power at H.W. freq.

Two cavity klystron amp $\rightarrow$ velocity modulated tube



high velocity e^- beam is formed, focused and sent down along glass tube through i/p cavity (buncher), field free drift space, and o/p cavity (catcher) to collector. Anode is +ve w.r.t cathode. e^- beam passes through gap-A consisting of two grids of buncher cavity separated by small disc and gap-B, which has two grids of catcher cavity separated by small disc. i/p, o/p taken from tube via resonant cavity with help of loops.

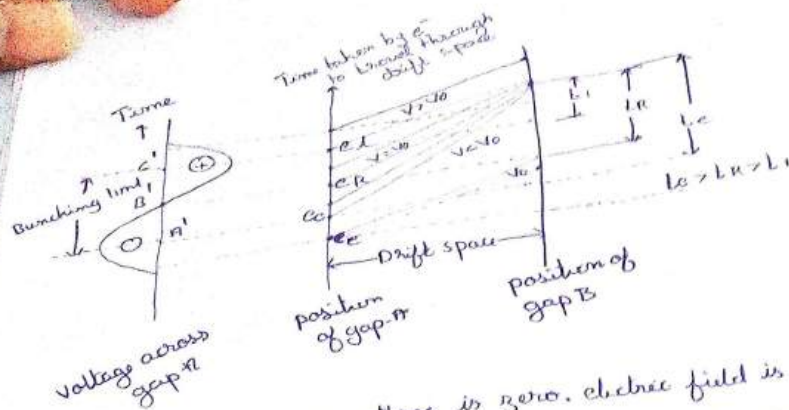
Operation

$\rightarrow$ RF signal to be amplified is given to i/p Buncher cavity and it produces alternating voltage across gap-A.

$\rightarrow$ now let us consider the effect of gap voltage on beam of e^- passing through gap-A.

It can be explained by apple-gate diagram.

$e^- \rightarrow 180^\circ \rightarrow$
 $e^+ \rightarrow 90^\circ \rightarrow$
Application:
 ① Used in Radio
 local oscillators
 rotatable microwave
 amp oscillators
 characteristics
 freq: 4 to 200
 : 1.0 MW +
 : 22-78 %
 10 % to 2
 5 GHz to 2
 Travelling



→ At pt B' alternative voltage is zero, electric field is zero (across gap II).
 e^- which passes through gap A at this instant is unaffected by RF signal.
 This e^- is called as early e^- and travels with an unchanged velocity
 $V_0 = \sqrt{2eV/m}$ $V = \text{anode to cathode } V$

→ At pt C' e^- leaves gap A later than e_R and called as e_L (late e^-).
 it has Max +ve RF voltage and travels towards gap B with velocity $v > V_0$.
 e_L overtakes e_R .

→ At pt A' e^- leaves gap A before e_R and called as e_e (early e^-).
 it has Max -ve RF voltage and travels with ^{reduced} velocity ($v < V_0$). e_e falls
 back and e_R catches e_e .

→ velocity of e^- varies based on RF i/p voltage resulting velocity modulation
 → e^- in b/w bunching limit bunch together as they travel towards
 drift space, (gap A to B). the stream of e^- pass through gap B, and
 produce oscillations in o/p cavity. e^- varies w/ cyc time, i.e. e^- beam
 contains ac current and is current modulated. Drift space converts
 velocity modulation to current modulation

Bunching occurs only once per cycle, with proper design, with little
 RF put applied to buncher cavity produce large beam of currents
 at catcher cavity with large pwr gain

- Applications of klystrons
- In UHF TV transmitters
 - Radar
 - Satellite comm. and station
 - As power osc if used as klystron osc.

Mathematical analysis

DC Voltage b/w cathode & anode V_0

v_0 velocity of e^- ; $V_s =$ RF i/p signal to be amplified
 $= v_0 \sin \omega t_1$

L drift space length

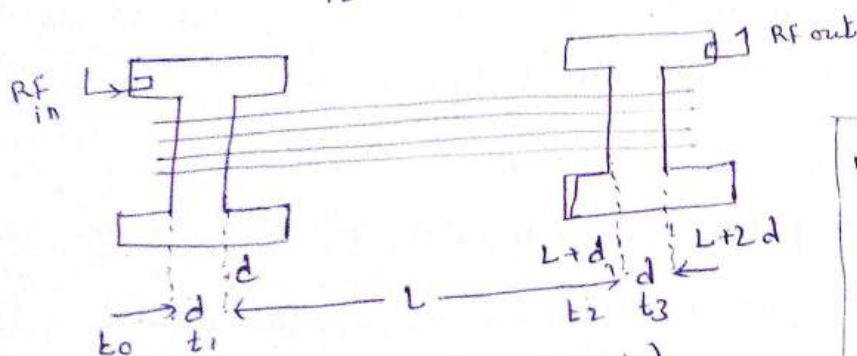
$$v_0 = \sqrt{\frac{2eV_0}{m}} = 0.593 \times 10^6 \sqrt{V_0} \text{ m/sec}$$

$$V_s = V_1 \sin \omega t$$

$V_1 =$ amp of signal $V_1 \ll V_0$ assume

energy of e^- at time of leaving buncher cavity is

$$\frac{1}{2} m v_1^2 = e(V_0 + V_1 \sin \omega t_1)$$



$$v_1^2 = \frac{2e(V_0 + V_1 \sin \omega t_1)}{m}$$

$$v_1 = \sqrt{\frac{2e(V_0 + V_1 \sin \omega t_1)}{m}}$$

$$= \sqrt{\frac{2eV_0}{m}} \cdot \sqrt{1 + \frac{V_1}{V_0} \sin \omega t_1}$$

$$v_1 = v_0 \left[1 + \frac{V_1}{V_0} \sin \omega t_1 \right]$$

expanding and neglecting higher powers of $\sin \omega t$, we get

$$v_1 = v_0 \left[1 + \frac{V_1}{2V_0} \sin \omega t_1 \right]$$

This is eq of velocity modulation

v_0 - velocity of e^- before RF
 v_1 - " " " after RF

v_1 - velocity of e^- after RF i/p

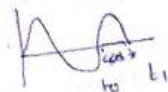
$$v_1 = \sqrt{\frac{2e(V_0 + V_s)}{m}}$$

$$v_1 = v_0 \left[1 + \frac{V_1}{2V_0} \sin \omega t_1 \right]$$

let θ_0 be phase angle of i/p signal at which acceleration of e^- takes place

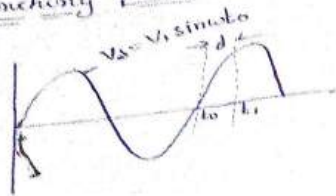
$$\theta_0 = \omega t_0 = \omega(t_1 - t_0)$$

$$\theta_0 = \frac{\omega d}{v_0}$$



$\rightarrow$ because $V_1 \ll V_0$ (assume)

Bunching process of electron beam



Max velocity at $+\pi/2$

$$v_{1, \max} = v_0 \left[1 + \frac{v_1}{2v_0} \right]$$

Min velocity at $-\pi/2$

$$v_{1, \min} = v_0 \left[1 - \frac{v_1}{2v_0} \right]$$

if bunching occurs at L_1 , dist from drift space from i/p grid
dist = v.e. time dist L_1

Then $L_1 = v_0(t_1 - t_0)$

The dist

$$L_1 \text{ at } t - \pi/2 = v_{\min}(t_1 - t - \pi/2)$$

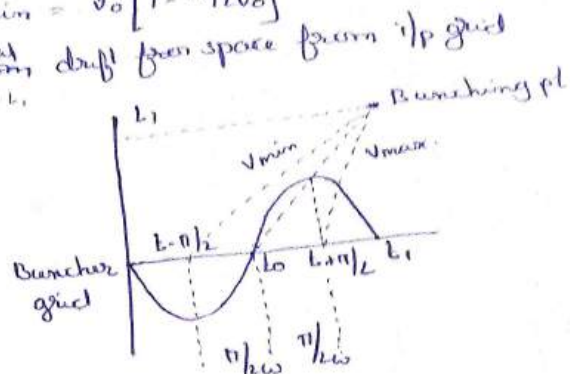
$$L_1 \text{ at } t + \pi/2 = v_{\max}(t_1 - t + \pi/2)$$

$$t - \pi/2 = t_0 - \pi/2\omega$$

$$t + \pi/2 = t_0 + \pi/2\omega$$

$$\text{So } L_1 \text{ at } t - \pi/2 = v_{\min}(t_1 - t_0 + \pi/2\omega) = v_0 \left[1 - \frac{v_1}{2v_0} \right] (t_1 - t_0 + \pi/2\omega)$$

$$L_1 \text{ at } t + \pi/2 = v_{\max}(t_1 - t_0 - \pi/2\omega) = v_0 \left[1 + \frac{v_1}{2v_0} \right] (t_1 - t_0 - \pi/2\omega)$$



Here we calculate drift space L_1 (dist b/w two gaps) where bunching occur
why we have to calculate drift space because its a space where e^- changes
its velocity modulation (ie velocity modulation occur at this place)

$$L_{\max} = v_0(t_1 - t_0) + v_0 \left[\frac{\pi}{2\omega} - \frac{v_1}{2v_0}(t_1 - t_0) - \frac{v_1}{2v_0} \cdot \frac{\pi}{2\omega} \right]$$

$$L_{\min} = v_0(t_1 - t_0) - v_0 \left[\frac{\pi}{2\omega} - \frac{v_1}{2v_0}(t_1 - t_0) - \frac{v_1}{2v_0} \cdot \frac{\pi}{2\omega} \right]$$

if dist has to be same from 3 diff position, $-\pi/2, 0, \pi/2$ the
second term should be zero

$$\frac{\pi}{2\omega} - \frac{v_1}{2v_0}(t_1 - t_0) - \frac{v_1}{2v_0} \cdot \frac{\pi}{2\omega} = 0$$

$$-(t_1 - t_0) = \frac{\pi}{2\omega} \left[\frac{v_1}{2v_0} - 1 \right] \cdot \frac{2v_0}{v_1}$$

$$-(t_1 - t_0) = \frac{\pi}{2\omega} - \frac{\pi v_0}{\omega v_1}$$

$$t_1 - t_0 = \frac{\pi}{\omega} \left(\frac{v_0}{v_1} - \frac{1}{2} \right)$$

$$t_1 - t_0 = \frac{\pi v_0}{\omega v_1} \therefore L_1 = v_0(t_1 - t_0)$$

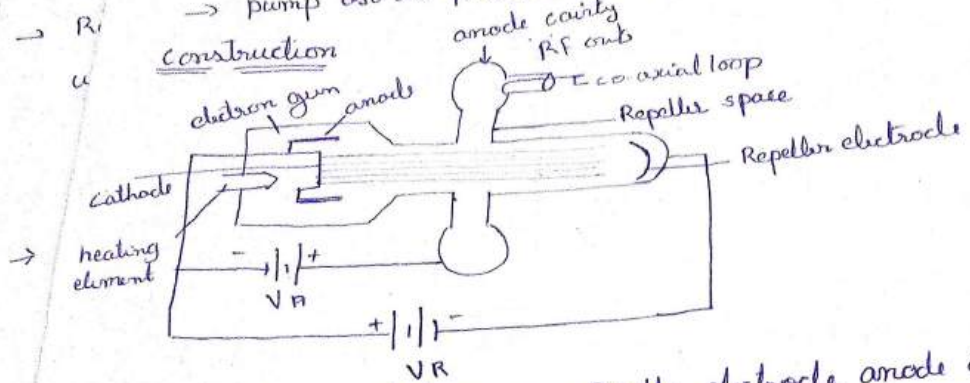
$$L_1 = v_0 \left[\frac{\pi}{\omega} \frac{v_0}{v_1} \right]$$

$$\frac{\pi}{2\omega} \left[\frac{v_1}{2v_0} - 1 \right] \cdot \frac{2v_0}{v_1}$$

$$\frac{\pi v_1}{2\omega v_0} - \frac{\pi v_0}{\omega v_1}$$

- It is and expla
 - The on
 - R
- Reflex klystron
- single cavity variable freq. H.W generator
 - mostly used in radar receivers
 - local osc in H.W receiver
 - portable H.W links
 - pump osc in parametric amp^r

Construction



anode act as a e⁻ gun

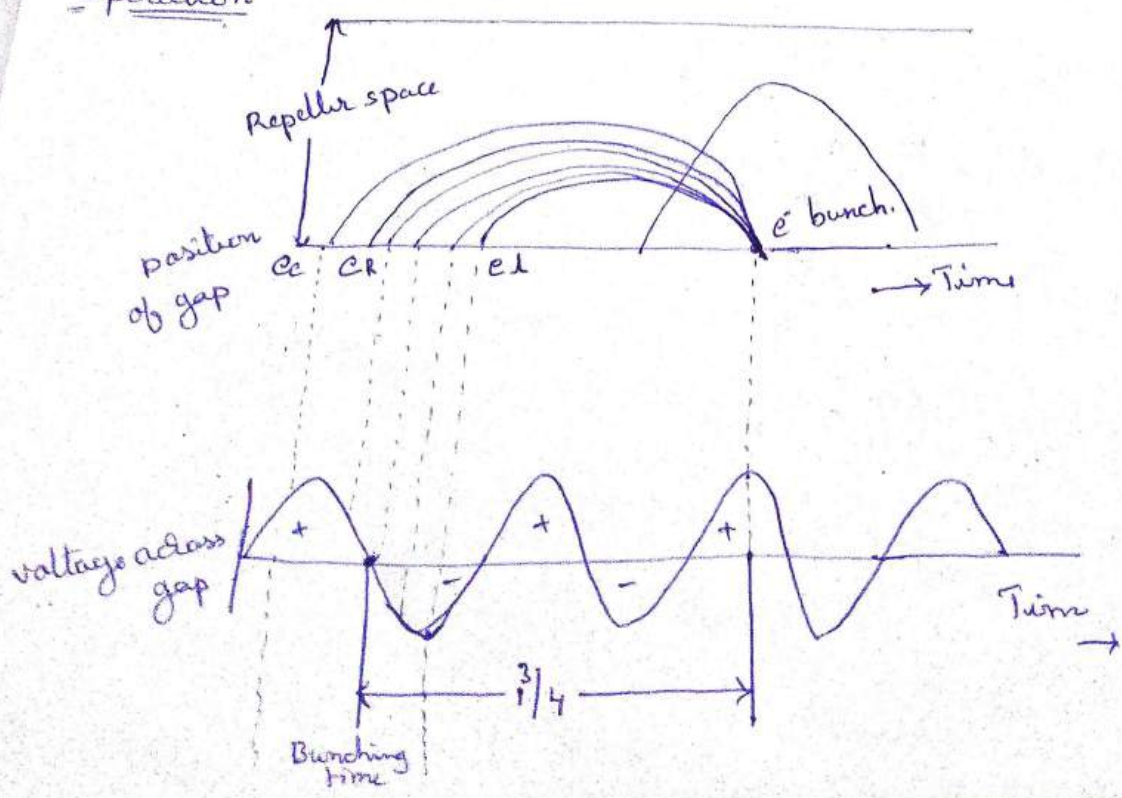
74.35

It consists of electron gun, Repeller electrode, anode cavity.

The electron gun emits e⁻ beam which accelerate towards anode cavity. (+ve potential)

After passing gap e⁻ beam moves towards repeller electrode which is at high -ve potential V_R. e⁻ never reach repeller electrode and return back towards gap. At suitable condition e⁻ give more energy to gap than they took from gap on their forward journey and oscillations are produced, sustained.

Operation



(4)

→ It is assumed oscillation are set up in tube initially due to noise and these oscillations are sustained by device operation. can be explained by applegate diagram.

→ The RF voltage that is produced across gap by cavity osc act on e^- beam to cause velocity modulation.

→ Reference e^- CR passes through gap when gap voltage is 0. so it is unaffected by gap voltage. This moves towards repeller electrode and reflect back due to -ve voltage of repeller. It returns and passes through gap 2nd time.

→ Early e^- passes through gap before CR with max +ve voltage. e^- experiences max +ve voltage across gap and is accelerated. It moves with greater velocity and penetrate deep into repeller space. e^- takes more time to return due to deep penetration. Hence e^- & CR appears at gap for 2nd time.

→ The late e^- passes through gap after CR exp more -ve voltage across gap and moves with retarded velocity. The return time is shorter as the penetration into repeller space is less and catches with e^- and CR forming a bunch. Bunches occur only once per cycle centered around CR and these bunches transfer max energy to gap to sustain osc.

→ For osc to be sustained, time taken by e^- to travel to repeller space and back to gap (transit time) should be optimum is correct value.

To get optimum value, we should know best possible time for e^- to leave the gap and best possible time for e^- to return the gap.

CR
 $e^- \rightarrow$
Application
① Used in
② local oscillator
③ rotatable H.F.
④ p oscillator
⑤ chara
⑥ : 4 to 2
⑦ 1.0 MW
⑧ 2.78
⑨ 1. to
⑩ 42 at
⑪ relia

→ The most suitable time departure time is obviously centered on reference e^- , at 180° pt of sine-wave voltage across gap.
It takes some energy to accelerate e^- and just as much is gained while retarding e^- . so energy required to accelerate and retard is same so energy is nil.

→ so the best possible time for e^- to return the gap voltage is when voltage across gap will apply max retarding to them.

[energy is spent while accelerating, but energy is gained while retarding].

→ Max retardation occur when gap voltage is Max +ve. so the best time for e^- to return to gap is at 90° pt of sine wave gap voltage.

→ e^- should return after $1\frac{3}{4}$ cycles, i

Returning after $1\frac{3}{4}$ cycles obviously satisfy these requirements

Optimum transit time should be

$$T = n + \frac{3}{4}, \quad n = 1, 2, 3, \dots$$

performance characteristics of Two cavity Reflex amp^r

→ freq: 250 MHz to 100 GHz (60 GHz nominal)

power: 10 kW - 500 kW (CW) 30 MW (pulsed)

power gain: 15 dB - 70 dB (60 dB nominal)

B.W: limited

Noise fig: 15-20 dB (sometimes > 25 dB)

Theoretical efficiency: 58% (30-40% nominal)

performance characteristics of reflex klystron

1. Freq range : 4 to 200 GHz.
2. o/p pow : 1.0 mW to 2.5 W
3. Theoretical η : 22.78%
4. practical η : 10% to 20%
5. Tuning range : 5 GHz at 2 watts to 30 GHz at 10 mW.

used re-entrant

HELIX TWTs

At H.W freq we have gain and B.W limitations. If gain $\uparrow$, B.W $\downarrow$.
To compensate this there are two methods

(i) Reentrant cavity $\rightarrow$ Resonant ckt [Re-entrant cavity]

(ii) slow wave structure $\rightarrow$ non resonant ckt.

* With slow wave structure it is possible to get $\uparrow$ gain and $\uparrow$ B.W.
and total arrangement is called Travelling wave tube (TWT) and also called as Broad band device.

* In case of reentrant cavities, RF field is stationary and hence no movement of wave, only e^- beam travels from cathode to anode.
Bt in TWT RF field is also moving ie propagating, so wave is propagating over here along with e^- beam.

* RF i/p given Helix and hence field is generated and accumulated at centre and wave propagates. so there is an interaction b/w e^- & field and hence field strength $\uparrow$ and more kinetic energy transfer takes place b/w e^- & field and gets amplified signal at o/p.
TWT is also called TWT amp^y.

Differences b/w Klystron and TWT

Klystron	TWT
→ Field stationary & only beam travels	→ Field travels along beam.
→ It is resonant type which uses re-entrant type	→ Non resonant type, which uses slow wave structure like Helix
→ Interaction of e with field occurs only at gaps of cavity	→ Interaction of e beam & RF field is continuous over entire length of tube
→ Wave is not propagating	→ wave is propagating
→ It is narrow band device	→ It is wide band device
→ each cavity operates independently	→ there is coupling effect b/w cavity
→ Low pow o/p	→ high pow o/p
→ very short life	→ long life
→ uses cavities cavities for both i/p and o/p	→ uses helix for i/p & o/p.

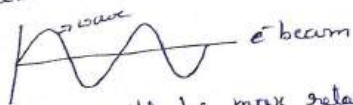
slow wave structures

significance:-

Re-entrant cavity uses resonating ckt which uses L and C . Depending upon L and C values, our resonating freq occur and there is a limitation on L and C values.

So we move for non-resonating ckt, where freq doesn't depend on L & C so to obtain high f resonating freq at u wave range we use slow wave structure and hence we get large B.W.

slow wave structures reduce wave velocity in certain direction so that e beam and signal can interact

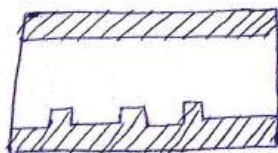


At +ve peak, there will be max retardation applied to e beam and hence velocity of e beam reduces (which is at large velocity) and phase velocity reduces by means of structure itself. and hence interaction takes place b/w e-beam and field, so kinetic energy transfer takes place from e to field.

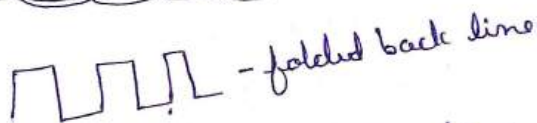
compared to, wave velocity, e velocity should be high. so high K.E transfer takes from e to field and hence signals also get amplified where as in -ve peak, e beam sees acceleration. so no. of e in -ve peaks are less.

Types:-

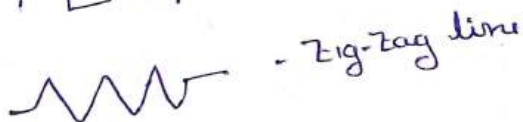
- i) Helix line
- ii) Folded - back line
- iii) Zig Zag line
- iv) Inter digital line
- v) Ridged wave guide



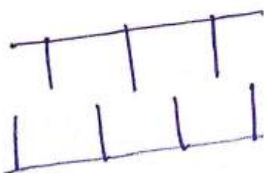
→ helix



- folded back line

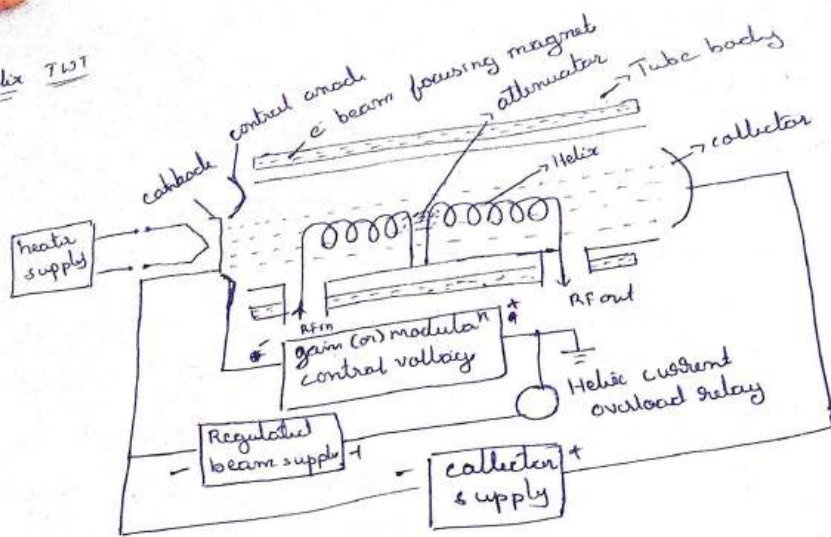


- Zig-Zag line



- Inter digital line

Helix T137



Construction

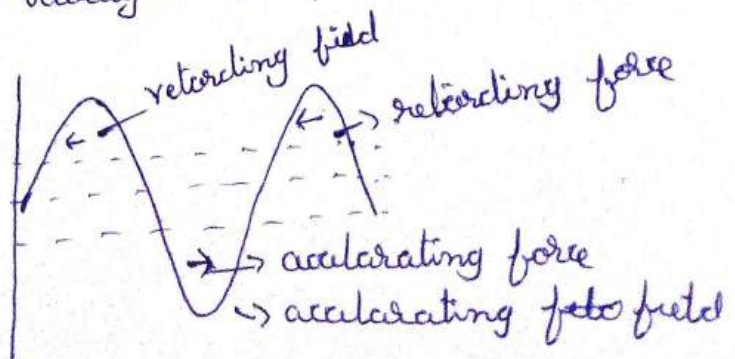
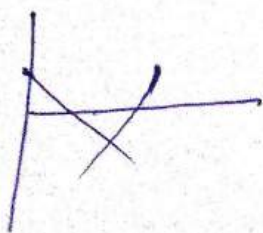
The construction is very similar to klystron. We need e gun to generate e beam and cavity klystron is replaced with slow wave structure which is helix line and (simply a conductor). We need magnet field arrangement (Two magnets) so as to make e beam always concentrate at centre. The RF i/p is given at starting pt of Helix and RF o/p is taken at last. There is an arrangement of attenuator at mid of Helix.

Working principle:-

- By applying voltage cathode generate e beam and focusing anode makes e move towards the structure.
- Helix is copper wire, weak signal to be amplified is given to RF i/p
- Some RF current is present in helix wire, so electric field is formed and it is concentrated at centre along axial direction, with some velocity wave propagates

(7)

- At the start all e^- travels with same velocity (v_0) and now e^- beam enters into helix, and based on field, which is concentrated at centre, velocity of e^- changes hence velocity modulation occur.
- If e^- propagates further, get current modulation and at end e^- bunch is formed.
- Some e^- are concentrated at centre of Helix because of electric field that is produced due to e^- beam focusing magnets. So e^- move through zero field and hence there is no change in velocity so no energy transfer takes place over here.
- There will be some e^- which moves towards max +ve field, and hence field provide max retardation to e^- and hence field strength $\uparrow$. Due to this e^- losses its velocity ($< v_0$) and max energy $\rightarrow$ is produced.
- e^- which move towards -ve field, provide max acceleration and hence velocity of $e^- \uparrow$ and field strength $\downarrow$. Due to this less energy transfer takes place.
- So we choose no: of e^- in such a way that there are more in +ve cycle and very less e^- in -ve field.
- There is a continuous interaction b/w e^- and field from starting to end. and at RF o/p, reduced velocity e^- are present which move towards collector.



$v_0 > \text{wave velocity}$

→ if velocity of $e^- >$ wave velocity, more e^- are focused towards retarding field and hence max K.E transfer takes place from e^- s to field

axial phase velocity is $v_p = v_c \times \frac{\text{pitch}}{\text{circumference } (2\pi r)}$

→ e^- bunch is formed around reference e^- .

Suppression of oscillations

→ Since helix structure is bidirectional, it can allow ~~wave~~ wave in both +ve & -ve direction. so there is a possibility of reflected wave at RF o/p. becoz of any mismatch impedance at o/p.

→ This reflected wave travel towards i/p hence gets oscillations. so to suppress osc we place attenuator in middle of helix line which encourages only forward wave and hence reflected wave can be eliminated.

Microwave Solid-state Devices

9

9.1 INTRODUCTION

Before the 1960s, vacuum tube technology was widely used in electronics, which increased the size and cost of the entire system. Later, the development in semiconductor technology led to the growth of solid-state devices, which were more reliable and had a longer life. These devices generally require low power and are very compact and lighter in weight.

In the recent past, tremendous research activities have taken place for the development of better, low-noise, high-frequency, and greater-bandwidth components to achieve lesser switching time and better performance characteristics. In this chapter, several microwave solid-state devices, including two terminal devices such as transferred electron devices (Gunn diodes), avalanche transit time devices (IMPATT, TRAPATT, and BARITT diodes), tunnel diodes, and three terminal devices such as bipolar junction transistors and field effect transistors (MESFETs and HEMTs) are discussed. This chapter also discusses the PIN diode, Schottky diode, Varactor diode, parametric amplifier, step-recovery diode, and crystal diode in detail.

9.2 NEGATIVE RESISTANCE PHENOMENON

There are various microwave solid state devices which show negative resistance characteristics. The real part of their impedance is negative over a range of frequencies. In a positive resistance, the current through the resistor and the voltage across it are in phase. The voltage drop across positive resistance is positive, and the power that is given by the product of voltage (V) and current (I) is dissipated in the resistor. In a negative resistance, the current and voltage are out of phase by 180° . Therefore, the voltage drop across it is negative, and a negative power is generated by the power supply associated with the negative resistance. In other words, negative resistance generates or supplies power (e.g. active solid state devices) to the external circuits and positive resistance absorbs power (e.g. passive solid state devices).

However, the reason for exhibiting negative resistance varies from one device to another. For instance, in tunnel diode the negative resistance is due to heavy doping where the majority carriers tunnel through the thin barrier. In an IMPATT, it is due to delay, which causes the current to lag behind the voltage; whereas in TEDs, it is due to differential negative electron mobility characteristic in materials such as GaAs and InP.

9.3 CLASSIFICATION OF SOLID-STATE DEVICES

Solid-state devices can be classified into two categories depending on the number of terminals these devices have. They are diodes and transistors. Depending on their principle of operation, the two terminal diodes are further classified as shown in Figure 9.1(a). Transferred Electron Devices (TED), tunnel diodes, and Avalanche Transit Time Devices (ATD) show negative resistance. The three terminal devices are also further divided into two types as shown in Figure 9.1(b). Microwave BJTs are bipolar devices, whereas microwave FETs are unipolar devices.

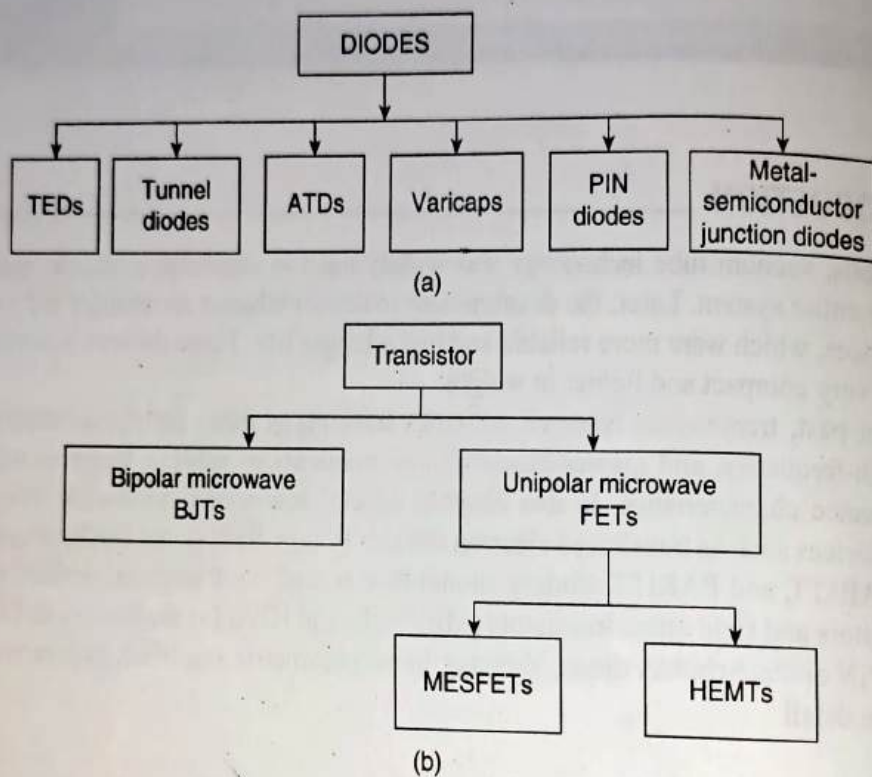


Figure 9.1 (a) Two terminal devices; (b) Three terminal devices

9.4 APPLICATIONS OF SOLID-STATE DEVICES

Microwave solid-state devices are becoming more important. These devices have been invented for various applications in microwave frequency regions such as frequency multiplication, signal detection, attenuation, generation of oscillation, switching, phase shifting, amplitude limiting, and amplification. Some of their applications are as follows:

- As microwave generators
- As amplifiers in satellite communications and also in other space applications
- As transmitters for millimeter communication systems
- In radio transmitters, such as CW Doppler radar
- In broadband linear amplifiers and in low-power amplifiers
- As a pumping source for parametric amplifiers
- In transponders
- In both combinational and sequential logic circuits
- In microwave receivers

9.5 TRANSFERRED ELECTRON DEVICES (TEDS)

Transferred Electron Devices (TEDs) are one of the important microwave devices. They are bulk devices that have no junction or gates as compared with microwave transistors, which operate with either junction or gates. Transferred electron devices are fabricated with compound semiconductor materials (e.g. gallium arsenide (GaAs), cadmium telluride (CdTe) and indium phosphide (InP)). They have two energy regions or valleys in the conduction band. The TEDs show a transferred electron effect in which transfer of electrons takes place from lower valley to upper valley in the conduction band. Most of the electrons will be in lower valley at low electric fields. When the electric field strength is increased to higher values, most of the electrons will be transferred into high-energy bands. In the higher-energy bands, the effective mass of electrons is larger than in lower-energy bands. Since the conductivity is directly proportional to mobility, the higher-energy band has lower conductivity than the lower-energy band. Hence, conductivity decreases with an increase in electric field strength. Thus, the current decreases with an increase in voltage, showing negative resistance. Gunn diode is an example of TEDs, and its theory is better explained in the next section.

A few differences between microwave transistors and TEDs are as follows:

- First, TEDs do not have junctions or gates as in the case of transistors.
- TEDs with smaller physical dimensions have a limited power output. In order to get a reasonable power output, the physical dimensions are to be made large as compared with a microwave transistor.
- TEDs are made of compound semiconductors from group III-V and II-VI elements (of periodic table) such as gallium arsenide (GaAs), indium phosphide (InP), and cadmium telluride (CdTe); whereas the majority of transistors are made from elements such as germanium or silicon.

9.6 GUNN DIODE

In 1963, Gunn observed a periodic variation of current passing through an N-type GaAs semiconductor. He found the following when a DC bias voltage is applied to the contacts of N-type GaAs or InP:

- Current first rises linearly from zero.
- Then, it begins to oscillate when a certain threshold is reached.
- The time period of oscillation is equal to the travel time of electrons from cathode to anode.

This is known as *Gunn effect* or *bulk effect*. The device that shows Gunn effect is known as *Gunn diode*. Gunn diodes are usually fabricated using N-type semiconductor materials (eg. GaAs, and InP); so, they should be associated with electrons rather than with holes. Gunn diode operations do not depend on junction properties. Even though it has no junction, it is called a *diode*, because it has two terminals (anode and cathode) attached to it. It is an active two-terminal solid-state device and is mainly used as a local oscillator in the microwave frequency range of 1 to 100 GHz. Gunn effect can be explained on the basis of domain formation, two-valley theory of Ridley-Watkins-Hilsum (RWH), or the transfer electron mechanism.

9.6.1 Operation and Characteristics of Gunn Diode

The Gunn diode is made up of a single piece of N-doped semiconductor material (e.g. GaAs, InP) with two thinner N^+ -doped layer contacts on opposite ends. The two N^+ layers are required to connect the anode and cathode leads.

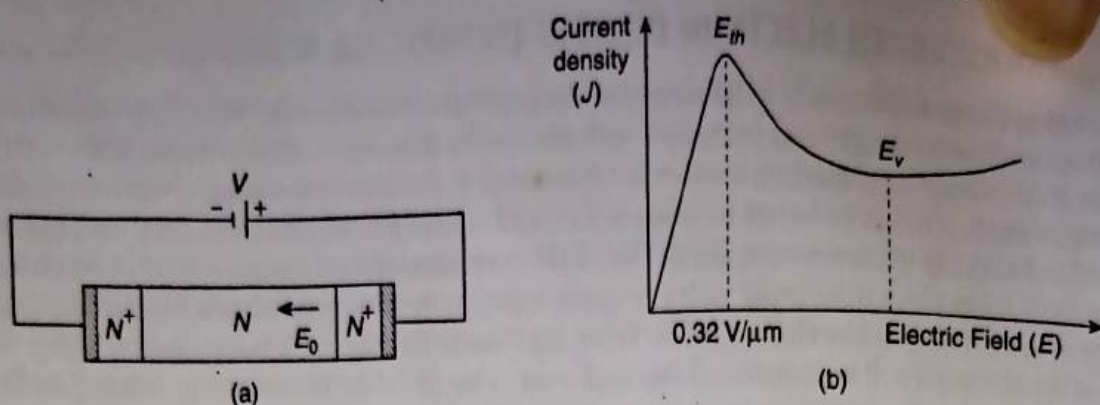


Figure 9.2 (a) Three-layer structure of Gunn diode; (b) J/E curve of a Gunn diode

When a DC voltage (V) is applied to the two terminals, an electric field (E_0) will be established across the piece of GaAs (just as in a resistor) (Figure 9.2(a)). Figure 9.2(b) shows how the current density (J) through the material varies with the electric field (E_0) across it. We know that the drift velocity (v_d), current density, and electric field have the following relations:

$$v_d = \mu \bar{E}$$

$$\bar{J} = nq\mu \bar{E}$$

Initially, when the electric field is increased, from the above relations it can be seen that the drift velocity and current density increase. Thus, the current density increases with an increase in the electric field, resulting in a positive resistance. This continues till the electric field reaches a value known as *threshold value* E_{th} (corresponding threshold voltage V_{th}). When the electric field is increased beyond the threshold value E_{th} , the Gunn effect takes place and the current density decreases, causing the device to exhibit negative resistance. This behavior is due to domain formation, which will be explained in the next section. This will continue till the field reaches a value known as E_v (corresponding valley voltage). When the voltage is increased beyond E_v , the current density increases. Thus, the device again exhibits positive resistance.

Negative resistance region: In common the current first rises linearly from zero with increasing electric field, however there is a region between the threshold electric field and valley electric field, where the current decreases as the electric field is increased. This is called the *negative resistance region*. The dynamic resistance, r , in this voltage range is given by

$$r = dV/dI, r < 0.$$

9.6.2 Domain Formation

When bias voltage (V) is applied across the diode terminals, an electric field (E_0) is established across the GaAs piece. As E_0 increases, electron drift velocity increases up to a certain potential (E_{th}). This is because the velocity (v_d) is proportional to the applied electric field. However, with a further increase in voltage, the mobility decreases, which causes the velocity to decrease, thereby causing the electrons to slow down. This slowing down results in a traffic jam, and more electrons pile up to form a charge layer or domain. This charge layer produces an electric field (E) such that it decreases the original field (E_0) on the left and increases the original field (E_0) to the right (Figure 9.3). As a result, the charge bunch gets pushed toward the anode on the right, forming a current pulse.

When this pulse moves to the anode, E_0 goes back to the original value, resulting in another traffic jam and another pulse formation, which causes oscillation.

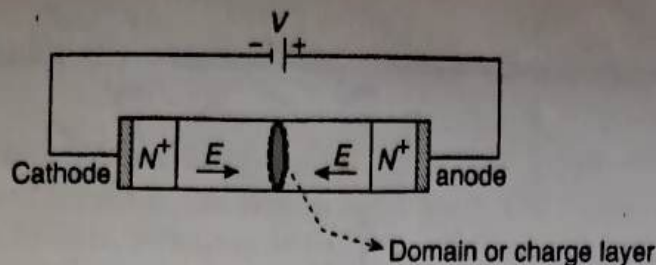


Figure 9.3 Domain formation in a Gunn diode

In this procedure, N-type GaAs releases the power that it has acquired during the domain formation in the direction of the anode. If conditions are appropriate, this release of power at anode will be utilized by the noise energy of suitable frequency existing in the Gunn diode configuration and gets amplified. The amplification of noise energy results in sustained oscillations. The frequency of oscillation depends on the length of the GaAs piece and concentration of electrons.

9.6.3 RWH Theory or Two-valley Theory of Gunn Diode

Gunn effect

The Gunn diode, which is made of *n*-doped semiconductor material (e.g. GaAs or InP), is characterized by having two valleys in their conduction bands with different mobility. The two-valley model is also called the *Ridley-Watkins-Hilsum (RWH) theory*. There are two regions in the conduction band of the N-type GaAs. These conduction band regions are known as *upper valley* and *lower valley*. The energy band and transfer of electrons between the two valleys are shown in Figures 9.4 and 9.5, respectively. The following can be noted about GaAs:

- In N-type GaAs, the valence band is filled with electrons, and the conduction band is partly filled.
- The forbidden energy gap between the valence band and the conduction band is about 1.43 eV.
- Electrons in the lower valley (n_1) exhibit a small effective mass (m_1) and very high mobility, μ_1 .
- Electrons in the upper valley (n_2) exhibit a large effective mass (m_2) and very low mobility, μ_2 .
- The two valleys are separated by a small energy gap, ΔE , of approximately 0.36 eV.
- At a low electric field, electrons remain in the lower valley and material behaves ohmically. When the electric field reaches a certain threshold value E_{th} , the electrons will be swept from the lower valley to the upper valley. If the rate of transfer of electrons from the lower to the upper valley is very high, the current will decrease with an increase in voltage. This leads to a decrease in the average electron mobility, μ with an increase in the field in a bulk semiconductor, thus resulting in equivalent negative resistance. This manifests itself as a bulk negative differential resistance.
- The average electron mobility, μ is given by

$$\mu = (n_1\mu_1 + n_2\mu_2)/(n_1 + n_2) \quad (9.1)$$

where n_1 = electron density in lower valley

n_2 = electron density in upper valley

μ_1 = mobility of electron in lower valley

μ_2 = mobility of electron in upper valley

Total carrier concentration = $n_1 + n_2$

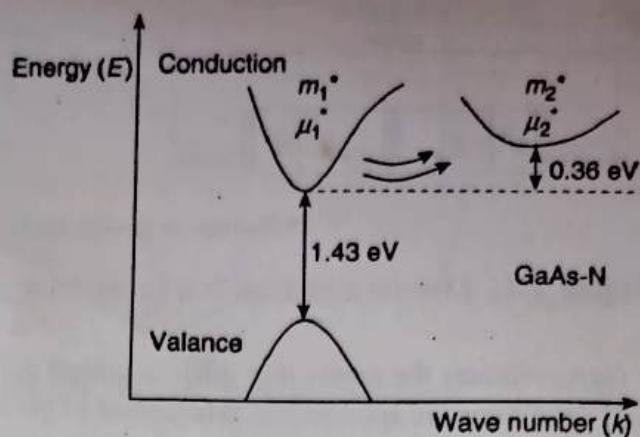


Figure 9.4 Energy band diagram for GaAs

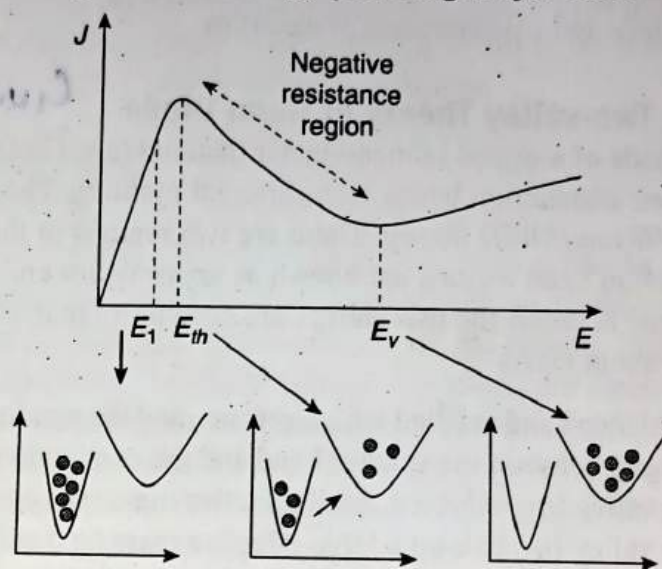


Figure 9.5 Inter-valley transfer

where m_1^* = effective mass of electron in lower valley

m_2^* = effective mass of electron in upper valley

The inter-valley transfer of electrons with regard to the E_{th} and E is illustrated in Figure 9.5. Three different ranges of electric field strength may be considered. The current (I_1) and the current density ($\bar{J}_1$) corresponding to the electric field $< (\bar{E}_{th})$ where all the electrons occupy the lower-energy valley, is given by

$$\bar{J}_1 = \mu_1 n_1 \bar{E}, I_1 = e A \mu_1 n_1 \bar{E} \left(\therefore \bar{J} = \frac{I}{A} \right)$$

The current (I_2) and the current density ($\bar{J}_2$) corresponding to the electric field $\bar{E}_{th} < \bar{E} < \bar{E}_v$, where the total number of electrons are distributed between the lower and upper valleys, is given by

$$\bar{J}_2 = e(n_1 \mu_1 + n_2 \mu_2) \bar{E}, I_2 = e(n_1 \mu_1 + n_2 \mu_2) A \bar{E}$$

The current (I_3) and the current density ($\bar{J}_3$) corresponding to the electric field $\bar{E} = \bar{E}_v$, where all electrons occupy the upper-energy valley, is given by

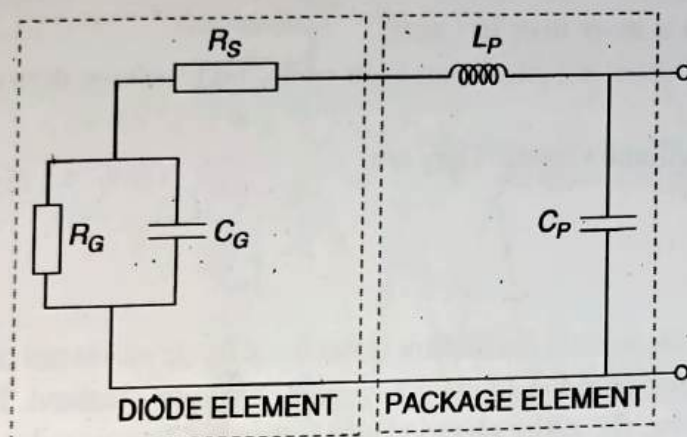
$$\bar{J}_3 = e \mu_2 n_2 \bar{E}, I_3 = e A \mu_2 n_2 \bar{E}$$

where $(\bar{J})$ = current density, $(\bar{E})$ = electric field, A = area of cross-section of device, and e = charge of electrons.

Performance characteristics: Gunn diode operates at 10-12 V of typical biased voltage, 300 mA of bias current, and the power output is 100 mW in the X band. The typical frequency range that it can operate is from 4 GHz to 94 GHz, efficiency is 10-20 %, CW power output is 250 mW at 15 GHz, and pulsed power is 5 W at 12 GHz.

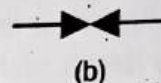
9.6.4 Equivalent Circuit of Gunn Diode

The scale of the parasitic impedances accredited to the package element decreases with the package dimension. The equivalent circuit of Gunn diode is shown in Figure 9.6(a).



C_P = Package Capacitance
 L_P = Bond Lead Inductance
 R_S = Series Resistance
 R_G = Gunn Diode Resistance
 C_G = Gunn Diode Capacitance

(a)



(b)

Figure 9.6 (a) Equivalent circuit Symbol; (b) Symbol

As a result, at lower microwave frequencies the stronger package styles are normally precise and for the higher microwave frequency the smaller parasitic impedance packages are suggested. Its symbol is as shown in Figure 9.6(b).

9.6.5 Basic Modes of Operation

Depending on the device characteristics and external circuitry, the Gunn diode can be made to oscillate in any one of the following four frequency modes:

- Gunn oscillation mode
- Limited space charge accumulation (LSA) mode
- Stable amplification mode
- Bias circuit oscillation mode

The modes of operations are classified based on the condition

$$n_o l > \frac{\epsilon_s v_d}{e |\mu_n|}$$

- v_d = electron drift velocity
- ϵ_s = semiconductor dielectric permittivity
- μ_n = electron mobility
- e = electron charge
- n_o = doping concentration
- l = device length

9.6.5.1 Gunn oscillation modes

This mode has the following features:

- The electric field is greater than the threshold.
- The product of frequency and length is about 10^7 cm/s.
- The product of doping and length is more than 10^{12} /cm².
- Due to the cyclic arrangement of either accumulation layer or the high voltage domain the Gunn diode is unstable.
- There are three types of Gunn oscillation mode. They are
 - Transit time domain mode
 - Delayed domain mode
 - Quenched domain mode.

Transit time domain mode: Transit time domain oscillation is the basic mode and is not depend on the external circuit. When a domain is quenched at the anode the current peaks are obtained. Then, another is nucleated near the cathode. At any time, the entire electric field across the device is above the threshold electric field. The frequency is given by

$$f = 1/\tau = v_d/l \text{ since domain transit time is } \tau = l/v_d$$

where v_d is the domain or drift velocity $= fl = 1 \times 10^7$ cm/s.

l is the effective length, and fl is the product of frequency and length

The oscillation time (τ_o) is equal to the transit time (τ). It is a low-power (< 2 W), low-efficiency mode, and the operating frequency is between 1 GHz and 18 GHz.

Delayed domain mode: This is also called an *inhibited mode*. In this mode, the domain is collected when the dc bias is less than the threshold electric field (E_{th}). The next domain can only be formed when the field again reaches the threshold. The oscillation period is greater than the transit time of the critical field (E_{th}), that is, $\tau_o > \tau$. The frequency of oscillation is determined by the resonant circuit. The efficiency of this mode is limited to 20%. The drift velocity (v_d) or $(f \times l)$ lies between 10^6 cm/s and 10^7 cm/s.

Quenched domain mode: In this mode, the domain collapses before it reaches the anode; that is, it is quenched before it is collected, hence the name *quenched mode*. As shown in Figure 9.7, the bias electric field reduces below the sustaining electric field (E_s) in the negative half cycle. When the bias electric field swings back again more than the threshold electric field, another domain is nucleated and the procedure repeats. The operating frequencies are higher than the transit time frequency (Figure 9.7). The maximum efficiency of this mode is 13%. The drift velocity (v_d) or $(f \times l) = 10^7$ cm/s.

9.6.5.2 Limited Space charge Accumulation (LSA) mode ($fl > 2 \times 10^7$ cm/s)

This mode operates by using a high-Q resonant cavity with current pulses from the Gunn diode. The Gunn diode is placed in a resonant cavity, which is tuned to a frequency of the LSA mode (f_o), so that the circuit operates like a negative resistance device and the domains do not have enough time to form. In this mode, the device can be biased to several times higher than E_{th} . When the input field increase

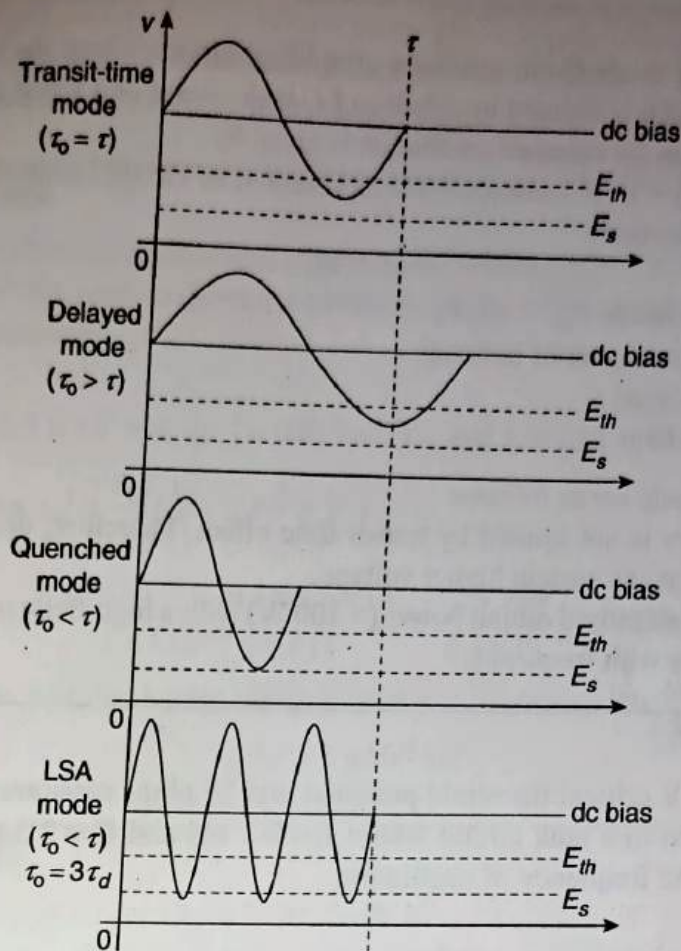


Figure 9.7 Various modes of Gunn diode

more than the threshold electric field, the device remains in the negative resistance region. Oscillation time is obtained by the external circuit, which is given by

$$\tau_0 = 2\pi\sqrt{LC} + \frac{L}{R(V_b/V_{th})} \quad (9.2)$$

where τ_0 is period in seconds

L is inductance in Henry

C is capacitance in farad

R is resistance in ohm

V_{th} is threshold potential

V_b is bias voltage

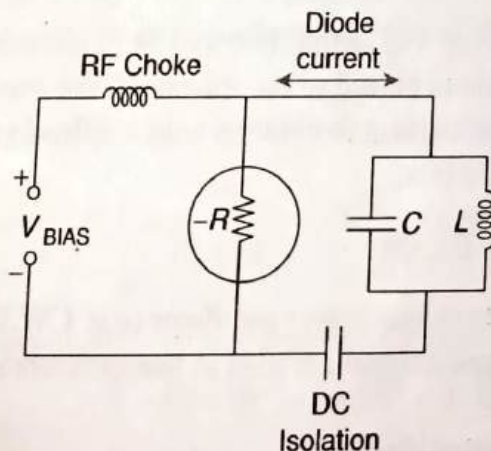


Figure 9.8 Gunn diode in LSA mode

Figure 9.8 shows the LSA mode Gunn oscillator simplified circuit. Gunn oscillator with LSA mode consists of Gunn diode which is shunted by either an LC tank circuit or a tuned cavity that behaves like a tank circuit. The condition for resonant oscillation is given by

$$(-G) = 1/R \geq \text{conductance represented by circuit losses } (G_0).$$

Where G = Negative conductance

$$-G \geq G_0$$

(9.3)

Output power of LSA mode $P_{\text{out}} = \eta I_0 V_0$

where η = conversion parameter of material

I_0 = operating current

V_0 = operating voltage

Advantages of LSA mode are as follows:

- Operating frequency is not limited by transit time effect. Therefore, device length can be made larger and the device can sustain higher voltage.
- This mode gives high-pulsed output power (> 100 W) with a high frequency (100 GHz). However, the power decreases with frequency.

EXAMPLE PROBLEM 9.1

A Gunn diode with 3.5 V critical threshold potential and 25 ohms resistance is connected to 3 V dc. If the diode is connected to a tank circuit where $L = 0.1$ nH and $C = 2.5$ pH, what is the period of oscillation? Also find the frequency of oscillation.

Solution

Given $V_b = 3$ V, $V_{\text{th}} = 3.5$ V, $R = 25$ ohms, $L = 0.1$ nH, and $C = 2.5$ pH

The period of oscillation is $\tau_o = 2\pi\sqrt{LC} + \frac{L}{R(V_b/V_{\text{th}})}$

$$= 2\pi\sqrt{2.5 \times 0.1 \times 10^{-21}} + \frac{0.1 \times 10^{-9}}{25(3/3.5)} = 9.9 \times 10^{-11} + 0.466 \times 10^{-11} \text{ s} = 1.366 \text{ picoseconds}$$

The frequency of oscillation is $f = 1/\tau_o = \frac{1}{1.366} \times 10^{11} \text{ Hz} = 7.32 \times 10^{12} \text{ Hz}$

9.6.5.3 Stable amplification mode

This mode is defined as the region where the frequency times length is about 10^7 cm/s and the doping time length is between 10^{11} cm^{-2} and 10^{12} cm^{-2} .

9.6.5.4 Bias circulation mode

In this mode, the product of frequency and length is very less. This mode occurs along with Gunn or LSA oscillations. When a bulk diode is biased to the threshold, the average current suddenly drops as Gunn oscillations begin. The drop in current at the threshold can lead to oscillations in the bias circuit that are typically from 1 kHz to 100 MHz.

9.6.6 Applications of Gunn Diode

1. It is used in Radar transmitters as low power oscillator (e.g. CW Doppler Radar, Police Radar).
2. Gunn diode oscillators with pulsed signal are used in transponders in industry telemetry systems and for air traffic control.
3. It is used in broadband linear amplifier.
4. It is used in sequential logic circuits and fast combinational circuits

5. It is used in microwave receivers as low and medium power oscillator.
6. It is used in parametric amplifier as pump sources.

EXAMPLE PROBLEM 9.2

A Gunn diode has the following parameters: electron drift velocity, $v_d = 2.5 \times 10^5$ m/s, negative electron mobility, $|\mu_n| = 0.015$ m²/Vs, relative dielectric constant, and $\epsilon_r = 13.1$. Determine the criterion for classifying the modes of operation.

Solution

The data given are $v_d = 2.5 \times 10^5$ m/s, $|\mu_n| = 0.015$ m²/Vs, and $\epsilon_r = 13.1$

$$\begin{aligned} \text{We have } n_0 l &> \frac{\epsilon_s v_d}{e |\mu_n|} \Rightarrow n_0 l > \frac{\epsilon_0 \epsilon_r v_d}{e |\mu_n|} (\because \epsilon = \epsilon_0 \epsilon_r) \\ &> \frac{8.85 \times 10^{-12} \times 13.1 \times 2.5 \times 10^5}{1.6 \times 10^{-19} \times 0.015} > \frac{2.9 \times 10^{-5}}{2.4 \times 10^{-21}} > 1.2 \times 10^{16} \end{aligned}$$

This means that the product of the doping concentration and the device length should be

$$n_0 l > 1.2 \times 10^{16} / \text{m}^2$$

EXAMPLE PROBLEM 9.3

Determine the conductivity of *n*-type GaAs Gunn diode if

Electron density $n = 10^{18}$ cm⁻³, electron density at lower valley $n_l = 10^{10}$ cm⁻³

Electron density at upper valley $n_u = 10^8$ cm⁻³, temperature $T = 300^\circ$ K

Solution

Given that electron density, $n = 10^{18}$ cm⁻³, $n_l = 10^{10}$ cm⁻³, $n_u = 10^8$ cm⁻³, and temperature $T = 300^\circ$ K.

We know that

$$\mu_l = 8000 \text{ cm}^2/\text{v.sec} = 8000 \times 10^{-4} \text{ m}^2/\text{v.s}, \mu_u = 180 \text{ cm}^2/\text{v.sec} = 180 \times 10^{-4} \text{ m}^2/\text{v.s}$$

We have conductivity,

$$\sigma = e(n_l \mu_l + n_u \mu_u) = 1.6 \times 10^{-19} (8000 \times 10^{-4} \times 10^{16} + 180 \times 10^{-4} \times 10^{14}) = 1.28 \times 10^{-3} \text{ } \Omega^{-1} \text{m}$$

EXAMPLE PROBLEM 9.4

A GaAs Gunn diode has an active region of 10 micrometers. If the electron drift velocity is 10^5 m/sec, calculate the natural frequency and the threshold voltage. The critical electric field is 3 kv/cm.

Solution

The given data are $v_d = 10^5$ m/sec = 10^7 cm/sec, $l = 10 \text{ } \mu\text{m} = 10 \times 10^{-4}$ cm, and $E_c = 3 \text{ kv/cm}$ $f = ?$, $V = ?$

$$\text{We have natural frequency, } f = \frac{v_d}{l} \therefore f = \frac{10^7}{10 \times 10^{-4}} = 10^{10} \therefore f = 10 \text{ GHz}$$

Critical voltage,

$$V = l \times E_c$$

Critical voltage = Active length $\times$ Critical electric field

$$\therefore V = 10 \times 10^{-4} \times 3 \times 10^3 = 3 \text{ V}$$

$\therefore$ Critical voltage,

$$V = 3 \text{ V}$$

Unit-3 Microwave components and applications

Microwave systems normally consist of several m.w components including source and load being connected to each other by waveguides or coaxial or transmission line system. All these components must be built with low standing wave ratio, lower attenuation, lower insertion losses and other desirable characteristics to achieve desired transmission of m.w signal.

Waveguide microwave junctions

At certain position in waveguide system, many times it become necessary to split all or part of m.w energy into particular directions. This is achieved by waveguide or H.W junctions. This junction may be used to combine two or more signals.

When i/p from m.w source is applied at port ① a part of it comes out of port ② another part out of port ③ some part out of port ④ and the remaining part may come out of port ① itself due to mismatch b/w port ① and m.w junction.

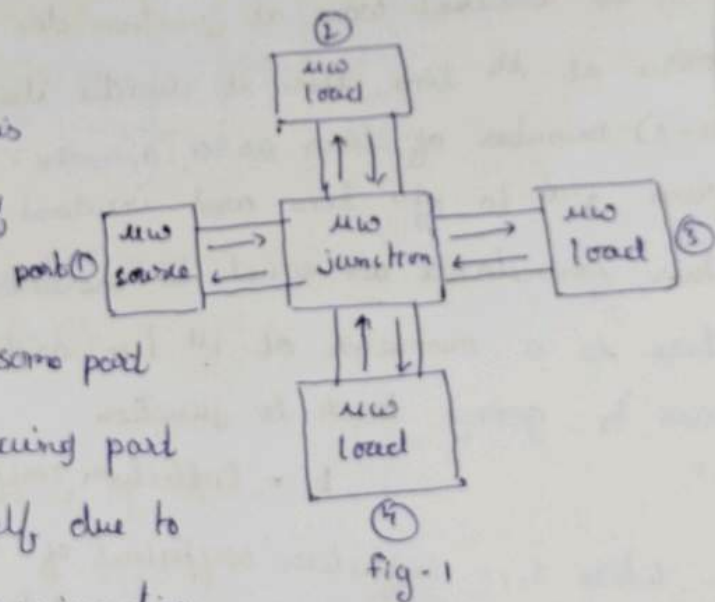


Fig-1

Scattering (or) S parameters

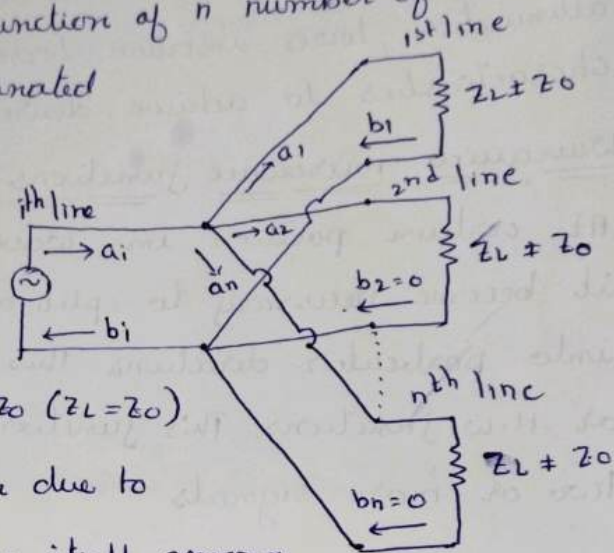
From fig ① if we apply i/p at one port, there are 4 o/p. Similarly if we apply i/p at all 4 ports we get 16 combinations, which are represented in matrix form and that matrix called as a

scattering matrix. It is square matrix which gives all the combinations of power relationship b/w various i/p and o/p port of H.W junction. The elements of this matrix are called scattering coefficients or scattering (s) parameters.

To obtain relationship b/w scattering matrix and i/p | o/p ports at different ports, consider a junction of n number of transmission lines where i th line is terminated

Case 1

Let first line be terminated in an impedance other than characteristic impedance (i.e. $Z_L \neq Z_0$) and all remaining lines equal to Z_0 ($Z_L = Z_0$)



if a_1 be incident wave at junction due to source at i th line, then it divides itself among $(n-1)$ number of lines as $a_1, a_2, \dots, a_n$. There will be no reflections from 2nd to n th line and incident waves are observed since their impedance are equal to characteristic impedance (Z_0). But, there is a mismatch at 1st line and hence there will be reflected wave b_1 going back to junction

$$b_1 = (\text{reflection coefficient}) a_1 = S_{11} a_1$$

where S_{11} = reflection coefficient of 1st line

1 = reflection from 1st line

i = source connected at i th line

Hence, the contribution to outward travelling wave in i th line given by

$$b_i = S_{ii} a_i \quad [\because b_2 = b_3 = \dots = b_n = 0]$$

Case 2: Let all $(n-1)$ lines be terminated in impedance other than Z_0 (ie $Z_L \neq Z_0$ for all lines)

Then, there will be reflection into junction from every line and hence total contribution to outward travelling wave in i th line is given by.

$$b_i = s_{i1}a_1 + s_{i2}a_2 + s_{i3}a_3 + \dots + s_{in}a_n$$

$i = 1$ to n since i can be any line from 1 to n

$$\therefore b_1 = s_{11}a_1 + s_{12}a_2 + \dots + s_{1n}a_n$$

$$b_2 = s_{21}a_1 + s_{22}a_2 + \dots + s_{2n}a_n$$

$$\vdots$$

$$b_n = s_{n1}a_1 + s_{n2}a_2 + \dots + s_{nn}a_n$$

In matrix form

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} \\ s_{21} & s_{22} & \dots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{n1} & s_{n2} & \dots & s_{nn} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$$[b] = [s][a] \rightarrow \text{incident wave or } i/p$$

$\hookrightarrow$ reflected waves or o/p

Properties of $[s]$ Matrix

1. $[s]$ is always a square matrix of order $(n \times n)$
2. $[s]$ is symmetric matrix i.e. $s_{ij} = s_{ji}$
3. $[s]$ is unitary matrix $[s][s]^* = [I]$.
- $[s]^*$ = complex conjugate of $[s]$
- $[I]$ = unit matrix of same order as that of $[s]$.
4. The sum of product of each term of any row (or column) multiplied by complex conjugate of corresponding term of

any other row (or column) is zero

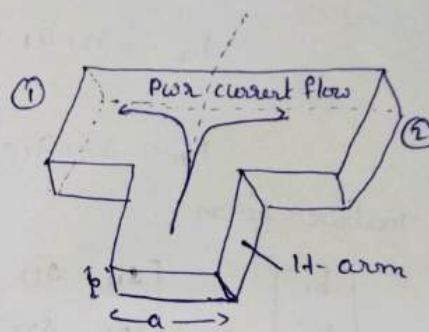
$$\sum_{i=1}^n S_{ik} S_{ij}^* = 0 \quad k \neq j \quad \begin{cases} k = 1, 2, 3 \dots n \\ j = 1, 2, 3 \dots n \end{cases}$$

5. if any of terminal or reference planes (say k^{th} port) are moved away from junction by electric dist $\beta_k l_k$, each of coefficients S_{ij} involving k will be multiplied by factor $e^{-j\beta_k l_k}$.

Microwave T-junctions

H-plane Tee junction

A H-plane Tee junction is formed by cutting rectangular slot along width of main waveguide and attaching another waveguide, side arm called



H-arm. The port ① and port ② of main waveguide are called collinear ports and port ③ is H-arm or side arm.

As all three arms of H-plane Tee lie in plane of magnetic field, the magnetic field divides itself into arms. Therefore it is also called a current junction.

The properties of H-plane Tee can be completely defined by its $[S]$ matrix

$$S = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

By properties of $[S]$ matrix

→ Because of plane of symmetry of max scattering

coefficient S_{13} and S_{23} must be equal

$$S_{13} = S_{23}$$

(3)

→ From symmetric property $S_{ij} = S_{ji}$

$$S_{12} = S_{21}; S_{23} = S_{32} = S_{13}; S_{13} = S_{31}$$

→ Since port is perfectly matched to junction

$$S_{33} = 0$$

$$\therefore [S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix}$$

→ From unitary property

$$[S][S]^* = [I]$$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & S_{13}^* \\ S_{13}^* & S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

multiplying we get

$$R_{1C1} = S_{11}S_{11}^* + S_{12}S_{12}^* + S_{13}S_{13}^* = 1$$

$$|S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 = 1 \quad \text{--- (1)}$$

$$R_{2C2} = |S_{12}|^2 + |S_{22}|^2 + |S_{13}|^2 = 1 \quad \text{--- (2)}$$

$$R_{3C3} = |S_{13}|^2 + |S_{13}|^2 = 1 \quad \text{--- (3)}$$

$$2|S_{13}|^2 = 1 \Rightarrow |S_{13}|^2 = 1/2 \Rightarrow \boxed{S_{13} = 1/\sqrt{2}} \quad \text{--- (4)}$$

$$R_{3C1} = S_{13}S_{11}^* + S_{13}S_{12}^* = 0$$

$$S_{13}(S_{11}^* + S_{12}^*) = 0$$

$$S_{13} \neq 0 \therefore S_{11} = -S_{12} \quad \text{--- (5)}$$

by comparing eq (1) and (2) we get

$$S_{11} = S_{22}$$

from eq (1) $|S_{11}|^2 + |S_{22}|^2 + |S_{13}|^2 = 1$

$$S_{11}^2 + S_{11}^2 + 1/2 = 1$$

$$2S_{11}^2 = 1/2 \Rightarrow S_{11}^2 = 1/4 \Rightarrow \boxed{S_{11} = 1/2}$$

from eq (5) $S_{12} = -1/2$; $S_{22} = 1/2$

$$\therefore S = \begin{bmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$

we know $[b] = [S][a]$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$b_1 = 1/2 a_1 - 1/2 a_2 + 1/\sqrt{2} a_3$$

$$b_2 = -1/2 a_1 + 1/2 a_2 + 1/\sqrt{2} a_3$$

$$b_3 = 1/\sqrt{2} a_1 + 1/\sqrt{2} a_2$$

case 1: i/p given at port 3, and no i/p at port ① and ②
 $a_3 \neq 0$; $a_1 = a_2 = 0$

$$\therefore b_1 = a_3/\sqrt{2}, b_2 = a_3/\sqrt{2} \text{ and } b_3 = 0$$

Let P_3 be pwr i/p at port ③. Then this pwr divides equally
b/w ports ① and ② in phase i.e. $P_1 = P_2$

$$P_3 = P_1 + P_2 = 2P_1 = 2P_2$$

The amt of pwr coming out of port ① and at port ② due

$$\begin{aligned} \text{to i/p at port ③} &= 10 \log_{10} P_1/P_3 = 10 \log_{10} P_1/2P_1 \\ &= 10 \log_{10} 1/2 \end{aligned}$$

$$= -10 \log_{10} 2 = -10 (0.3010) \approx -3 \text{ dB.}$$

Hence pwr coming out of port ① or port ② is 3 dB down with
respect to i/p pwr at port ③. Hence H-plane Tee is 3-dB splitter

Case 2:

$$a_1 = a_2 = a; a_3 = 0$$

$$b_1 = a/2 - a/2 + 1/\sqrt{2} a_3 = 0$$

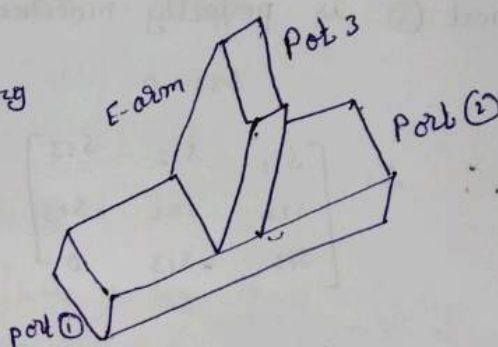
$$b_2 = -a/2 + a/2 + 1/\sqrt{2} a_3 = 0$$

$$b_3 = 2a/\sqrt{2}$$

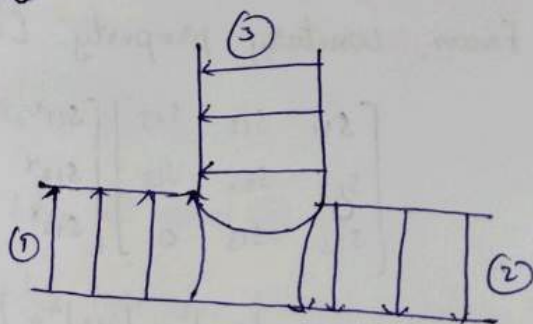
i.e. The o/p at port (3) is addition of two i/p at port (1) and port (2) and these are added in phase.

E-plane Tee

A rectangular slot is cut along broader dimension of long waveguide and side arm is attached. Port (1) and (2) are collinear arms and port (3) is E-arm.



When TE_{10} mode is made to propagate into port 3, the z.o/p at port (1) and (2) will have a phase shift of 180° . Since electric field lines change their direction when they come out of port (1) and (2), it is called E-plane Tee.



The scattering matrix of E-plane Tee can be used to describe its properties.

→ $[S]$ is a 3×3 matrix since there are 3 ports

$$S = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

→ The scattering coefficient

$$S_{23} = -S_{13}$$

→ From symmetric property $S_{ij} = S_{ji}$

$$S_{12} = S_{21}; S_{13} = S_{31}; S_{23} = S_{32}$$

→ if port ③ is perfectly matched to junction

$$S_{33} = 0$$

$$\therefore S = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix}$$

→ From unitary property $[S] \cdot [S]^* = [I]$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & -S_{13}^* \\ S_{13}^* & -S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 C_1 = |S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 = 1 \quad - (1)$$

$$R_2 C_2 = |S_{12}|^2 + |S_{22}|^2 + |S_{13}|^2 = 1 \quad - (2)$$

$$R_3 C_3 = |S_{13}|^2 + |S_{13}|^2 = 1 \quad - (3)$$

$$\boxed{S_{13} = 1/\sqrt{2}}$$

$$R_3 C_1 = S_{13} \cdot S_{11}^* - S_{13} S_{12}^* = 0$$

$$S_{11} = S_{12} \quad - (4)$$

$$\text{from eq. (1)} \quad |S_{11}|^2 + |S_{12}|^2 + 1/2 = 1 \Rightarrow 2|S_{11}|^2 = 1/2$$

$$\boxed{S_{11} = 1/\sqrt{2}}$$

From eq (2)

$$1/4 + |s_{22}|^2 + 1/2 = 1$$

$$s_{22} = 1/2$$

$$S = \begin{bmatrix} 1/2 & 1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix}$$

$$[b] = [S][a]$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$b_1 = 1/2 a_1 + 1/2 a_2 + 1/\sqrt{2} a_3$$

$$b_2 = 1/2 a_1 + 1/2 a_2 - 1/\sqrt{2} a_3 ; b_3 = 1/\sqrt{2} a_1 - 1/\sqrt{2} a_2$$

Case 1: $a_1 = a_2 = 0 ; a_3 \neq 0$

$$\text{then } b_1 = 1/\sqrt{2} a_3 \quad b_2 = -1/\sqrt{2} a_3 ; b_3 = 0$$

An i/p at port ③ equally divides b/w ① and ② but introduces a phase shift of 180° b/w two o/p. Hence E-plane Tee also act as 3dB splitter

Case 2 $a_1 = a_2 = a ; a_3 = 0$

$$b_1 = a ; b_2 = a ; b_3 = 0$$

equal i/p at port ① and ② result in no o/p at port ③

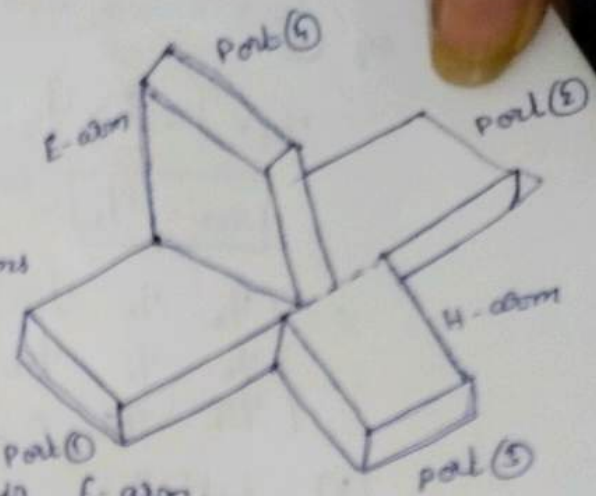
Case 3 $a_1 \neq 0 ; a_2 = 0 ; a_3 = 0$

$$b_1 = a_1/2 ; b_2 = a_1/2 ; b_3 = -a_1/\sqrt{2}$$

Magic Tee

Here rectangular slots are cut both along width and breadth of long waveguide and side arms are attached as shown in fig

port ① and ② are collinear arms port ③ in H-arm, and port ④ is E-arm.



Scattering matrix can be obtained as follows

$$\rightarrow [S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix}$$

→ Because of H-plane Tee section

$$S_{23} = S_{13}$$

→ Because of E-plane Tee section

$$S_{24} = -S_{14}$$

→ Because of geometry of junction an i/p at port ③ cannot come out of port ④ since they are isolated ports and vice versa

$$S_{34} = S_{43} = 0$$

→ from symmetric property

$$S_{12} = S_{21}; S_{13} = S_{31}; S_{23} = S_{32}$$

$$S_{34} = S_{43}; S_{24} = S_{42}; S_{41} = S_{14}$$

→ if port ③ and ④ are perfectly matched to junction

$$S_{33} = S_{44} = 0$$

(6)

$$\therefore S = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{23} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

→ From unitary property $[S][S]^* = [I]$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{23} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* & S_{14}^* \\ S_{12}^* & S_{22}^* & S_{23}^* & -S_{14}^* \\ S_{13}^* & S_{13}^* & 0 & 0 \\ S_{14}^* & -S_{14}^* & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 C_1 = |S_{11}|^2 + |S_{12}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1 \quad \text{--- (1)}$$

$$R_2 C_2 = |S_{12}|^2 + |S_{22}|^2 + |S_{13}|^2 + |S_{14}|^2 = 1 \quad \text{--- (2)}$$

$$R_3 C_3 = |S_{13}|^2 + |S_{13}|^2 = 1 \Rightarrow S_{13} = 1/\sqrt{2} \quad \text{--- (3)}$$

$$R_4 C_4 = |S_{14}|^2 + |S_{14}|^2 = 1 \Rightarrow S_{14} = 1/\sqrt{2} \quad \text{--- (4)}$$

comparing eq (1) and (2) we get

$$S_{11} = S_{22}$$

From eq (1) we get

$$|S_{11}|^2 + |S_{12}|^2 + 1/2 + 1/2 = 1$$

$$|S_{11}|^2 + |S_{12}|^2 = 0 \quad \boxed{S_{11} = S_{12}}$$

$$\therefore S_{11} = S_{12} = S_{22} = 0$$

This means port (1) and (2) are also perfectly matched to junction. ing
so A junction where all four ports are perfectly matched to
junction is called magic Tee.

$$S = \begin{bmatrix} 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{bmatrix}$$

$$[b] = [S][a]$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

$$b_1 = 1/\sqrt{2}(a_3 + a_4); \quad b_3 = 1/\sqrt{2}(a_1 + a_2)$$

$$b_2 = 1/\sqrt{2}(a_3 - a_4); \quad b_4 = 1/\sqrt{2}(a_1 - a_2)$$

case 1; $a_3 \neq 0, a_1 = a_2 = a_4 = 0$

we get $b_1 = a_3/\sqrt{2}; b_2 = a_3/\sqrt{2}; b_3 = b_4 = 0$

This is the property of H-plane Tee

case 2; $a_4 \neq 0, a_1 = a_2 = a_3 = 0$

$$b_1 = a_4/\sqrt{2}; \quad b_2 = -a_4/\sqrt{2}; \quad b_3 = b_4 = 0$$

This is the property of E-plane Tee

case 3; $a_1 \neq 0, a_2 = a_3 = a_4 = 0$

$$b_1 = 0; \quad b_2 = 0; \quad b_3 = a_1/\sqrt{2}; \quad b_4 = a_1/\sqrt{2}$$

ie when power is fed into port ① nothing comes out of port ② even though they are collinear ports. Hence port ① and ② are called isolated ports. similarly an i/p at port ② cannot come out at port ①. Similarly E and H ports are isolated ports

case 4: $a_3 = a_4$; $a_1 = a_2 = 0$

$$b_1 = 1/\sqrt{2} (2a_3); \quad b_2 = b_3 = b_4 = 0$$

This is nothing but additive property. equal i/p at port ③ and ④ result in o/p at port ①

case 5: $a_1 = a_2$, $a_3 = a_4 = 0$

$$b_1 = 0 = b_2 = b_4; \quad b_3 = 1/\sqrt{2} (2a_1)$$

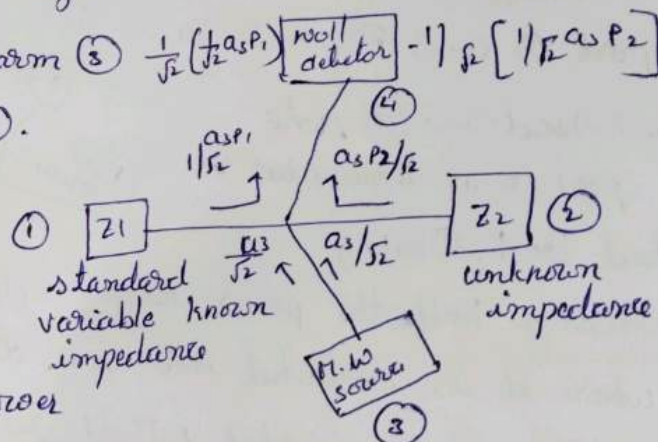
equal i/p at port ① and ② results in o/p at port ③ and no o/p at port ① ② & ④.

Applications of magic tee

1a) Measurement of impedance: Magic tee is used in form of bridge.

M.W source is connected in arm ③ $\frac{1}{\sqrt{2}} (\frac{a_3}{\sqrt{2}})$ null detector in arm ④ $\frac{1}{\sqrt{2}} [1/\sqrt{2} a_3 P_2]$ and null detector in arm ④.

The unknown impedance is connected in arm ② and a standard variable known impedance in arm ①. Using properties of magic tee, the power



from M.W source (a_3) gets divided equally b/w arms ① and ② $a_3/\sqrt{2}$ (to the unknown impedance and standard variable known impedance). These impedances are not equal to characteristic impedance (Z_0) and hence there will be reflection from arm ① & ② if P_1 and P_2 are reflection coefficient, powers $P_1 a_3/\sqrt{2}$ and $P_2 a_3/\sqrt{2}$ enter magic tee junction from arm ① and ②. The resultant wave into arm can be calculated as

The net wave reaching null detector

$$= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} a_3 P_1 \right) - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} a_3 P_2 \right) = \frac{1}{2} a_3 (P_1 - P_2)$$

For perfect balancing of bridge

$$\frac{1}{2} a_3 (P_1 - P_2) = 0$$

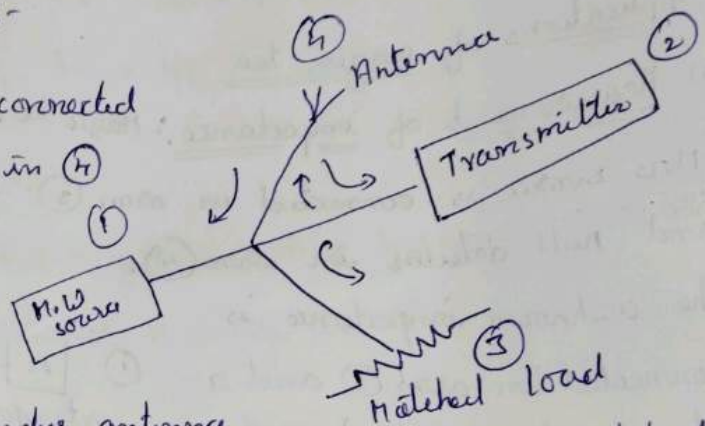
$$P_1 - P_2 = 0 \text{ or } P_1 = P_2$$

Thus unknown impedance can be measured by adjusting standard variable impedance till bridge is balanced and both impedance become equal.

(b) Magic Tee as a Duplexer:-

The Transmitter and receiver are connected in port ① and ②, antenna in ④

matched load in ③, where and port ③ is terminated in matched load. During



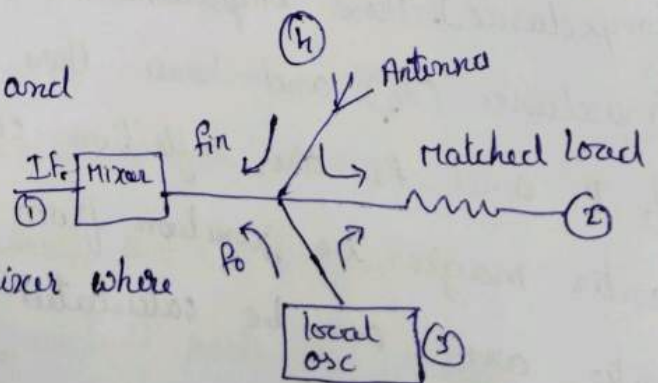
transmission half the power reaches antenna from where it is radiated into space, other half reaches the matched load where it is absorbed without reflection. No transmitter power reaches receiver since port ① and ② are isolated ports. During reception, half of the received power goes to receiver and the other half to transmitter.

(c) Magic Tee as a mixer

It can be used as mixer where signal and local osc are fed into ② and ③ respectively.

Half of local osc power and half of received power from antenna goes to mixer where they are mixed to generate IF freq.

$$IF = f_{in} \sim f_0$$



Microwave attenuator

A device used to control amount of microwave power transferred from one pt to another pt on a microwave transmission system is called microwave attenuator.

It can attenuate power either by absorption (or) reflection

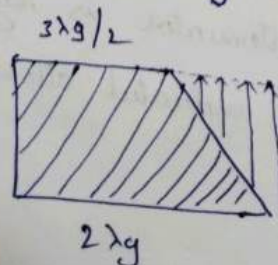
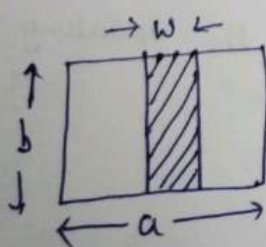
These are commonly used for

- Measuring power gain or loss in dB
- For providing isolation b/w instruments
- For reduce power i/p to a particular stage to prevent overloading
- calibrating o/p accurately in measuring instruments

Types of attenuators

- Fixed attenuator
- variable attenuator

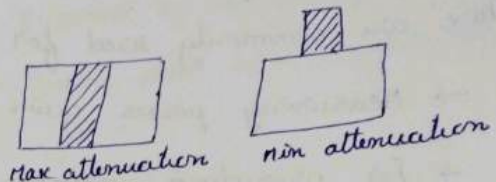
Fixed attenuators : They are used where fixed amt of attenuation is needed. If such a fixed attenuator absorbs all the energy entering into it we call it as a waveguide terminator. This normally consists of short section of waveguide with a tapered plug of absorbing material at end. The tapering is done for providing a gradual transition from waveguide medium to absorbing medium thus reducing reflection occurring at media interface



Variable attenuator :- It provides continuous or step size variable attenuation for rectangular waveguide. These attenuators can be flap type or vane type. For circular waveguide rotary type is used.

Flap type attenuator

It consists of resistive element as disk inserted into longitudinal slot cut along centre of wider dimension of waveguide.

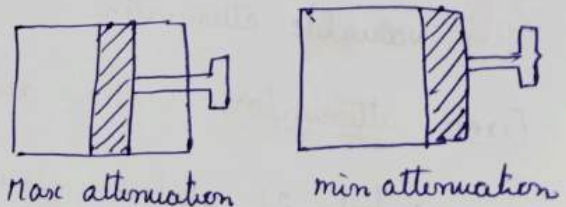


The flap is mounted on hinged arm allowing it to descend into centre of waveguide. The degree of attenuation is determined by depth of insertion of flap.

Vane type attenuator

This basically consists of glass vane with a coating of aquadag or carbon similar to fixed vane attenuator.

The vane positioned at centre of waveguide can be moved laterally from centre, where it provides attenuation to edge, where the attenuation is



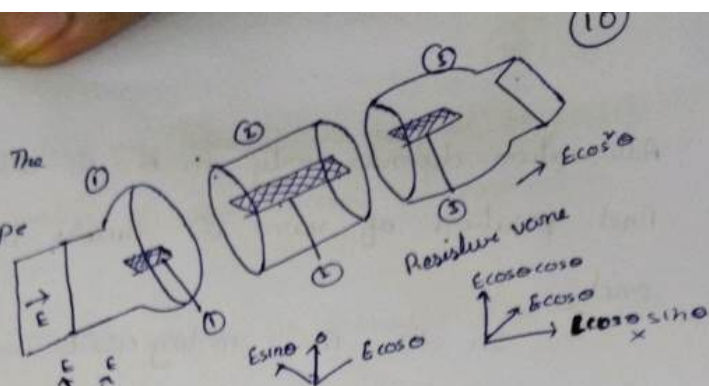
considerably reduced since electric field lines are concentrated at the centre of waveguide. The vane is tapered at both the ends to match attenuator to waveguide. tapered length must be equal to $\lambda_g/2$.

Rotary vane precision attenuator

precision is word related to measurement. It is very close to accuracy.

Rotary vane precision attenuator is highly used device with an accuracy of $\pm 2.1\%$ of indicated attenuation over the operating frequency range.

consists of three vanes. The centre vane is rotating type placed in central section of circular waveguide and it is tapered at both ends.



The other two vanes are in rectangular section.

Vane 1 and 3 are fixed type. Vane 2 is movable, both ends are tapered to provide matching.

Vane 1 prevents horizontal polarization, only vertical polarized waves are propagated through it. Vanished field induces currents within the vane.

Elimination completely depends on arrangement of vane, and transition from rect to circular shape.

In case of central vane, it is moving with an angle θ , because of this we get components $E \sin \theta$, $E \cos \theta$. But because of arrangement of vane, $E \sin \theta$ eliminated.

Because of transition from circular to rectangular in case of vane 3, $E \cos \theta$ again divided into 2 components.

Horizontal polarized wave is eliminated. So only $E \cos^2 \theta$ component is available at end of RWH.

Power transferred from one end to another end is

$$\begin{aligned} P_{\text{tan}} &= \frac{1}{R} (E \cos^2 \theta)^2 \\ &= \frac{E^2}{R} \cos^4 \theta \\ &= P_{\text{in}} \cos^4 \theta \end{aligned}$$

$$\frac{P_{\text{tan}}}{P_{\text{in}}} = \cos^4 \theta \Rightarrow \text{Attenuation} = \cos^4 \theta$$

Attenuation depends only on θ so attenuation is precise
 And position of vane ② decides amount of powr at
 end.

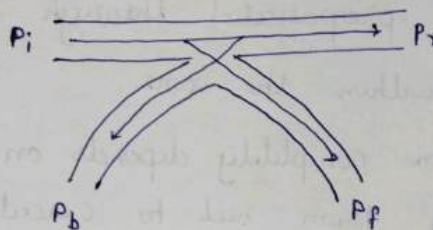
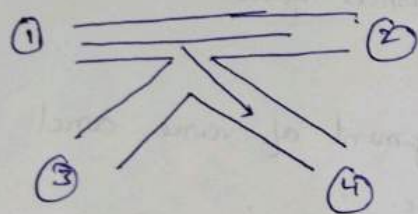
$$\text{In dB } A = -10 \log \cos^4 \theta \quad (-ve \text{ sign due to attenuation})$$

$$= -40 \log \cos \theta$$

Attenuation is freq independent

When all 3 vanes are alligned their planes 90° to the
 direction of electric field, there is no attenuation

Directional coupler



Directional coupler can sample a small amt of m.w powr for
 measurement purpose. They are designed to measure incident and
 reflected powr and SWR values. They can be unidirectional and
 also bidirectional. It is 4 port wave guide junction consisting
 of primary main wave guide and secondary auxiliary wave guide

With matched terminations at all ports the properties of ideal
 directional coupler can be summarized as follows.

- A portion of powr travelling from port ① to port ② is
 coupled to port ④ but not port ③
- A portion of powr travelling from port ② to port ① is coupled
 to port ③ but not port ④

A portion of power incident on port ③ is coupled to port ② but not to port ① and portion of power incident on port ④ is coupled to port ①

The performance of directional coupler is usually defined in terms of parameters which are defined as

$$\text{coupling factor} = 10 \log P_i / P_f$$

$$\text{Directivity} = 10 \log P_f / P_b$$

$$\text{isolation} = 10 \log P_i / P_b$$

Coupling factor: def as ratio of incident power (P_i) to forward coupled power (P_f)
 $C = 10 \log_{10} P_i / P_f$

It is measure of how much incident power is being sampled

Directivity: def as ratio of forward power to backward power

$$D = 10 \log_{10} P_f / P_b$$

It is measure of how well direction coupler distinguish b/w forward and reverse travelling power.

Isolation: def as ratio of incident power to backward power

$$I = 10 \log_{10} P_i / P_b$$

It describes the directional propagation of directional coupler

For typical directional coupler

$$C = 20 \text{ db}, D = 60 \text{ db}$$

$$C = 20$$

$$10 \log_{10} P_i / P_f = 20$$

$$P_i / P_f = 10^2$$

$$P_f = P_i / 100$$

$$D = 60 = 10 \log P_f / P_b$$

$$P_f / P_b = 10^6$$

$$P_b = P_f / 10^6 = P_i / 10^8 [P_f = P_i / 100]$$

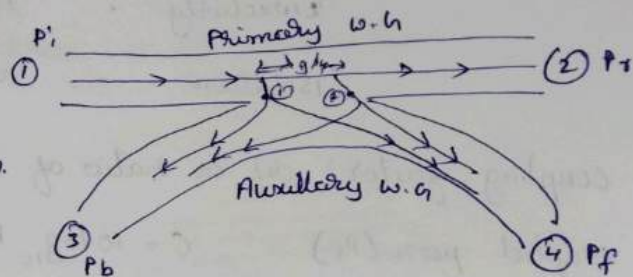
Since P_b very small, $[1/10^8] P_i$, the power coming from port 3 can be neglected

Types of directional coupler

- Two hole directional coupler
- Four hole " "
- Bethe hole or single hole DC
- short-slot DC
- Loop DC

Two hole directional coupler

It consists of two waveguides the main an auxiliary with two tiny holes common b/w them.



- Two holes are at dist $\lambda/4$
- The two leakages of holes (1) and (2) are both in phase at position of 2nd hole and hence they are add up contributing P_f
- But two leakages are out of phase by 180° at position of 1st hole and therefore they cancel each other making $P_b = 0$ (ideally)
- The magnitude of power coming from the two holes depends upon dimension of two holes.
- The no. of holes can be one or more than two. The degree of coupling is determined by size and location of holes in waveguide walls.
- Although a high degree of directivity can be achieved at a fixed freq. it is quite difficult over a band of freq.

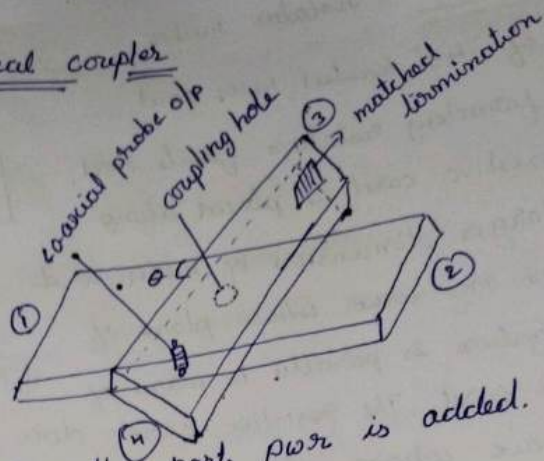
Bethe or single hole Directional coupler

By using this single hole directional coupler we are getting high directivity, Here to get $P_3 = 0$

Here we are adjusting θ in such a way that, at one port

the power gets cancelled and at another port power is added.

port ③ must be terminated with matched load, so we are placing matched termination. It absorbs any wave. As port ④ we are taking o/p, there is an arrangement called co-axial probe using it we collect our o/p by connecting to C.A.O.



Isolator Microwave devices which make use of Faraday rotation are

1. cryator
2. isolator
3. circulator

Isolator : An isolator is two port device which provides very small attenuation for transmission from port ① to ② but provides more attenuation for transmission from port ② to ①. In most m.w generator the o/p will be fluctuated with changes in load impedance. This is due to mismatch of o/p to load resulting reflected waveform. But these reflected waves should not be allowed to reach m.w generator, which will cause amplitude and freq instability of m.w generator.

construction: Isolator makes

use of 45° twisted R.W.G. and

45° faraday rotation ferrite rod.

A resistive card is placed along

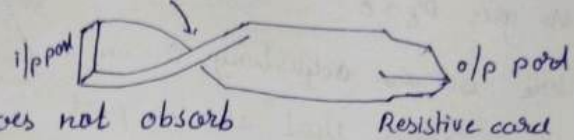
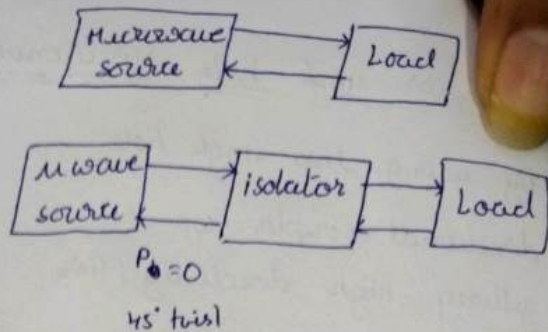
the larger dimension of R.W.G. so it

absorbs any wave whose plane of

polarization is parallel to plane of

resistive card. The resistive card does not absorb

any wave whose plane of polarization is $\perp$ to its own plane.



operation

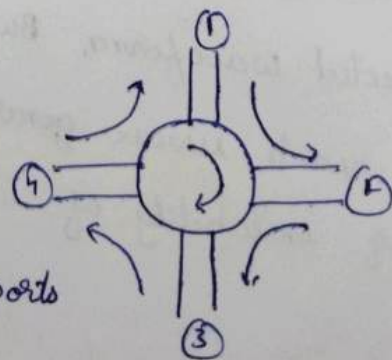
A TE_{10} wave passing from port ① through resistive card and is not attenuated. After coming out of card the wave get shifted by 45° because of anti clk wise shift twist and then by another 45° in clk wise direction because of ferrite rod. And hence comes out of port ② with out any attenuation. This wave -

TE_{10} wave fed from port ② just pass from resistive card since plane of polarization is $\perp$ to resistive card, the wave is shifted by 45° due to ferrite rod in clk wise direction and further wave rotated by 45° in clk wise direction due to twist in waveguide. Now plane of polarization will be parallel to resistive card and hence it is absorbed by resistive card resistive card and o/p of port 1 will be zero. This power is dissipated in card as heat.

Circulator

A circulator is four port device which has peculiar property that each terminal is connected only to next clk clk wise terminal i.e. port ① to port ② not to ③ and ④.

Although there is no restriction on no. of ports four ports are commonly used



They are useful in parametric amp^r, tunnel diode, amplifiers and duplexer in radars

construction working

Suppose if circulator works in clockwise direction

* When i/p is at port ① o/p is from port ② Remaining ports are isolated

When i/p is at port ②, o/p is from port ③ Remaining ports are isolated

Sectional coupler S-matrix

$$S = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \quad - (1)$$

four ports of DC are perfectly matched to jux

$$S_{11} = S_{22} = S_{33} = S_{44}$$

From symmetrical property

$$S_{12} = S_{21}; S_{31} = S_{13}; S_{24} = S_{42}$$

$$S_{43} = S_{34}; S_{23} = S_{32}; S_{41} = S_{14}$$

For ideal 3 dB DC the back power is zero $P_b = 0$

Hence there is no coupling b/w port 1, 3 and port 2, 4

$$S_{13} = S_{31} = 0 \quad S_{24} = S_{42} = 0$$

eq (1) can be written as

$$S = \begin{bmatrix} 0 & S_{12} & 0 & S_{14} \\ S_{12} & 0 & S_{23} & 0 \\ 0 & S_{23} & 0 & S_{34} \\ S_{14} & 0 & S_{34} & 0 \end{bmatrix}$$

using relation $[S][S]^* = I$

$$\begin{bmatrix} 0 & S_{12} & 0 & S_{14} \\ S_{12} & 0 & S_{23} & 0 \\ 0 & S_{23} & 0 & S_{34} \\ S_{14} & 0 & S_{34} & 0 \end{bmatrix} \begin{bmatrix} 0 & S_{12}^* & 0 & S_{14}^* \\ S_{12}^* & 0 & S_{23}^* & 0 \\ 0 & S_{23}^* & 0 & S_{34}^* \\ S_{14}^* & 0 & S_{34}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$|S_{12}|^2 + |S_{14}|^2 = 1 \quad - (2)$$

$$|S_{12}|^2 + |S_{23}|^2 = 1 \quad - (3)$$

$$|S_{23}|^2 + |S_{34}|^2 = 1 \quad - (4)$$

$$|S_{14}|^2 + |S_{34}|^2 = 1 \quad - (5)$$

$$|821| \quad S_{12}S_{23}^* + S_{14}S_{34}^* = 0 \quad (R_1 C_3) \Rightarrow |S_{12}||S_{23}^*| = |S_{14}||S_{34}^*| \quad - (6)$$

$$S_{12}S_{14}^* + S_{23}S_{34}^* = 0 \quad (R_2 C_4) \Rightarrow |S_{12}||S_{14}^*| = |S_{23}||S_{34}^*| \quad - (7)$$

$$S_{23}S_{12}^* + S_{34}S_{14}^* = 0$$

$$S_{14}S_{34}^* \neq 0$$

As mul eq (6) and (7)

$$|s_{12}|^2 |s_{14}| |s_{23}| = |s_{14}| |s_{23}| |s_{34}|^2$$

$$\therefore |s_{12}| = |s_{34}|$$

lly from other eq, $|s_{14}| = |s_{23}|$

on assuming a positive and real variable 'p' such that

$$s_{12} = s_{34} = p = s_{34}^*$$

on assuming a positive and real variable q such that

$$s_{14} = s_{23} = jq \quad (q \neq 0)$$

$$\text{Since } s_{12} \cdot s_{23}^* + s_{14} s_{34}^* = 0$$

$$s_{12} s_{23}^* + s_{23} s_{12} = 0 \quad [\because s_{12} = s_{34} = s_{34}^*]$$

$$s_{12} [s_{23}^* + s_{23}] = 0$$

$$p [s_{23}^* + s_{23}] = 0$$

$$\text{Since } p \neq 0 \quad s_{23}^* = -s_{23}$$

$$s_{23}^* = -jq \quad [\because s_{23} = jq]$$

Hence scattering parameters of 3db DC are given as

$$S = \begin{bmatrix} 0 & s_{12} & 0 & s_{14} \\ s_{12} & 0 & s_{23} & 0 \\ 0 & s_{23} & 0 & s_{34} \\ s_{14} & 0 & s_{34} & 0 \end{bmatrix} = \begin{bmatrix} 0 & s_{12} & 0 & s_{14} \\ s_{12} & 0 & s_{14} & 0 \\ 0 & s_{14} & 0 & s_{12} \\ s_{14} & 0 & s_{12} & 0 \end{bmatrix}$$

$$\begin{aligned} b_3 &= 0.5b_1/\sqrt{2} + 0.6b_2/\sqrt{2} \\ b_4 &= 0.5b_1/\sqrt{2} - 0.6b_2/\sqrt{2} \\ b_1 &= 1/\sqrt{2} + 0.8b_4/\sqrt{2} \\ b_1 - 0.8b_4/\sqrt{2} &= 1/\sqrt{2} \\ b_2 &= 1/\sqrt{2} - 0.8b_4/\sqrt{2} \\ b_2 + 0.8b_4/\sqrt{2} &= 1/\sqrt{2} \end{aligned}$$

$$[S][S^*] = [I]$$

$$\begin{bmatrix} 0 & S_{21} & 0 & 0 \\ 0 & 0 & S_{21} & 0 \\ 0 & 0 & 0 & S_{21} \\ S_{21} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & S_{21}^* \\ S_{21}^* & 0 & 0 & 0 \\ 0 & S_{21}^* & 0 & 0 \\ 0 & 0 & S_{21}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_{ic1} = |S_{21}|^2 = 1 \Rightarrow S_{21} = 1$$

$\therefore$ S matrix of circulator is

$$S = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

S-matrix parameters of Isolator

S-matrix has size $n \times n$ $n = \text{no. of ports}$

$$n = 2$$

$$\therefore 2 \times 2 \Rightarrow [S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

$\rightarrow$ As port 1 and port 2 are perfectly matched then

$$S_{11} = S_{22} = 0$$

$\rightarrow S_{12}, S_{21}$ - port i/p $\rightarrow$ port 2 o/p

$$S_{12} = 0 \quad [\text{since no reflection coefficient}]$$

$\downarrow$
when i/p applied at port 1, 100% o/p is delivered at port 2

$$S_{21} = 1$$

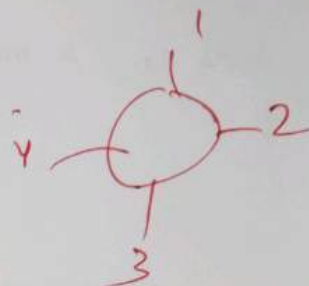
From property of Isolator $S_{12} = 0$

Scattering matrix of s-matrix circulator

circulator are n-port lossless and non-reciprocal device matched at all ports in which power flow occur from port 1 to port 2, port 2 to port 3 P_{n-1} to P_n to 1 (clockwise)

Size of s-matrix is $n \times n$, where n represent no. of ports.

$$S = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix}$$



when all ports are perfectly matched

$$S_{11} = S_{22} = S_{33} = S_{44} = 0$$

consider the ports are reciprocal

$$S_{21} = S_{32} = S_{43} = S_{14}$$

Rest of s-matrix parameters are zero

$$S_{12} = S_{13} = S_{23} = S_{24} = S_{31} = S_{34} = S_{41} = S_{42} = 0$$

$\therefore$ The s-matrix becomes

$$S = \begin{bmatrix} 0 & 0 & 0 & S_{21} \\ S_{21} & 0 & 0 & 0 \\ 0 & S_{21} & 0 & 0 \\ 0 & 0 & S_{21} & 0 \end{bmatrix}$$

from unitary property $[S][S]^* = I$

is transverse of s matrix is

$$S = \begin{bmatrix} 0 & S_{21} & 0 & 0 \\ 0 & 0 & S_{21} & 0 \\ 0 & 0 & 0 & S_{21} \\ S_{21} & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore [S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ S_{21} & 0 \end{bmatrix}$$

$$\left| \begin{array}{cc|cc} 0 & S_{21} & 0 & 1 \\ 0 & 0 & S_{21}^* & 0 \end{array} \right|$$

from unitary property

$$[S][S^*] = I$$

transverse of S matrix is

$$\begin{bmatrix} 0 & S_{21} \\ 0 & 0 \end{bmatrix}$$

$\therefore [S][S^*] = I$ becomes

$$\begin{bmatrix} 0 & S_{21} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ S_{21}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$|S_{21}|^2 = 1$$

$$\therefore S_{21} = 1$$

As there is maximum attenuation from port 2 to port 1

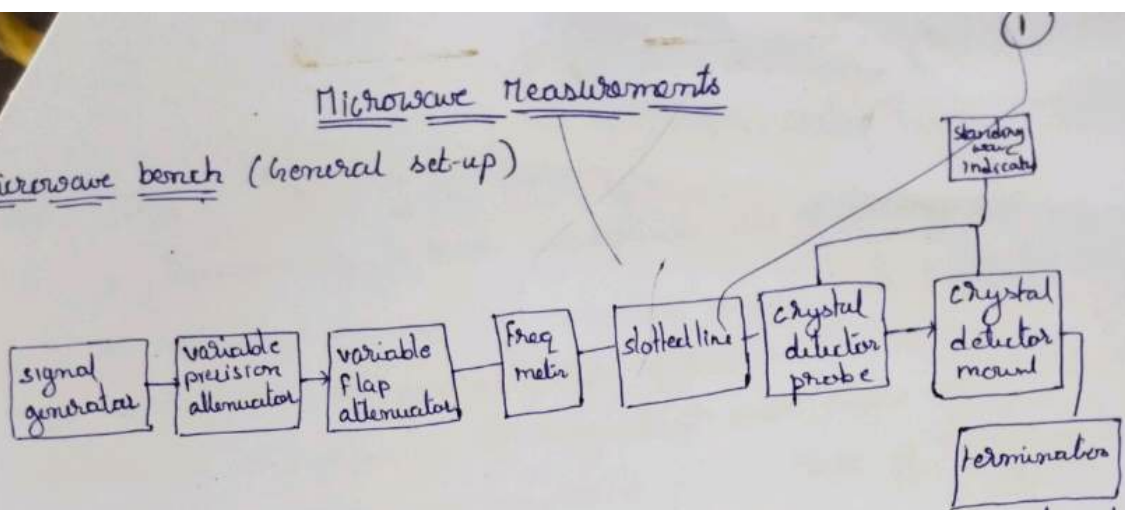
$$S_{21} = 1$$

$\therefore S$ matrix of Isolator is

$$S = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

Microwave Measurements

Microwave bench (General set-up)



- The signal generator is a microwave source whose o/p is of order of milliwatts. It could be gunn diode oscillator, a backward wave oscillator or reflex klystron tube. It provide either continuous wave or sq wave modulated at audio rate which is normally 1 kHz.
- The precision attenuator can provide 0 to 50 dB attenuation abv its insertion loss.
- The variable flap attenuator is used, whose calibration can be checked against readings of precision attenuator
- A freq meter is used for direct reading of freq that consists of single cylindrical cavity which can be adjusted to resonance
- A slotted line is coupled to waveguide
- The crystal detector, inserted in E probe of slotted line is contained in crystal detector mount at end of waveguide gun is used to detect modulated signal
- SWR is (standing wave ratio indicator) is basic basically a sensitive tuned voltmeter that provides direct reading of SWR. or its

Slotted line

In n.w transmission line or waveguide, the electromagnetic field is considered as sum of incident wave from generator and reflected wave to generator. The reflection indicates a mismatch or discontinuity. The magnitude and phase of reflected wave depends on amplitude and phase of reflecting impedance.

The standing waves obtained are measured to know the waveguide imperfections which is necessary to have a knowledge on impedance mismatch for effective transmission.

Slotted line consists of slotted section of waveguide, where measurement has to be done. It has travelling probe carriage, to let the probe get connected whenever necessary, and facility for attaching and detaching instrument.

In waveguide, a slot is made at center of broad side, axially. A movable probe connected to crystal detector is inserted into of waveguide.

Operation

- The o/p of crystal detector is proportional to sq of i/p voltage applied
- The movable probe permits convenient and accurate measurement but as probe is moved along, its o/p is proportional to standing wave pattern, which is formed inside the waveguide.
- variable attenuator is employed to obtain accurate results.

$$\text{The o/p VSWR} = \sqrt{V_{\max}/V_{\min}}$$

In order to obtain low freq modulated signal on oscilloscope a slotted line with tunable detector is employed.

A slotted line with tunable detector can be used to measure

→ VSWR (voltage standing wave ratio)

→ standing wave pattern

→ Impedance

→ Reflection coefficient

→ Return loss.

Tunable detector

The tunable detector is detector mount which is used to detect low freq square wave modulated microwave signal.

To provide a match b/w uwave transmission system and detector mount, a tunable stub is used. There are 3 diff types of tunable stubs.

→ Tunable waveguide detector

→ Tunable co-axial "

→ " probe "

There are fixed stubs like

→ fixed broad band tuned probe

→ fixed waveguide matched detector mount

Measurement of power

Measurement of power is divided into 3 categories.

(a) Measurement of low power (0.01 mW - 10 mW)

- Bolometer technique

(b) Measurement of medium power (10 mW - 1 W)

- Calorimetric Technique

(c) Measurement of high power (> 10 mW)

- calorimetric with meter

Measurement of low microwave power

→ Devices such as bolometers and thermocouples whose resistance changes with applied power are capable of measuring low microwave powers.

→ Bolometers are most widely used.

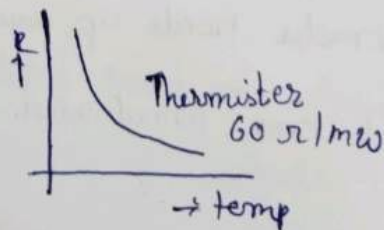
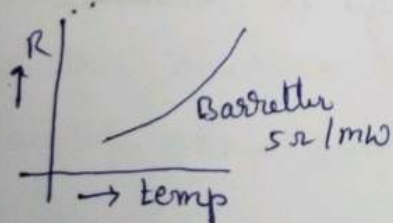
→ It is simple device whose temp Resistance varies with temp

→ These are two types: ① Barretters → positive temp coefficient

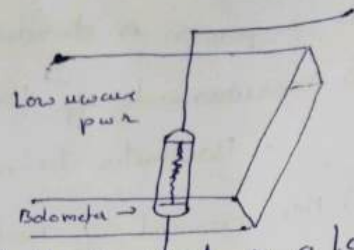
② Thermister → negative

→ In Barretters Resistance ↑ with ↑ temp. It consists of short length of fine platinum wire mounted in cartridge, like an ordinary fuse. It is very delicate device.

→ Thermistors are semiconductor materials



→ A bolometer such as crystal diode is square law device and it produces a current that is proportional to applied power



→ Bolometer is mounted inside waveguide, where bolometer itself is used as a load, with operation resistance R_1 .

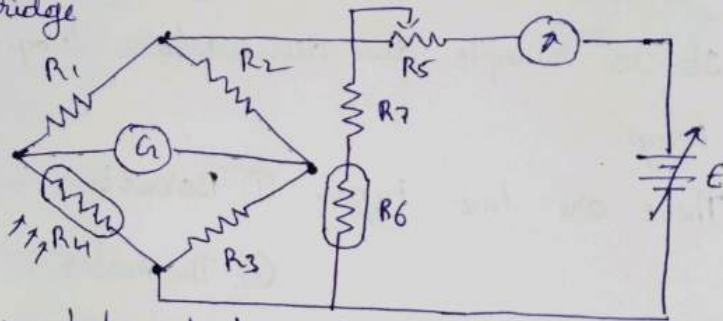
→ Now low microwave power which is to be measured is applied

→ Some power is absorbed in bolometer load and dissipated as heat and the resistance changes to R_2 . This change in resistance is proportional to microwave power which can be measured using bridge.

→ Inaccuracy is introduced due to bolometer non-linear characteristics

→ In balanced bolometer bridge

technique, the bolometer itself is made to be one of arms of bridge



→ Initially, the bridge is balanced by adjusting R_5 ,

→ Let the voltage of battery be E , at balance.

→ The microwave power is now applied and this power gets dissipated in bolometer.

→ The bolometer heats up and it changes its resistance... the bridge becomes unbalanced.

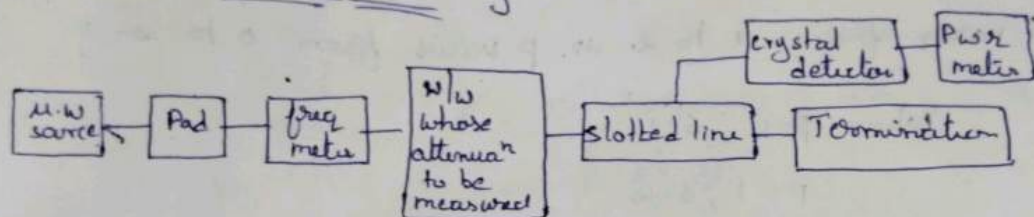
(4)

The applied dc power is changed to E_2 to get back the balance and this change in dc battery voltage ($E_1 \sim E_2$) will be proportional to microwave power.

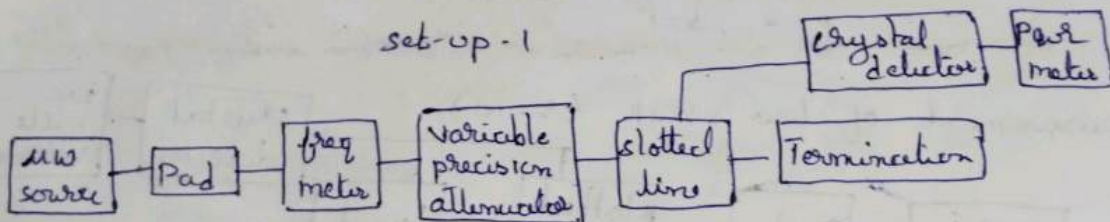
→ Alternately the detector 'Cr' can be directly calibrated in terms of microwave power so that when bridge is unbalanced, the detector reads microwave power directly.

→ Here R_6, R_7 are temp compensation ckt.

Attenuation Measurement by RF substitution method



set-up - 1

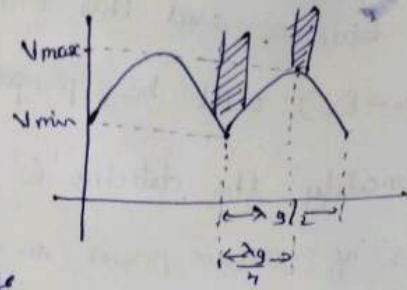


set-up 2

- This method overcomes drawback of power ratio method since we measure attenuation at single power position.
- This method involves measuring o/p power 'P' by including the N/w whose attenuation is to be measured in set-up 1.
- In set-up 2 this N/w is replaced by precision calibrated attenuator which can be adjusted to obtain same power P as measured in set-up 1.
- Under this condition the attenuation read on precision attenuator would give attenuation of N/w directly.

Measurement of voltage standing wave ratio (VSWR)

Any mismatched load leads to reflected waves resulting in standing waves along the length of line.



The ratio of max to min voltage gives VSWR

$$S = V_{\max}/V_{\min} = \frac{1+P}{1-P}$$

where P = reflection coefficient = $P_{\text{reflected}}/P_{\text{incident}}$

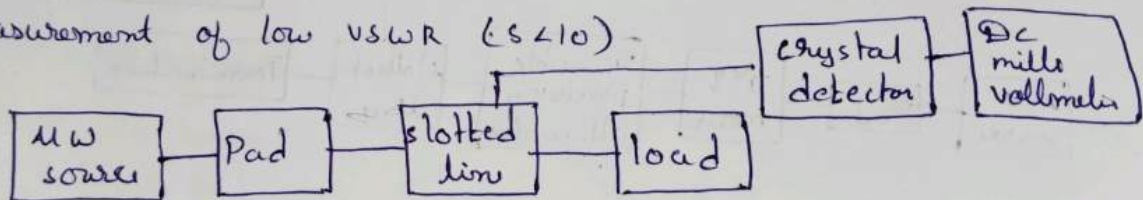
S varies from 1 to ∞ as P varies from 0 to ∞

when $P=0$, $S=1$

$P=1$, $S=\infty$

$P=\infty$, $S=1$

Measurement of low VSWR ($S \leq 10$)



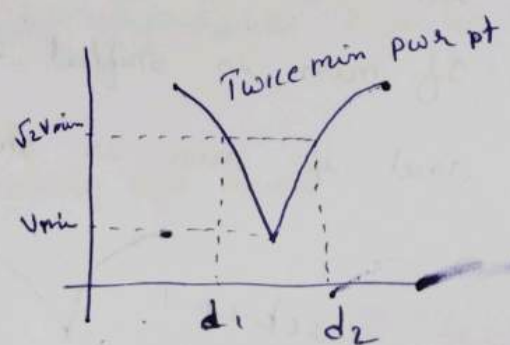
- Values of VSWR not exceeding 10 are very easily measured with abv bench. and can be read off directly on VSWR meter.
- The measurement basically consists of simply adjusting attenuator to give adequate reading on meter, which is DC millivolt meter.
- The probe on slotted waveguide is moved to get max reading on meter (corresponding to $V_{\max}$)
- The attenuation is now adjusted to get full scale reading. This full scale reading is noted.

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- Next probe on slotted line is adjusted to get min reading on meter (corresponding to V_{min}). The ratio of first reading to second reading (V_{max}/V_{min}) gives VSWR.
- The meter itself can be calibrated in terms of VSWR.
- The probe carriage is moved to give max deflection on VSWR meter by adjusting pad.
- This full scale deflection ^(FSD) corresponds to VSWR of 1.
- As an example, a FSD of 10mV corresponds to VSWR of 1.
- The travelling probe is adjusted to get min reading on meter. if this corresponds to 5mV, then $VSWR = 10mV/5mV = 2$.
- if it is 3.3mV, $VSWR = 3$, if it is 2.5mV, $VSWR = 4$, if $VSWR = 1$ then it is 10mV.
- Such a calibrated VSWR meter gives expanded scale upto VSWR of 2 but for $VSWR > 10$, the meter measurement will not be accurate for $VSWR > 10$.

Measurement of High VSWR ($S > 10$)

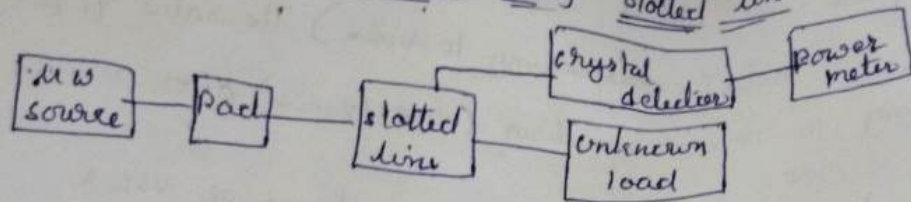
For VSWR's > 10 , we use double min method. In this method, probe is adjusted to get min reading. The probe is then moved to pt where



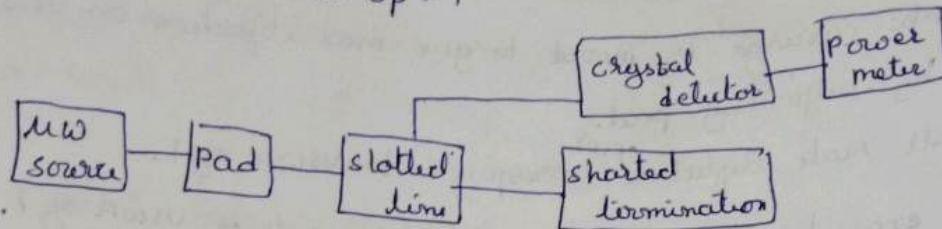
power is twice min. Let this position be d_1 . The probe is then moved to twice the power to other side of min (d_2)

$$\therefore VSWR = \lambda g / \pi (d_2 - d_1).$$

Measurement of impedance using slotted line

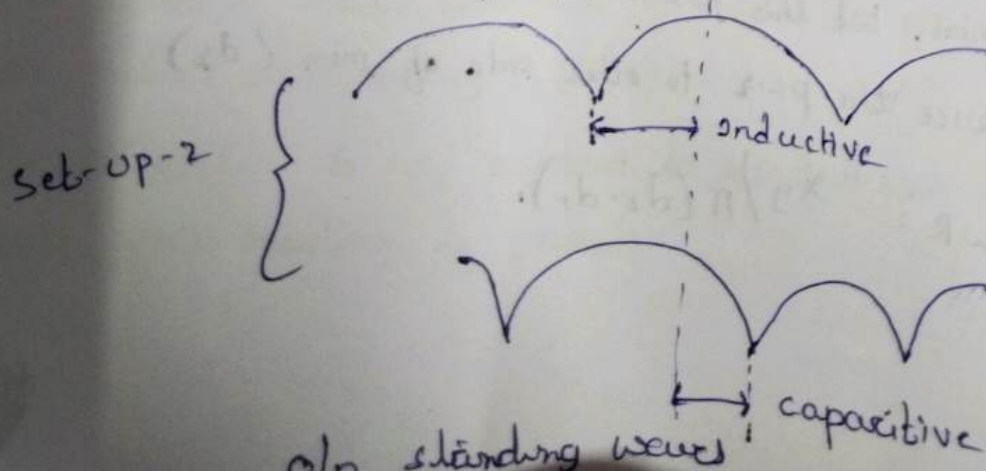
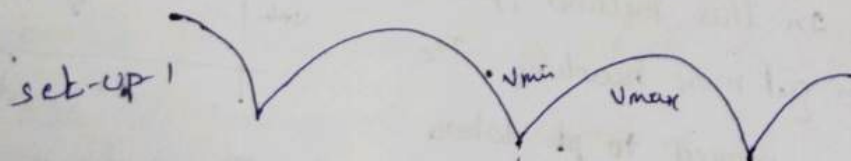


Set-up - 1

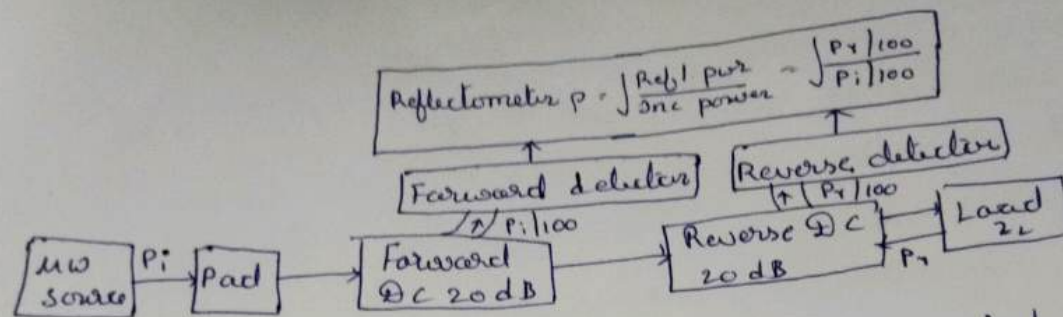


Set-up - 2

- Incident and reflected waves will be proportional to mismatch of load under test resulting standing waves
- Using slotted waveguide and unknown load the position of V_{max} and V_{min} can be determined
- Now load Z_L is replaced by short ckt and shift in min is measured.
- If min is shifted to left, then impedance is inductive and if min is shifted to right, it is capacitive,



Measurement of impedance using Reflectometer



- The reflectometer indicates magnitude of impedance but not phase angle, whereas a slotted line waveguide measurement gives both
- Here two directional couplers are used to sample reflected power P_r and incident power P_i from load.
- Both Directional coupler are identical except their direction. The r
- The magnitude of reflection coefficient P can be directly obtained on reflectometer from which impedance can be calculated.

From reflectometer reading we have

$$P = \sqrt{|P_r|/P_i}$$

Knowing P we can calculate VSWR and impedance by

$$S = \frac{1+P}{1-P} \quad \text{and} \quad \frac{Z - Z_0}{Z + Z_0} = P$$

Z_0 is known impedance, Z unknown impedance

UNIT-5

Requirement of optical source in optical communication

The advancement in i.c tech led to design of optical sources with high efficiency and with very ^{low} losses. It improved quality of o/p.

These sources constitute desired physical and performance characteristics necessary for efficient fiber optic system. These includes

- size compatibility of source to a low loss optical fiber. The sources can be LED or Laser which launch light to fiber.
- Compatibility of ~~laser~~ launching adequate power so that it can compensate fiber attenuation and losses connection losses and detect signal at receiver
- Optimized wavelength that reduces fiber losses and dispersion
- A narrow spectral width to reduce dispersion
- It supports direct modulation of optical o/p power

Optical sources are independent to environment changes. These are more accurate compared to electrical devices.

The basic light sources used in fiber optic communication system are

- * Semiconductor LED
- * Laser diode

Factors affecting the performance of optical source

Power required

(The electric power required for functioning of injection carriers and LED is same as with usual current and voltage drop with magnitude in range of 20 to 300 mW and 1.5 to 2.5 V respectively)

The optical power coupled into fiber from LED is 10 to 20 dB below than that of obtained with typical injection laser

Thus power required is major factor in choice of source and its performance

Linearity

Linearity of optical o/p power against current is an imp factor to estimate performance of optical source.

The non linearity of sources such as LED causes severe distortion thus affecting performances of system

Thermal behaviour

The variation of threshold current with the device junction temp can cause major affect to performance of optical source.

Response

Speed of response play an imp role in performance of sources. Spontaneous emission from LED and stimulated emission from injection lasers specify speed of response. To achieve highest speed it is necessary to minimise switch-on-delay.

Spectral width

The finite spectral width of optical source cause pulse broadening thus affecting source performance.

Optical source of narrow spectral width are utilized.

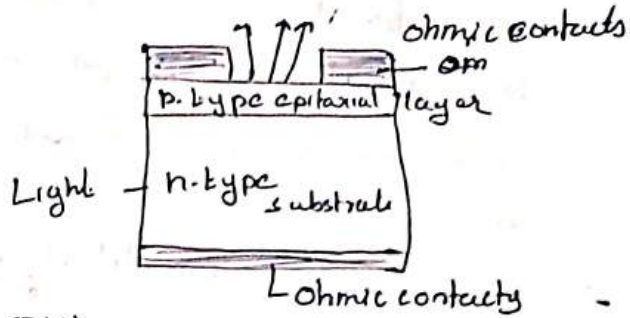
LED structure

→ Planar LED

→ Dome LED

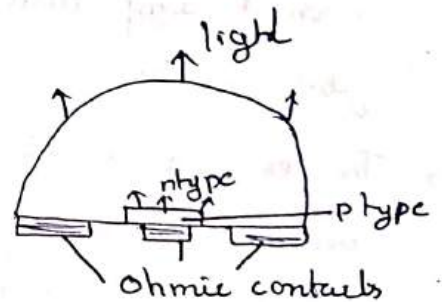
Planar LED

- It is basic type of LED structure
- It is constructed using liquid or vapour phase epitaxial processes.
- In this process a junction is formed over the surface of GaAs substrate by ^{incorporating} diffusing a p-type diffusion into n-type substrate
- In planar LED, forward current flows through entire junction which produces Lambertian spontaneous emission. Hence light is emitted from all surfaces.
- Due to total internal reflection, only some amt of light leaves ~~surface~~ structure. As a result planar LED has low radiance.



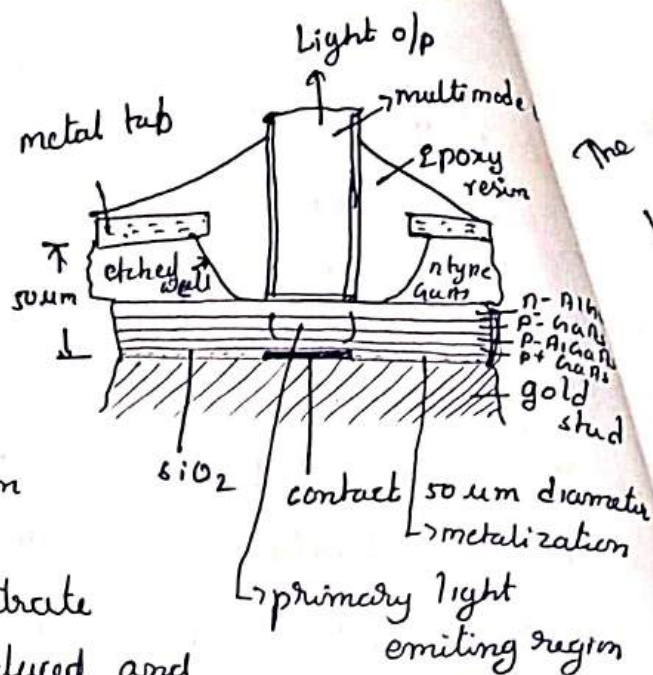
Dome LED

- In Dome shape LED, a hemisphere of n-type GaAs is created around a diffused p-type region.
- The diameter of dome is chosen in such a way that total amt of light reaching the surface within the critical angle of GaAs-air interface should be maximum.
- Therefore, the external power efficiency of dome LED is greater than planar LED
- It has low radiance



Surface emitting LED

- It is also called as butt type or front emitting LED
- In order to acquire high radiance the emission of light must be restricted to small active region within the device
- Hence, a well is etched in GaAs substrate such that absorption of ^{emitted} light is reduced and fiber is placed into it for transmission of radiant light
- Because of large band gaps, the internal absorption becomes negligible and has high reflection coefficient at crystal. This increases forward radiation
- In surface emitting LED, the wavelength of emission ranges from 0.8 to 0.9 μm .
The diameter and thickness of active region is 50 μm and 2.5 μm
- Thermal impedance is low, the coupling of fiber light into fiber is easy.
- The emitted light has isotropic pattern with 120° half power beam width.
- The isotropic pattern is called lambertian pattern
- The brightness of light source is equal in all direction but power reduces by factor $\cos \theta$. θ is angle b/w viewing direction and normal axis



3

The power coupled into multimode step index fiber can be written as

$$P_c = \pi(1-\gamma)AR_0(NA)^2$$

P_c - power coupled into fiber

γ - Fresnel reflection coefficient

A - Emission area of source

R_0 - Radiance of source

NA - numerical aperture

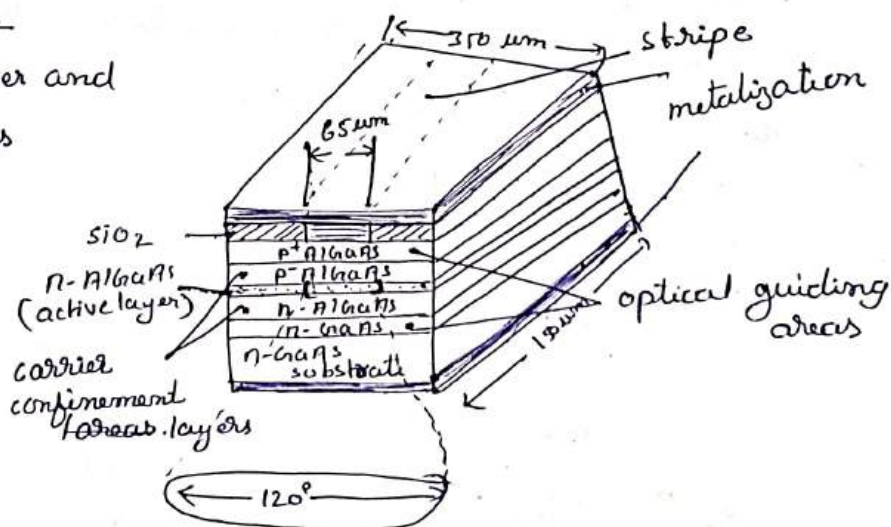
The coupling power relies on following aspects

- Distance and alignment b/w emission area and fiber
- Medium b/w emitting area and fiber
- Emission pattern

By adding epoxy epoxy resin in etched well, the mismatch of refractive index can be minimised and efficiency of external power is improved.

Edge Emitting LED

- It contains active layer and a pair of guiding layers
- The refractive index of guiding layers is greater than that of coating material (SiO_2) and is smaller than that of active layer.



- one face of LED is coated with a reflector and other end face coated with antireflector, so that radiation pattern emits from one face (i.e. from edge coated with antireflector).
- A waveguide channel is formed in LED to guide the optical radiation towards fiber core.
- The contact stripes having diameter 50-70 μm are used to attain proper matching of fiber core of diameter 50-100 μm .
- The emission pattern in edge emitter is highly directional than in surface emitter.
- It efficiently couple the optical power into fiber of small NA.

Internal Quantum efficiency

- The efficiency of LED depends on no. of injected excess electrons with emission of photons.
- The excess of electrons combine radiatively or non-radiatively.
- The ratio of radiative recombination rate to total recombination rate is def as internal quantum efficiency

$$\eta_{\text{int}} = \frac{R_r}{R_r + R_{nr}}$$

η_{int} - Internal quantum efficiency

R_r - Radiative recombination rate

R_{nr} - non-radiative recombination rate

if $\tau_r = n/R_r$ → radiative combination lifetime

$\tau_{nr} = n/R_{nr}$ → non radiative " "

then $\eta_{int} = \frac{n/\tau_r}{\frac{n}{\tau_r} + \frac{n}{\tau_{nr}}} = \frac{1}{1 + \frac{\tau_r}{\tau_{nr}}}$

$$= \frac{\tau_{nr}}{\tau_{nr} + \tau_r} = \frac{\tau}{\tau_r}$$

where $\frac{1}{\tau} = \frac{1}{\tau_r} + \frac{1}{\tau_{nr}}$

τ = bulk recombination rate

If amt of current I is introduced into LED then total no: of recombination per second is given as

$$R_r + R_{nr} = I/q$$

then $\eta_{int} = \frac{R_r}{\frac{I}{q}} = \frac{R_r \cdot q}{I}$

The major factor that reduces internal quantum efficiency is non-radiative recombination, because:

- optical absorption in active region (self absorption)
- carrier recombination at heterojunction interface
- The energy released during electron-hole recombination is transferred to another carrier in form of kinetic energy.

External quantum efficiency

The photodiode's capability to convert light energy to electrical energy is referred quantum efficiency

External quantum efficiency is def as ratio of photons emitted from LED to no: of internally generated photons

The external quantum efficiency can be determined as

$$\eta_{\text{ext}} = \frac{1}{4\pi} \int_0^{\phi_c} T(\phi) (2\pi \sin \phi) d\phi$$

where

$T(\phi)$ - Fresnel Transmission coefficient

ϕ - Incidence angle

For normal incidence i.e. $\phi = 0$

$$T(0) = \frac{4n_1 n_2}{(n_1 + n_2)^2}$$

consider a case in which inner medium refractive index $n_1 = n$ and outer medium as air (i.e. $n_2 = 1$)

$$T(0) = \frac{4n}{(n+1)^2}$$

Then approximate expression for external quantum efficiency

is $\eta_{\text{ext}} = \frac{1}{n(n+1)^2}$

Advantages of LED

Simple design :- The fabrication of LED is very simple. It is of small size and weight which facilitates ease to transportation

Reliability :- LEDs are less sensitive to degradation and

Simpler circuitry :- The drive circuitry of LED is simple because of low value of drive currents. They are easy to interface with other ckt's

Linearity :- The variation of light o/p is linear with device current

cost :- Typical LEDs are low cost and easily available

operation :- LEDs are fast operating devices, generates more light

Lifetime : The life of LEDs is very long.

Dependence of on temperature :-

The o/p current may $\uparrow$ on $\uparrow$ the Temp, but will not stop functioning. LEDs generate less heat

Disadvantages

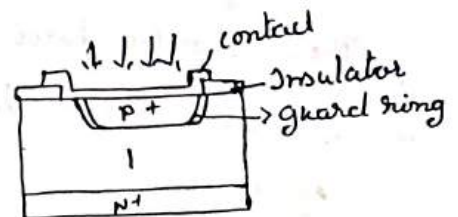
- LEDs drive more current leading to frequent replacement of battery in low power battery operated devices.
- It has low luminous efficiency
- high quality LED are expensive
- LEDs are ~~fragile~~ fragile i.e. easily broken
- require more power for ~~fast~~ fast functioning
- The slight dispersed light radiated by LED may couple with light in optic fiber creating a problem.

Avalanche Photodiode

- It is used where very sensitive light detection is needed.
- It operates under high reverse bias condition. This enables avalanche multiplication of holes and e^- created by photon / light impact.

Avalanche diode operation

- Light enters un-doped region of avalanche photodiode and causes generation e^- hole pairs.
- Under the action of electric field e^- s migrate towards avalanche region.



Performance parameters	Surface LED	Edge emitting LED
------------------------	-------------	-------------------

- How electric field causes their velocity to increase to the extent that collisions with crystal lattice create further hole e^- pairs.
- In this way single e^- created by light in un-doped region may result in many more being created.
- The structure of avalanche diode is more complicated.
- An n-type guard ring is required around p-n junction to minimise electric field around edge of junction.
- It is also found that current gain is dependent not only on bias applied, but also thermal fluctuations. As a result it is necessary to ensure the devices are placed on an adequate heat sink.

Photo detector

- It is an essential component of optical fiber communication system
- Its fun is to convert received optical signal into electrical signal which is amplified before further processing.
- The requirements for photo detector are
 - They should have high sensitivity at operating wavelength
 - It should have high fidelity to reproduce original waveform effectively
 - It should produce max electrical signal for given amt of optical power

6

- It should have short response time to obtain a suitable B.W
- Dark currents, & leakage currents, shunt conductance should be low
- The physical size of detector should be small for effective coupling
- Det. detector should not have require excessive bias voltage or currents
- It must be highly reliable and must be capable of continuous stable operation at room temperature
- It should be economical.

Pin Photodiodes

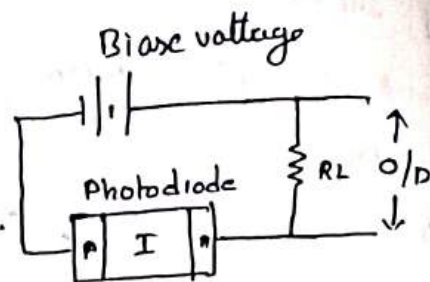
1. The range of wavelength (λ) when Si is used as semiconductor is 400-1100nm
2. The dark current (I_p) for pin photodiode ranges from 1nA to 10nA
3. The range of rise time (τ_r) is 0.5ns to 1ns
4. The B.W of pin photodiode ranges from 0.3 GHz to 0.7 GHz
5. The biasing voltage (V_b) of pin photodiode is 5V

Avalanche Photodiode

1. The range of wavelength (λ) when Si is used as the semiconductor is given as 400-1100nm
2. The dark current (I_p) for APD ranges from 0.1 to 1nA
3. The range of rise time (τ_r) is given as 0.1ns to 2ns
4. The gain B.W of APD ranges from 100 GHz to 400 GHz.
5. The biasing voltage (V_b) of APD ranges from 150V to 400V.

PIN photo detector

→ It consists of p and n regions. The width of depletion region can be increased by adding intermediate intrinsic region.



→ As the intrinsic region has high resistance, a small reverse bias is good enough to increase width of depletion region, so that it extends into n-layer.

→ Adv of pin diode is charge separation in active region is larger, which leads to smaller junction capacitance.

→ The electric field in intrinsic region is uniform and has magnitude E_0 .

$$E_0 = \frac{N_a q x_p}{\epsilon_0 K}$$

q - magnitude of electronic charge

x_p - width of depletion region in p side

K - dielectric const

N_a - acceptor density.

→ The carrier lifetimes are denoted by τ_n and τ_p for n and p. and diffusion lengths as L_n and L_p . So

$$L_n = (D_n \tau_n)^{1/2}$$

D_n, D_p are e^- and hole diffusion

$$L_p = (D_p \tau_p)^{1/2}$$

coefficients

→ The total cutoff wavelength can be calculated depending on bandgap energy (E_g) of material

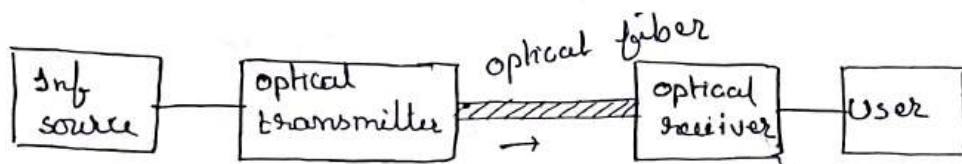
$$\lambda_c = \frac{hc}{E_g} \quad (\text{or}) \quad \lambda_c = \frac{1.24}{E_g}$$

pt to pt link

- The simplest transmission link is pt to pt line that has a transmitter on one end and receiver on other.
- This type of link places least demand on optical fiber technology and thus sets the basic for examining more complex architectures.
- The design of optical fiber link involves many interrelated variables among fiber, source and photodetector operating characteristics, so that actual link design and analysis may require several iteration before they are completely satisfied.
- since performance and cost constraints are very imp factor in optical fiber communication links, the designer must carefully choose components to ensure that desired performance level is maintained.

The following key system requirements are needed in analyzing a link

- * The desired transmission distance
- * The data rate (or) channel bandwidth
- * The bit error rate



The components that used in optical link are

1. Multimode or single-mode optical fiber
 - (a) core size
 - (b) core-refractive index profile
 - (c) Bandwidth (or) dispersion
 - (d) Attenuation
 - (e) numerical aperture (or) mode field diameter

2. LED (or) LASER diode optical source

- (a) Emission wavelength
- (b) Spectral line width
- (c) output power
- (d) Effective radiating area
- (e) Emission pattern
- (f) Number of emitting modes

3. Pin or avalanche photodiode

- (a) Responsivity
- (b) operating wavelength
- (c) speed
- (d) sensitivity

→ Two analyses are carried out to ensure that desired system performance can be met. They are

- * link power budget
- * The system rise-time budget analysis.

- In link power budget analysis, first determine the power margin b/w optical transmitter o/p & minimum receiver sensitivity needed to establish a specified Bit error rate (BER)
- This margin can then be allocated to connector, splice and fiber losses, plus to any additional margins req. for other components, possible component degradation, transmission line impairments or temp effects.
- If choice of components did not allow the desired transmission dist, then components are changed (or) amplifiers are used at link.
- Once a link power budget has been established, the designer can perform a system rise time analysis to ensure that all over system performance has been met.

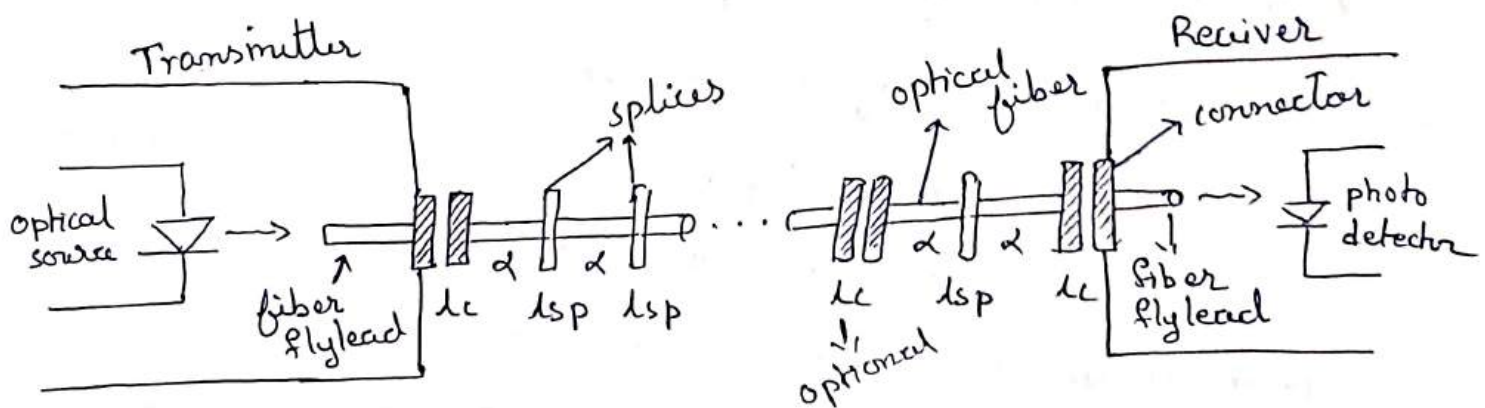
System considerations

- In carrying out a link power budget, we first decide at which wavelength to transmit and then choose components that operate in this region.
- If the distance is not too far away, we may decide to operate in 770- to -910 nm region.
- If transmission dist is long, then we operate at o-band through u-band region.
- If the components have been over- or under specified, a design iteration may be needed.
- First we choose photodetector followed by optical source and see how far data can be transmitted over a particular fiber before an amplifier is needed to boost up power level of optical signal.
- In choosing photodetector, we need to determine minimum optical power that must fall on photodetector to satisfy the BER. the designer need to take into account any cost and complexity constraints.
- A pin photodiode is simpler, more stable with changes in temperature, and less expensive than avalanche photodiode.
- Biasing voltage of pin photodiode is 5V, whereas for avalanche photodiode 40V to 100V.

- But avalanche photodiode is more sensitive than pin photodiode
- The for optical source system parameters involved in deciding b/w the use of LED and laser diodes are
- * dispersion
 - * data rate
 - * transmission dist
 - * cost
- The spectral width of laser o/p is much narrower than that of LED. This is important in 770 to 910 nm region,
- where spectral width of LED and dispersion characteristics of multimode fiber limits data rate to 150 Mb/s
- For higher data rate upto 2500 Mb/s, a laser must be used at these wavelength
- At wavelength around 1.3 μm , where signal dispersion is very low and bit rate is 1500 Mb/s can be achieved with LED in multimode fiber
- A single mode fiber can provide higher rates over longer distances
- Laser diodes couple more power into fiber than LED,
- If we consider cost and complexity along with device ageing, LED is far better than Laser.

- For optical fiber we can choose either single mode or multimode, which could have step index or graded index based on type of light source used and on amount of dispersion that can be tolerated
- LED is tend to be used with multimode, and the optical power that can be coupled into fiber depends on
 - * core-cladding index difference Δ
 - * numerical aperture of fiber
- As $\Delta \uparrow$ coupled power increases which in turn results in increase of dispersion
- A tradeoff must be made b/w the optical power that can be launched into fiber and maximum tolerable dispersion

Link power budget



The losses occur at connector (lc), splices (lsp) and in fiber (α)

→ The optical power received at photodetector depends on amt. light coupled into fiber and losses occurring in fiber, connectors and splices.

→ The link loss budget is derived from sequential loss contribution of each element in link.

Each of these loss element is expressed in (db) as

$$\text{loss} = 10 \log P_{\text{out}} / P_{\text{in}}$$

where P_{in} and P_{out} are optical powers entering and leaving in loss element

→ In addition to link loss contribution contributors, a link power margin is normally provided to know system

* component ageing

* temperature fluctuation

* losses arising from components

→ A link margin of 6-8 db is generally used.

→ The link loss budget simply consider the total optical power loss P_T that is allowed b/w light source & photodetector and allocates this loss to cable attenuation, connector loss, splice loss and system margin.

amt
reacts

If P_s is optical power emerging from end of fiber attached to light source or from
if P_R is receiver sensitivity then

$$P_T = P_s - P_R \\ = 2L_c + \alpha L + \text{system margin}$$

L_c - connector loss

α - fiber attenuation

L - transmission dist.

system margin = 6 dB.

We assume connectors are present at ends only.

We incorporate splice loss into cable loss

LED	LASER Diode
→ Low efficiency	→ High efficiency
→ slow response time	→ fast response time
→ Lower data transmission rate	→ higher data transmission rate
→ Broad o/p spectrum	→ narrow o/p spectrum
→ Incoherent beam	→ coherent o/p beam
→ Low launch power	→ high launch power
→ higher distortion at o/p	→ Less distortion
→ suitable for shorter transmission dist	→ suitable for longer transmission distance
→ higher dispersion	→ Lower dispersion
→ less temperature dependent	→ more temp dependent
→ simple construction	→ construction is completely complicated